Dusk Audio

Plugin User Manual

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Free, professional audio plugins for VST3, LV2, and AU.

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Multi-Comp

Overview

Multi-Comp is a single plugin that switches between eight different compressor flavors plus a four-band multiband. Each mode emulates a distinct hardware design, with its own attack and release behavior, knee shape, and saturation character. The eight modes are Vintage Opto, Vintage FET, Classic VCA, Vintage VCA (Bus), Studio FET, Studio VCA, Digital, and Multiband.

Use it the way you would use any compressor: control dynamic range, glue a bus, tame transients, add audible color. The mode you pick determines how it sounds doing those jobs. Vintage Opto and Vintage FET lean into their character; Studio FET, Studio VCA, and Digital stay out of the way; Vintage VCA (Bus) is purpose-built for mix-bus glue; Multiband splits the signal so you can compress problem frequencies without touching the rest.

Multi-Comp is not a transparent peak limiter (use Digital mode with a low ratio if you need that), and it is not a creative effect plugin (use the Creative preset for the closest thing to one). It is a working compressor that covers most of what a mixing engineer reaches for.

Quick Start

1. Drop Multi-Comp on a track that needs dynamic control. Vocals, bass, and drum bus are good first targets.
2. Set the **Mode** dropdown at the top. If you do not know which mode to pick, start with **Vintage Opto** for vocals and bass, **Classic VCA** for drums, **Vintage VCA (Bus)** for the mix bus.



3. Pull the **Threshold** (or **Peak Reduction** in Opto mode) until you see 3 to 6 dB of gain reduction on the meter. The meter sits to the right of the mode-specific controls. 4. Set **Ratio** to 4:1 for most sources. Ratios above 8:1 are limiting territory. 5. Listen. If the compression sounds choked, lengthen the **Release** (or pick a slower release time on Bus mode). If it sounds soft and lazy, shorten the **Attack**. 6. Use the global **Mix** knob (top of the plugin) to dial in parallel compression. 100% is fully compressed; pulling toward 0% blends the dry signal back in.

You should hear the loud parts come down in level while the quiet parts stay where they are. If you hear pumping on bass-heavy material, raise **SC HP Filter** to 80 to 120 Hz so the compressor stops reacting to the kick fundamental.

Workflows

Lead vocal with Vintage Opto

Source: A pop or rock lead vocal, recorded around -18 dBFS RMS with peaks near -6 dBFS. **Goal:** Even level across the verse and chorus, no audible pumping, the singer still feels alive.

Settings:

- **Mode:** Vintage Opto
- **Peak Reduction:** 50% (this is the sweet spot on the hardware)
- **Gain:** start at 50% (unity), raise to taste once compression is dialed in
- **Limit Mode:** Off
- **SC HP Filter:** 60 Hz
- **Mix:** 100%



Figure 1: Vintage Opto with Peak Reduction at 50%

Why this works. Vintage Opto's optical cell has a program-dependent attack and release, so it does not need you to tune times. It naturally rides level on a vocal. Peak Reduction at 50% gives 4 to 6 dB of gain reduction on a typical lead. The 60 Hz sidechain HP keeps low-frequency rumble from triggering the compressor.

If the vocal still feels uneven, push Peak Reduction toward 70%. If it starts breathing or pumping, drop to 35%.

Drum bus glue with Vintage VCA (Bus)



Figure 2: Bus Compressor with Classic Drum Glue settings

Source: Stereo drum bus with kick, snare, hats, and overheads, summed at around -10 dBFS peak. **Goal:** A cohesive drum kit with the kick and snare punching through, gentle level riding on busy fills.

Settings:

- **Mode:** Vintage VCA (Bus)
- **Threshold:** -15 dB
- **Ratio:** 4:1
- **Attack:** 30 ms (the slowest setting)
- **Release:** Auto
- **Makeup:** 3 dB
- **SC HP Filter:** 90 Hz

- **Bus Mix:** 100%

Why this works. The slowest attack (30 ms) lets the kick and snare transients pass through before the compressor clamps down, which is what makes the drums punch. Auto release adapts to the program material so you do not get pumping on busy passages. The 90 Hz sidechain HP prevents the kick fundamental from over-driving the compressor on every hit.

This is the configuration behind the "Classic Drum Glue" factory preset. If you want more obvious glue, drop Threshold to -18 dB. For a softer, more transparent version, use Ratio 2:1.

Snare punch with Classic VCA

Source: Single snare track, hit hard, peaks at -3 dBFS. **Goal:** Tighter snare with the crack preserved.

Settings:

- **Mode:** Classic VCA
- **Threshold:** -18 dB
- **Ratio:** 4:1
- **Attack:** 15 ms
- **Release:** 80 ms
- **Output:** +3 dB
- **Over Easy:** Off

Why this works. 15 ms is slow enough to pass the initial transient (the "crack") and clamp on the body of the hit. Faster attack (5 ms or less) kills the snap and makes the snare sound dull. 80 ms release recovers in time for the next hit at typical rock tempos.

For a softer, less obvious result, turn **Over Easy** on. This switches the knee from hard to soft (around 6 dB wide), so compression eases in instead of slamming.

De-essing a vocal with Multiband

Source: Lead vocal with harsh sibilance in the 6 to 9 kHz range. **Goal:** Tame the "sss" without dulling the vocal.

Settings:

- **Mode:** Multiband
- Use the **ON** strip toggles to turn off Low, Low-Mid, and Hi-Mid bands. Leave High enabled.
- **Crossover 3:** 6000 Hz (drag the rightmost crossover fader down from 8 kHz)
- **High Threshold:** -28 dB
- **High Ratio:** 6:1
- **High Attack:** 1 ms
- **High Release:** 80 ms
- **MB Mix:** 100%

Why this works. With three bands disabled, the multiband splitter collapses so the High band covers everything from 6 kHz up. A fast attack (1 ms) catches sibilance before it gets loud. Ratio 6:1 is aggressive, but the narrow band limits the audible damage. Watch the High band's gain reduction meter; aim for 4 to 8 dB on the loudest "sss" and near zero on everything else.



Figure 3: Multiband mode default 4-band layout



Figure 4: Multiband collapsed to 1 active band

This same pattern works for kick mud (enable just the Low band, set Crossover 1 around 150 Hz) and for harsh cymbals (just the High band, Crossover 3 around 8 kHz).

Parameter Reference

Global controls (top of the UI)

- **Mode:** Picks the compressor flavor. Eight choices. Mode-specific controls below the drop-down change to match.
- **Bypass:** Reports zero latency to the host while bypassed (no PDC offset on parallel buses).
- **Mix:** 0 to 100%, default 100%. The dry/wet blend at the plugin output. Pull below 100% for parallel compression.
- **Stereo Link:** 0 to 100%, default 100%. How tightly the two channels share gain reduction. 100% means both channels react to the loudest channel; 0% is fully independent (dual mono). Lower values widen perceived stereo on busy material.
- **Link Mode:** Stereo, Mid-Side, or Dual Mono. Mid-Side processes the M and S channels independently; useful for tightening center elements without affecting width.
- **Auto Makeup:** Off or On. When On, automatic level compensation tries to keep output loudness similar to input.
- **SC HP Filter:** 0 to 500 Hz, default 0 (Off). Highpass on the sidechain only (does not affect output). Use 60 to 120 Hz on bass-heavy material to stop low-frequency pumping.
- **Lookahead:** 0 to 10 ms, default 0. Delays the audio by the lookahead time so the detector sees transients before they hit the compressor. Adds latency to the host.
- **Oversampling:** Off, 2x, or 4x, default 2x. Reduces aliasing from the saturation stage. 4x costs about twice the CPU of 2x.
- **External Sidechain:** Off or On. Routes a separate sidechain input from the host (DAW-specific routing required).
- **SC Listen:** Off or On. Sends the sidechain signal to the output for monitoring; turn off before printing.

Mode-specific notes

- **Vintage Opto:** Peak Reduction (0 to 100%) is the only compression control. Limit Mode toggles between gentle (off) and aggressive (on). The release models the T4B optical cell's dual-decay behavior, including the slow "afterglow" that keeps a residual grip on sustained material after the main release has recovered, so long notes settle rather than jumping straight back to unity.
- **Vintage FET:** Ratio is fixed in steps (4:1, 8:1, 12:1, 20:1, All), with the ratio curves re-derived from measurements of the real hardware rather than the datasheet ideal. Threshold (-60 to 0 dB) sets where compression begins. All-Buttons enables the famous extreme compression with non-linear character; Curve Mode picks Modern (the original algorithm) or Measured (matches hardware capture). Transient (0 to 100%) blends raw transient through the compressor. The FET harmonic content scales with gain reduction, so it stays clean when barely working and grows dirtier as it clamps harder.
- **Classic VCA:** Wide threshold range (-38 to +12 dB) and ratio up to 120:1 (limiting). Over Easy switches to soft knee.
- **Vintage VCA (Bus):** A faithful stereo-linked bus compressor modeled on the classic British-style master-buss design. Discrete attack (0.1, 0.3, 1, 3, 10, 30 ms) and release (0.1, 0.3, 0.6, 1.2 s, Auto). The two channels share a single stereo-linked detector so the stereo image stays

stable under gain reduction, and the output carries gain-reduction-scaled harmonics for the characteristic bus "glue." 30 ms attack plus Auto release is the classic glue setting.
- **Studio FET:** Same control set as Vintage FET (including the -60 to 0 dB Threshold) but voiced cleaner with less harmonic content.
- **Studio VCA:** Continuous controls, ratio up to 10:1, voiced for transparent dynamics work.
- **Digital:** Continuous everything, with adjustable Knee width (0 to 20 dB) and Adaptive Release (program-dependent). Use this when you want precision.
- **Multiband:** Four bands (Low, Lo-Mid, Hi-Mid, High) with three Linkwitz-Riley 4th order crossovers. Each band has its own threshold/ratio/attack/release/makeup, plus solo and ON. Disabling a band collapses the splitter so the band's frequency range merges into the nearest enabled neighbor (a real two-band or three-band layout, not just a bypass).

Tips and Traps

- **Auto Makeup hides level changes.** When you compare presets or A/B settings, Auto Makeup can mask whether you actually like the compression or just the level boost. Turn it off when auditioning.
- **Oversampling costs CPU.** 2x is the default and is plenty for most material. Drop to Off for tracking; raise to 4x only on a final mix bus where the saturation is doing audible work.
- **Mode switching does not preserve compression amount.** Each mode has its own threshold scale (Opto's Peak Reduction at 50% is not the same as Digital's threshold at -20 dB). Re-set the compression amount when you switch modes.
- **Lookahead adds latency.** The host applies plug-in delay compensation (PDC), so your other tracks stay aligned, but live monitoring through the plugin will sound late. Bypass during tracking.
- **Multiband bands cannot drop below two enabled.** The plugin enforces a minimum of two active bands. Trying to disable a third band locks the toggle.
- **The Bus Compressor's "Auto" release is the magic setting** for glue. If you can't hear the difference between Auto and 0.6 s, listen during a sparse-to-busy transition; that's where Auto earns its keep.

Presets Explained

Multi-Comp ships with 13 factory presets across six categories. Each is a starting point, not a finished sound; expect to nudge threshold and makeup once it is on your source.

Vocals

- **Smooth Opto Vocal.** Vintage Opto, Peak Reduction 50%, SC HP at 60 Hz. The classic optical vocal sound; reach for this on a lead vocal that needs level control without character changes.
- **Vocal Presence.** Vintage FET, Ratio 4:1, fast attack (~0.5 ms), 60 ms release, SC HP at 100 Hz. Adds the FET-style aggression and presence; good for rock and indie vocals where you want the compressor heard.
- **Modern Pop Control.** Studio FET, Ratio 8:1, very fast attack (0.3 ms), 120 ms release, Auto Makeup on. Tight, transparent peak control for pop and contemporary genres.

Drums

- **Classic Drum Glue.** Vintage VCA (Bus), Ratio 4:1, 30 ms attack, Auto release, SC HP at 90 Hz. The standard drum-bus setting; let the kick and snare punch through, glue the rest.



Figure 5: Preset menu open showing the 6 categories

- **Room Nuke (FET All).** Vintage FET with All-Buttons mode, threshold -24 dB, slightly slower attack to develop the famous plateau. Use on parallel room mics for the smashed-room effect; not a primary compressor.
- **Snare Snap.** Classic VCA, Ratio 4:1, 15 ms attack, lets the snare crack pass before clamping. Apply per-snare, not on the bus.

Bass

- **Rock Bass Anchor.** Vintage FET with a slower attack to keep sub frequencies clean.
- **Vintage Pinned Bass.** Vintage Opto with maximum Peak Reduction for the classic Motown "pinned" feel.

Guitars

- **Acoustic Strum Tamer.** Classic VCA with a fast attack to control strumming spikes.
- **Funk Rhythm Guitar.** Vintage FET with a fast release; emphasizes the "up" stroke.

Mix Bus

- **British-style Glue.** The 4:1, 10 ms attack, Auto release, 4 dB GR setting; the canonical mix-bus compressor sound.
- **Gentle Master.** Studio VCA at Ratio 1.5:1; a low-ratio, modern mastering-bus starting point.

Creative

- **EDM Pump (115-130 BPM).** Heavy compression with a release tuned to quarter notes in the 115 to 130 BPM range; the "side-chain pump" effect without an actual sidechain.

Troubleshooting

I hear nothing. Check the Bypass button at the top, check Mix is not at 0%, and confirm your DAW track output is not muted. If on Multiband mode, the GR meter shows the loudest band; if all bands show zero GR even on a loud signal, your Threshold is set above the input level.

The compressor is pumping on bass-heavy material. Raise SC HP Filter to 80 to 120 Hz. The detector stops reacting to the kick fundamental.

The CPU usage is high. Drop Oversampling from 4x to 2x or Off. Disable True Peak detection if it is on (it adds a 4x or 8x detector path). Multiband at 4x oversampling on a busy bus is the worst-case combination.

The audio sounds late through the plugin. Lookahead and Digital mode's internal lookahead both delay output. Your host applies PDC for playback alignment, but live monitoring is still delayed. Bypass during tracking.

The mode dropdown changes my sound dramatically. That is intentional. Each mode has its own threshold scale, attack and release behavior, and saturation profile. After switching, re-set the compression amount on the new mode's controls.

Multi-Q

Overview

Multi-Q is three EQs in one plugin. The mode switch at the top picks between **Digital** (an 8-band parametric with per-band dynamic EQ), **British** (a four-band console EQ with Brown and Black character options), and **Tube** (a vintage passive tube-style EQ with the classic "boost and cut at the same frequency" interaction). Each mode is voiced and operated differently; switching modes is a creative choice as much as a workflow choice.

Use Digital for surgical work and dynamic frequency control. Use British for tonal shaping with console color (it is the same engine as 4K-EQ, integrated). Use Tube for vintage warmth and the unique passive-EQ curves that interact between bands. The real-time spectrum analyzer above the curve display works across all three modes.

It is not a corrective de-noise tool, and it is not a mastering limiter; pair it with a compressor or limiter for level work. It is a clean, capable EQ that swaps personalities at will.

Quick Start

1. Insert Multi-Q on the source you want to shape. Pick a mode at the top: **Digital**, **British**, or **Tube**.
2. The main display shows your EQ curve overlaid on a real-time spectrum. Click and drag a band's control point to set frequency (horizontal) and gain (vertical).
3. Mouse-wheel over a control point to adjust Q (band width).
4. Right-click a control point for band-specific options (filter shape, dynamic EQ activation in Digital mode, Brown/Black switch in British mode).
5. Use the **Bypass** button to A/B against unprocessed audio.
6. The **A/B** preset slots at the top let you compare two settings: load Multi-Q, set up your EQ on slot A, click **B** to switch to a fresh slot, build a different EQ, then toggle between A and B by clicking the letter.

You should hear the source change tonal balance immediately. If the band you are dragging produces no audible change, check the Q (very high Q with low gain is nearly inaudible) and confirm the band is enabled (in Digital mode, the band-strip toggles below the curve let you disable individual bands).

Workflows

Digital mode: surgical resonance notch

Source: A snare drum with a ringy 380 Hz resonance. **Goal:** Remove the ring without dulling the snare.

Settings:

- **Mode:** Digital
- Choose any unused parametric band. Drag its control point to 380 Hz.
- **Filter shape:** Bell (default for parametrics)
- **Gain:** -8 dB
- **Q:** 6 (narrow)



Figure 6: Multi-Q in Digital mode

Why this works. A narrow bell at the offending frequency cuts only the resonance and leaves the rest of the snare unchanged. To find the exact frequency, set Gain to +8 dB temporarily, sweep the band's frequency, and listen for where the ringing gets worse; that is your target. Then flip Gain to -8 dB.

For an even more surgical version, switch the band's filter shape to **Notch** (right-click). A notch is sharper than a high-Q bell and removes the resonance entirely.

Digital mode: dynamic de-essing

Source: A vocal with sibilance peaks in the 6 to 8 kHz range. **Goal:** Reduce the sibilance only when it gets loud, leave the rest of the vocal alone.

Settings:

- **Mode:** Digital
- Pick the High Shelf band or any parametric band. Drag to 7000 Hz.
- **Filter shape:** Bell
- **Gain:** 0 dB (the static cut is zero; dynamics will do the work)
- **Q:** 1.5

Right-click the band to access **Dynamic EQ** controls:

- **Threshold:** -28 dB
- **Range:** -8 dB (max cut applied when threshold is exceeded)
- **Attack:** 1 ms
- **Release:** 80 ms



Figure 7: Dynamic EQ controls on a band

Why this works. The static gain stays at 0, so non-sibilant content passes through untouched. When sibilance pushes the band's energy above -28 dB, the dynamic EQ cuts up to 8 dB at that frequency. Fast attack catches the transient; release back to zero in 80 ms. The "Multiband De-Ess" factory preset uses this approach.

British mode: vocal sweetening with console character

Source: A lead vocal that needs presence and warmth without surgical cuts. **Goal:** Tonal balance with audible analog character.

Settings:

- **Mode:** British
- **EQ Type:** Brown (E-series) for the warmer character
- **HPF:** On, 80 Hz (cleans up rumble)
- **LF Gain:** -3 dB at 300 Hz, Bell mode
- **HM Gain:** +4 dB at 3500 Hz, Q 0.7
- **HF Gain:** +2 dB at 8000 Hz, Shelf
- **Saturation:** 15%

Why this works. British mode in Multi-Q runs the same filter engine as 4K-EQ, so the "Vocal Presence" approach from 4K-EQ applies directly. The 300 Hz cut clears boxiness, the 3.5 kHz boost adds intelligibility, the 8 kHz shelf adds air. With Brown (E-series) selected, Saturation at 15% adds low-weighted transformer-core harmonics that thicken the vocal's body and help it sit in a mix without surgical work. For a more aggressive character, switch to Black mode.



Figure 8: Multi-Q in British mode



Figure 9: Multi-Q in Tube mode with boost+cut at 60 Hz

Tube mode: vintage bass with the boost-and-cut trick

Source: Bass guitar that needs both fullness and clarity. **Goal:** Add low-frequency body and tighten the muddy lower-mids.

Settings:

- **Mode:** Tube
- **LF Boost:** 60 Hz, +4 dB
- **LF Attenuation:** 60 Hz, +3 dB (boost AND cut at the same frequency)
- **HF Boost:** 5000 Hz, +2 dB
- **Tube Drive:** 25%

Why this works. The vintage passive EQ trick is to boost and cut at the same low frequency. The interaction of the two passive curves creates a unique shape: more energy at the boost peak (around 60 Hz) AND a slight dip just above it (where the boost rolls off and the cut kicks in). The result is a punchy low end that does not muddy the mids. Tube Drive at 25% adds the harmonic warmth of the simulated 12AX7 stage.

This is not a digital trick; it mimics how real passive tube EQs behave. The "Vintage Bass Trick" factory preset shows this exact approach.

Parameter Reference

Mode-independent (always visible)

- **Mode:** Digital, British, or Tube. Each mode has its own parameter set; the visible controls change when you switch.
- **Bypass:** Reports zero latency to the host while bypassed.
- **Output Gain:** Final makeup gain.
- **Auto Gain Compensation:** Keeps output level matched to input as you boost and cut.
- **Spectrum Analyzer Pre/Post:** Pre shows the input signal; Post shows the EQ output.
- **Oversampling:** 2x or 4x. Reduces aliasing in saturation stages (British and Tube modes).
- **A/B Slots:** Two complete parameter snapshots accessible from the top toolbar. Useful for comparing two different EQ approaches on the same source.

Digital mode

8 fully parametric bands (HPF at the low end, Low Shelf, four mid parametrics named Low / Lo-Mid / Mid / Hi-Mid, High Shelf, LPF at the high end). Each band has:

- **Frequency:** 20 Hz to 20 kHz (range varies by band).
- **Gain:** -24 to +24 dB.
- **Q:** 0.1 to 36 (effectively notch-narrow at the high end).
- **Filter shape:** Bell, Shelf, Pass, Notch, or Band Pass (choices vary by band).
- **Enabled:** Per-band toggle in the strip below the curve.
- **Dynamic EQ:** Threshold (-60 to 0 dB), Attack (0.1 to 100 ms), Release (10 to 1000 ms), Range (-24 to +24 dB). Right-click a band to expand the dynamic controls.

The HPF and LPF have selectable slopes (12, 18, or 24 dB/oct). The mid parametrics support Notch and Band Pass shapes for surgical work.

British mode

Same controls as 4K-EQ:

- **HPF / LPF:** Frequency and enabled toggles.
- **Four bands** (LF, LM, HM, HF): gain, frequency, Q (LM/HM only); LF and HF have Bell/Shelf switches.
- **EQ Type:** Brown or Black. Brown (E-series) drives the saturation stage from an E-series transformer-core model whose drive is weighted toward the low frequencies, so pushing Saturation thickens and adds weight to the low end rather than coloring the whole band evenly. Black is the cleaner console voicing.
- **Saturation:** 0 to 100%.
- **M/S Mode:** Off or On.

British mode is built on the 4K-EQ engine and shares its filter topology, so the 4K-EQ chapter's parameter notes apply. The one difference is the Brown/E-series transformer saturation above, which is voiced specifically for Multi-Q's British mode; see the 4K-EQ chapter for the filter details.

Tube mode



Figure 10: Tube mode controls

- **LF Boost:** Frequency selectable (typically 20, 30, 60, 100 Hz), gain 0 to +12 dB.
- **LF Attenuation:** Frequency selectable, gain 0 to +12 dB (cut amount; passive design).
- **HF Boost:** Frequency selectable (3, 5, 10, 16 kHz), gain 0 to +12 dB, with **Bandwidth** control (broad or narrow).
- **HF Attenuation:** Frequency selectable, gain 0 to +12 dB.
- **Mid Dip / Peak:** A center-frequency band with selectable curve shape; useful for vocal presence shaping.
- **Tube Drive:** 0 to 100%. Drives the simulated 12AX7 triode stage. 0 is clean; 30 to 50% adds audible second-harmonic warmth.

Boost and Attenuation at the same frequency interact in the passive style: the curve becomes neither pure boost nor pure cut but a unique interaction shape.

Tips and Traps

- **Modes do not share band settings.** A 4 kHz boost in Digital mode is not present when you switch to British, even though the curves of both can pass through 4 kHz. Each mode has its own parameter set; the mode switch is a hard switch, not a re-mapping.
- **Right-click for hidden options.** Filter shape, dynamic EQ, M/S routing per-band, all live in the right-click menu. Beginners often miss them and miss half the plugin's capability as a result.
- **The boost-and-cut trick is mode-specific.** Only Tube mode has that interaction. In Digital or British modes, boosting and cutting at the same frequency cancels out (or nearly so).
- **A/B is your friend on long EQ sessions.** Set A as your starting point. Build B to compare. If neither sounds right, click A again to reset. The A/B slots are persistent within the session; they do not save with presets.
- **British mode shares the 4K-EQ filter engine.** The EQ curves match, so a 4K-EQ preset you like can be recreated in Multi-Q's British mode by hand. The one thing that differs is Brown/E-series saturation, which is voiced with a low-weighted transformer-core drive here; matched EQ settings will track, but heavy saturation will sound a touch fuller in the lows.
- **The spectrum analyzer is a guide, not a sound.** Pre/Post toggle shows you what changed; do not let the pretty curve trick you. Listen first, look at the spectrum to confirm.

Presets Explained

Multi-Q ships with around 50 factory presets across roughly 8 categories. They are organized by source (Vocals, Drums, Guitars, Bass, Mix Bus), by mode (each mode contributes presets), and by use case (Mastering, Creative, Dynamic, M/S). Listing every one would balloon this manual; instead, here are the themes you should know about.

Vocals presets

Multiple presets across all three modes. Digital-mode "Vocal Presence" uses static EQ for clarity and air. British-mode "Console Vocal Chain" applies the 4K-EQ Vocal Presence approach plus saturation. Tube-mode "Warm Vocal (Tube)" uses LF boost-and-cut for body and Tube Drive for harmonic warmth. Pick by what character you want; all three are valid starting points for a lead vocal.

Drums presets

Per-source: "Punchy Kick", "Snare Crack", "Overhead Clarity", and "Rock Drums" (a British-mode bus preset). Digital-mode kick presets often pair a low-shelf boost with a narrow mid cut for click. British-mode drum presets add saturation. Tube-mode drum presets are rare; tube character is usually too colored for kit work.

Mastering presets

"Mastering Surgical" uses Digital with narrow bells; "Mastering Air" lifts the top end with shelves; "Mastering Wide" uses M/S processing for stereo width. "Vintage Air" and "Console Warmth" sit in British and Tube modes respectively for analog flavor on a master.

Dynamic-EQ presets (Digital mode only)

"Multiband De-Ess" applies dynamic cut at sibilance frequencies. "Dynamic Bass Control" tames bass build-up dynamically. "Resonance Tamer" is a starting point for cutting unwanted resonances that only show up at higher levels. These presets demonstrate per-band dynamic EQ; right-click a band on any of them to see how the dynamics are wired.

Creative presets

"Telephone Effect" (narrow band-pass for radio voice), "Lo-Fi Warmth" (heavy filtering with British-mode saturation), "Stereo Width" (M/S width tricks). Use these as effects rather than starting points for serious mixing.

Vintage presets (Tube mode)

"Vintage Bass Trick" demonstrates the boost-and-cut at the same frequency. "Vintage Air" shelves the top end gently. "Warm Vocal (Tube)" combines LF boost-and-cut with Tube Drive. These are the showcases for what Tube mode can do that the other modes cannot.

If you want to study the presets, open one and right-click each band to inspect its settings. The preset names suggest use cases; the band-by-band layout shows the technique.

Troubleshooting

The plugin sounds the same in all three modes. Confirm you are actually switching modes (the curve display changes when you do; the band labels change too). With all bands flat, the difference between modes is only the saturation and the filter topology of any active filters. To hear the difference, boost or cut a band in each mode and compare.

The spectrum analyzer shows different curves than the EQ shape. That is intentional. The analyzer shows actual frequency content; the EQ curve shows the filter response. They are different views. Pre/Post toggle on the analyzer compares input to output.

Right-click does nothing. On Linux, some hosts intercept right-click for their own menus. Try Ctrl+click or check your host's preferences for plugin context-menu handling.

My British mode preset sounds different from a 4K-EQ preset with the same settings. The filter curves are the same, so a difference usually comes from one of three things: British mode's Brown/E-series saturation uses a low-weighted transformer-core drive that differs from 4K-EQ, so match the Saturation amount and EQ Type; or check that both plugins are at the same sample rate; or confirm Auto Gain is either on for both or off for both.

4K EQ 2

Overview



Figure 11: 4K EQ 2 with Vocal Presence loaded

4K EQ 2 models the equalizer and high-pass/low-pass filters of a classic British E-series channel strip. The Brown and Black modes are separate voicings, not cosmetic themes. Each mode uses its own measured frequency, gain, Q, shelf, filter, interaction, nonlinearity, and overload behavior.

Use it for broad tonal decisions where a console EQ is faster than a surgical parametric equalizer: shaping drums, moving a vocal forward, removing low-end clutter, or adding a restrained shelf to a mix bus. The LMF and HMF sections remain fully adjustable when a narrower correction is needed, but this is not a dynamic EQ or a linear-phase mastering tool.

The modeled EQ path has a fixed amount of native nonlinear color. There is no Drive control. Raise Input when you want to hit the modeled path harder, then trim Output for a fair comparison. The preamp and dynamics portions of the original channel strip are intentionally outside this plugin.

Quick Start

1. Insert 4K EQ 2 and click **INIT**. Leave Input and Output at 0 dB, Auto Gain off, and the filters at OUT while you learn the bands.
2. Start in **Brown** mode. Raise one gain control to about +6 dB, sweep its frequency until the target becomes obvious, then reduce the gain to the amount the mix needs.

3. Use **LMF Q** or **HMF Q** to set bandwidth. Lower values make a wider move; higher values focus the boost or cut.
4. Use the LF or HF button to switch that band between **SHELF** and **BELL**.
5. Turn the HPF away from OUT to remove low-frequency buildup. Turn the LPF away from OUT only when the source needs deliberate top-end restriction.
6. Switch to **Black** and compare at the same settings. Black uses a different control law and proportional-Q mid bands, so it can feel tighter as the gain move grows.
7. If you raise Input for more nonlinear character, lower Output by roughly the same amount. Enable Auto Gain only as a convenience while auditioning.

The LF, LMF, HMF and HF frequency knobs are in Hz: the mark a knob points at, the value bubble and a typed value are the frequency the band plays. For a bell that is its centre. For a shelf it is the corner: from about 3 dB of boost or cut, roughly half to two-thirds of it is in at that frequency, and nearly all of it two octaves further out. The bubble does not change when you turn a gain knob. The band itself moves slightly with gain, as the console's does: a bell by at most about 2.3%, a shelf's corner by at most about 4% at 3 dB of gain or more. The HPF and LPF knobs are in Hz too: the frequency where the filter is 3 dB down, the same in Brown and Black. Their marks sit where the console's printed dial marks do.

Workflows

Vocal presence without losing body

Source: Lead vocal peaking around -10 dBFS with low-frequency proximity build-up. **Target:** Reduce rumble and boxiness while moving consonants forward.

- Brown mode
- HPF at 80 Hz
- LF +3 dB at 100 Hz, shelf
- LMF -3 dB at 300 Hz, Q 1.3
- HMF +4 dB at 3.5 kHz, Q 0.7
- HF +2 dB at 8 kHz, shelf
- Input and Output at 0 dB; Auto Gain off

This is the **Vocal Presence** factory preset. The HPF clears material below the useful vocal range, the LMF cut reduces boxiness, and the wide HMF/HF boosts add intelligibility and air. If the result is sharp, lower the HMF gain first rather than moving every frequency.

Kick with weight and beater definition

Source: Close kick microphone with useful sub energy but too much 200 Hz body. **Target:** Add low-end weight and a readable attack in a dense mix.

- Brown mode
- HPF at 30 Hz
- LF +4 dB at 50 Hz, shelf
- LMF -2.5 dB at 200 Hz, Q 0.8
- HMF +3 dB at 2 kHz, Q 1.5
- HF flat
- Input +2 dB and Output -2 dB if additional path color is wanted

The broad LF move supplies weight, the LMF cut makes room above it, and the narrower HMF boost adds beater definition. Start from **Kick Punch** and move the LF frequency toward 70 or 80 Hz when the drum's fundamental sits higher.

Black-mode drum bus shaping

Source: Stereo drum bus with enough level but soft low-end impact and a cloudy low-mid range. **Target:** Tighter punch with a modest top lift.

- Black mode
- Filters at OUT
- LF +4 dB at 70 Hz, shelf
- LMF -3 dB at 350 Hz, Q 0.6
- HMF +3 dB at 3.5 kHz, Q 1.0
- HF +2.5 dB at 10 kHz, shelf
- Input 0 dB; Output 0 dB; Auto Gain off

This is **Drum Bus Punch**, the factory bank's Black-mode preset. Black's proportional-Q mid behavior keeps larger moves more focused. If cymbals become forward, reduce the HF gain before changing the HMF presence move.

Acoustic guitar cleanup and detail

Source: Close-miked steel-string guitar with low-body buildup and muted pick detail. **Target:** Clear space below the instrument and recover articulation.

- Brown mode
- HPF at 80 Hz
- LF -2 dB at 100 Hz, shelf
- LMF +2 dB at 200 Hz, Q 0.7
- HMF +3 dB at 2.5 kHz, Q 0.9
- HF +4 dB at 12 kHz, shelf

Load **Acoustic Guitar** as the starting point. If the guitar is already thin, return LF to 0 dB before reducing the HPF. If finger noise dominates, lower the HMF move while keeping the HF shelf for openness.

Restrained mix-bus sweetening

Source: Balanced mix with headroom and no major corrective problem. **Target:** Small tonal moves that remain useful after level matching.

- Brown mode
- Filters at OUT
- LF +1 dB at 50 Hz, shelf
- LMF -1 dB at 600 Hz, Q 0.5
- HMF +0.5 dB at 4 kHz, Q 0.6
- HF +1.5 dB at 15 kHz, shelf
- Output -0.5 dB
- Input 0 dB; Auto Gain off

This is **Master Bus Sweetening**. Compare with Bypass at the same perceived level. On a finished mix, a 0.5 dB change is meaningful; avoid compensating for an unsuitable curve by driving the modeled path harder.

Parameter Reference

Header and presets

- **Preset browser:** Recalls 14 factory programs and saved user presets. Factory recall owns the sound-shaping parameters but leaves oversampling and analyzer preferences alone. A common mistake is expecting a recalled preset to restore the previous oversampling choice.
- **INIT:** Restores all preset-owned parameters to their defaults. It does not alter Bypass, oversampling, or analyzer state. Use it for a flat starting point.
- **SAVE:** Writes the current sound-shaping state to the user preset library. On macOS and Linux, presets live under `~/.config/DuskAudio/FourKEQ2/presets`; Windows uses the application-data directory. Invalid or unsupported preset files are ignored by the browser.
- **OS:** Cycles 1x, 2x, and 4x; default 4x. Higher factors reduce aliasing from the nonlinear path but use more CPU and add latency. At 88.2 or 96 kHz the effective maximum is 2x; at 176.4 or 192 kHz it is 1x.
- **BROWN / BLACK:** Selects the complete console voicing; default Brown. Brown uses constant-Q mid bands. Black applies proportional Q as boost or cut grows.
- **GRAPH:** Shows or collapses the response graph. The graph range menu offers +/-6, +/-12, +/-18, +/-30 dB, and a warped view.
- **FFT:** Enables or disables the live spectrum trace. Disable it when you want the lowest possible UI overhead.
- **PRE / POST:** Selects the analyzer tap before or after the EQ. It changes only the display, not the audio path.

Filters

- **HPF:** OUT by default, then 16 to 350 Hz, the frequency where the filter is 3 dB down. Moving away from OUT enables the filter. Brown and Black use different measured slopes. Use it to remove rumble; the common mistake is setting it by number alone without listening to the source's fundamental.
- **LPF:** OUT by default, then 15.201 down to 3 kHz, again the 3 dB point. Use it for deliberate bandwidth restriction or high-frequency cleanup. The control runs downward in frequency as it turns clockwise, matching the panel layout.

EQ bands

- **LF Gain:** -15 to +15 dB, default 0 dB. **LF Frequency:** 30 to 450 Hz, default 200 Hz. Like the other band frequencies, it is the Hz the band plays. **SHELF/BELL:** shelf by default. Use the shelf for broad weight and the bell for a localized low-frequency move.
- **LMF Gain:** -15 to +15 dB, default 0 dB. **LMF Frequency:** 200 Hz to 2.5 kHz, default 1 kHz. **LMF Q:** 0.5 to 3.0, default 1.5. Use lower Q for broad body changes and higher Q for a focused low-mid correction.
- **HMF Gain:** -15 to +15 dB, default 0 dB. **HMF Frequency:** 600 Hz to 7 kHz, default 3 kHz. **HMF Q:** 0.5 to 3.0, default 1.5. Use it for presence, attack, or upper-mid harshness.
- **HF Gain:** -15 to +15 dB, default 0 dB. **HF Frequency:** 1.5 to 16 kHz, default 8 kHz. **SHELF/BELL:** shelf by default. Use the shelf for air and the bell when the desired change is centered below the extreme top end.

At exactly 0 dB, each EQ section bypasses its recursive filter state. This keeps the neutral path free of the subsonic numerical residue that can result from running nominally unity biquads.

Master and metering

- **Input:** -12 to +12 dB, default 0 dB. It sets the level entering the modeled EQ path and therefore the nonlinear behavior. Do not treat it as an output loudness control.
- **Output:** -12 to +12 dB, default 0 dB. Use it to restore level after Input changes or deliberate EQ boosts.
- **Bypass:** Off by default. It uses a short smoothing transition to avoid a click. Compare at matched level.
- **Auto Gain:** Off by default. It estimates compensation from the active EQ curve. It is useful while exploring, but a spectrum-based estimate cannot guarantee equal perceived loudness for every source.
- **IN / OUT meters:** Peak meters at the panel edges. The numeric hold refreshes more slowly than the bars so short peaks remain readable.

Tips and Traps

- Brown and Black put a band at the same frequency for the same knob setting. They differ in Q, gain law, band interaction and color.
- Up to version 1.0.5 the band frequency parameters were positions on the console's printed dial, which played a different frequency from the number: HMF stopped at about 6.4 kHz in Brown mode, and the Brown HF shelf never went past about 6 kHz. Those parameters are still there, named "LF Frequency (Legacy Dial)" and so on, so older sessions and their automation sound as they did. A band follows whichever of its two frequency parameters was set last. New automation belongs on "LF Frequency", "LM Frequency", "HM Frequency" and "HF Frequency", which are in Hz.
- The HPF and LPF frequency parameters were dial positions too: an HPF set to 80 was 3 dB down at about 26 Hz in Brown mode. They work the same way as the bands, as "HPF Frequency (Legacy Dial)" and "LPF Frequency (Legacy Dial)" beside "HPF Frequency" and "LPF Frequency" in Hz. A filter from an older session can play outside its knob's range, for example a Brown LPF above 15.2 kHz; the knob then rests at its end stop.
- A band from an older session can play a frequency outside its knob's range, for example a Black LF shelf above 450 Hz. The knob then rests at its end stop and the bubble shows the frequency the band plays. Turning the knob moves the band onto the knob's range.
- A large EQ boost can hit the modeled rail even when the DAW input meter looks safe. Lower Input or the band gain if the overload is not intentional.
- Auto Gain is a convenience, not part of the modeled channel-strip signal path. Factory presets turn it off so their level and curve recall deterministically.
- Oversampling changes reported latency. Avoid changing it while recording or while a host is not prepared to refresh plugin delay compensation.
- The old Drive and M/S parameter indices may still appear in generic host views for compatibility. They are hidden or inert and are not part of the UI. So do the six legacy dial frequencies, "Legacy Dial Bands" and "Legacy Dial Filters", which record which bands and filters follow them.

Presets Explained

- **Vocal Presence:** Brown vocal cleanup and forward presence. Adjust the HPF first for the singer, then the 3.5 kHz HMF gain.
- **Kick Punch:** Low shelf, low-mid cut, and 2 kHz attack. Move LF frequency to the drum's fundamental.
- **Snare Crack:** 250 Hz body, 5 kHz crack, and an 8 kHz HF bell. Lower HMF if the snare becomes brittle.
- **Drum Bus Punch:** The single Black factory preset, with tighter low-mid and presence shaping. Reduce HF first on cymbal-heavy buses.
- **Bass Warmth:** 80 Hz weight with a 400 Hz cut and 10 kHz LPF. Move the LPF out for bright bass parts.
- **Bass Guitar Polish:** 60 Hz weight, 250 Hz cleanup, and upper-mid definition. Adjust HMF for finger or pick noise.
- **Acoustic Guitar:** HPF cleanup plus 2.5 kHz detail and a 12 kHz shelf. Return LF to flat on thin recordings.
- **Piano Brilliance:** Low foundation, 500 Hz cleanup, and two presence lifts. Reduce HF on close, bright piano recordings.
- **Bright Mix:** 60 Hz and 10 kHz shelves around a 2.5 kHz dip. Keep each move small on a finished mix.
- **Glue Bus:** A broader mix-bus contour centered on 100 Hz, 3 kHz, and 10 kHz. Use Output for level matching.
- **Telephone EQ:** 300 Hz HPF, 3 kHz LPF, and 1 kHz LMF boost. Adjust the LMF gain to control how nasal the result becomes.
- **Air & Silk:** HMF at 7 kHz plus a 15 kHz shelf. Lower HMF before HF when sibilance rises.
- **Master Sheen:** Small 5 kHz and 16 kHz lifts. Start here when the mix needs only top-end polish.
- **Master Bus Sweetening:** Four sub-1.5 dB mastering moves with -0.5 dB output. Bypass frequently and keep the change only if it survives level matching.

Troubleshooting

The EQ appears to do nothing. Confirm Bypass is off and at least one band gain is away from 0 dB, or an HPF/LPF knob is not OUT. The HPF and LPF remain bypassed only while their knobs read OUT.

A band frequency knob sits at its end stop. The band follows a legacy dial setting from a session saved with version 1.0.5 or earlier, and the frequency it plays is outside the knob's range. The bubble shows that frequency. Turn the knob to bring the band back onto the knob's scale.

The HPF or LPF knob sits at its end stop. The filter follows a legacy dial setting from a session saved with version 1.0.5 or earlier, and its 3 dB point is outside the knob's range. The bubble shows it. Turn the knob to bring the filter back onto the knob's scale.

In an LV2 host, a band set through its legacy dial moves after a reload. After a preset is applied, a write to a band's or filter's "(Legacy Dial)" parameter, from a generic parameter view or from automation an older version recorded there, is not kept when the session reloads: the band or filter goes back to its Hz frequency. Applying the same preset again does not bring it back to the preset's frequency either; the band stays on the dial. An LV2 plugin cannot write its own input ports, and hosts such as Ardour do not implement the request that would let it, so the host keeps the value the preset gave the parameter that records which bands follow their dials. Set the

band on its knob or with its Hz parameter ("LF Frequency" and so on) instead, and it keeps that frequency through a reload.

The plugin becomes louder or distorts after an Input change. Lower Output by the same amount for comparison. If it still overloads, lower Input or the largest EQ boost. Auto Gain compensates the curve estimate, not every nonlinear level change.

A saved preset is missing from the browser. Check that the preset directory is writable. Files with malformed numeric fields, unsupported format versions, or unsupported frequency domains are rejected rather than partially loaded.

4K-EQ

Overview



Figure 12: 4K-EQ main UI

4K-EQ is a 4-band parametric equalizer modeled on the British large-format console channel strip. The four bands (Low, Low-Mid, High-Mid, High) cover the full audio spectrum, with separate high-pass and low-pass filters at the input. Two character modes ("Brown" and "Black") swap the EQ curves between the E-series and British G-series designs.

Use it where you would use a console EQ: tonal shaping during the mix, gentle pre-master sweetening, broad corrective work on individual sources. The bands are musical rather than surgical; if you need to notch out a single resonant peak, reach for Multi-Q's Digital mode instead.

The British-style saturation stage adds a small amount of harmonic color when driven, and stays clean when the input gain is conservative. Auto-gain compensation keeps the output level matched to input as you boost and cut, so an A/B comparison reflects tone changes only.

Quick Start

1. Insert 4K-EQ on the source you want to shape. The interface shows four band sections (LF, LM, HM, HF) with HPF on the far left and LPF on the far right.
2. Pick a character with the **EQ Type** switch at the top: **Brown** for the warmer, slightly soft E-series sound, **Black** for the more aggressive G-series with extended highs and lows.

3. Sweep one band's **Frequency** while you boost its **Gain** by about 6 dB. Listen for the area you want to enhance, then back the gain off to 2 to 4 dB.
4. Use the **Q** knob (on LM and HM) to widen or narrow the affected range. The default Q (0.7) is broad and musical; raise toward 2 for surgical work.
5. The **HPF** and **LPF** are off by default. Toggle them on to clean up rumble (HPF at 60 to 100 Hz on most sources) or tame harshness (LPF only when needed).
6. Leave **Auto Gain Compensation** on while you work. It keeps levels matched so your ears focus on tone, not loudness.

You should hear the source change tonal balance without changing apparent level. If you do hear a level change, double-check Auto Gain is on and that you have not pushed any band past 12 dB.

To dial in an exact setting, double-click any knob to type a value directly instead of dragging.

Workflows

Vocal presence and clarity



Figure 13: Vocal Presence preset loaded

Source: A lead vocal that sits in the mix but needs more presence and air. **Goal:** Lift the singer forward without harshness.

Settings:

- **EQ Type:** Brown
- **HPF:** On, 80 Hz
- **LF Gain:** -3 dB at 300 Hz, Bell Mode On
- **HM Gain:** +4 dB at 3500 Hz, Q 0.7
- **HF Gain:** +2 dB at 8000 Hz, Shelf
- Other bands flat.

Why this works. The HPF at 80 Hz removes proximity rumble and headphone bleed without thinning the body. The LF cut at 300 Hz tames the boxy buildup that lives there in close-mic'd vocals. The HM boost at 3.5 kHz adds intelligibility (the consonant range). The 8 kHz shelf adds air. The Brown character keeps the boost gentle. This matches the "Vocal Presence" factory preset.

If the vocal sounds harsh after this, drop HM Q to 0.5 (wider, softer boost) or move the HM frequency down to 2.5 kHz. If it still sounds dull, try Black mode for a touch more clarity.

Kick drum punch

Source: Close-mic'd kick drum. **Goal:** Solid low-end thump and audible click for cut-through.

Settings:

- **EQ Type:** Brown
- **HPF:** On, 30 Hz (removes sub-rumble below the fundamental)
- **LF Gain:** +4 dB at 50 Hz, Shelf
- **LM Gain:** -2.5 dB at 200 Hz, Q 0.8
- **HM Gain:** +3 dB at 2000 Hz, Q 1.5
- **HF Gain:** flat
- **Saturation:** 10%

Why this works. The 50 Hz shelf adds weight; bell mode at the same frequency would only add to a narrow range, and you want the whole low end lifted. The 200 Hz cut removes "boxy" buildup that competes with the bass. The 2 kHz bell with a tight Q (1.5) brings out the beater click. A small amount of console saturation (10%) thickens the body without obvious distortion. This matches the "Kick Punch" factory preset.

For a thumpier kick, raise LF center to 70 Hz. For more click, raise HM gain to +5 dB.

Mix bus glue

Source: Stereo mix bus, near final balance. **Goal:** Add console color and gentle high-end sheen without changing the mix.

Settings:

- **EQ Type:** Black (G-series for slightly tighter low end)
- **LF Gain:** +2 dB at 60 Hz, Shelf
- **HM Gain:** -2 dB at 2500 Hz, Q 0.8
- **HF Gain:** +2.5 dB at 10000 Hz, Shelf
- **Saturation:** 20%
- **Auto Gain Compensation:** On

Why this works. A wide 2 dB shelf at 60 Hz adds weight without muddying. A gentle 2 dB scoop at 2.5 kHz pulls the harshness band back, which can fight a mix when many instruments live in

that range. The 10 kHz shelf adds gloss. 20% saturation gives the bus the harmonic glue of a real console without obvious distortion. This is the "Bright Mix" factory preset's approach. The "Glue Bus" preset uses similar settings with a slightly higher saturation and a touch more low end.

Parameter Reference

Filters (HPF and LPF)

- **HPF Frequency:** 20 to 500 Hz, default 20. The corner frequency of the high-pass filter (18 dB/oct). Use 60 to 120 Hz on vocals and most instruments to remove rumble.
- **HPF Enabled:** Off by default. Truly bypassed when off (no phase or magnitude effect on the signal).
- **LPF Frequency:** 3 to 20 kHz, default 20 kHz. The corner frequency of the low-pass filter (12 dB/oct).
- **LPF Enabled:** Off by default. Use sparingly; over-aggressive low-pass filtering dulls a track.

Low frequency band (LF)

- **LF Gain:** -20 to +20 dB, default 0. Hardware spec is ±15 dB on Brown and ±18 dB on Black, but the plugin lets you go further when you need to.
- **LF Frequency:** 30 to 480 Hz, default 100. Center frequency for bell mode, corner frequency for shelf mode.
- **LF Bell Mode:** Off (shelf) by default. Shelf mode boosts or cuts everything below the frequency; bell mode affects only a band around the frequency.

Mid bands (LM and HM)

- **LM Gain / HM Gain:** -20 to +20 dB. Always bell-shaped (no shelf mode on the mid bands).
- **LM Frequency:** 200 to 2500 Hz, default 600.
- **HM Frequency:** 600 to 7000 Hz, default 2000.
- **LM Q / HM Q:** 0.4 to 4.0, default 0.7. Lower Q is wider and more musical; higher Q is narrower and more surgical.

High frequency band (HF)

- **HF Gain:** -20 to +20 dB.
- **HF Frequency:** 1500 to 16000 Hz, default 8000. Black mode extends further to 16 kHz; Brown is more focused around 7 kHz.
- **HF Bell Mode:** Off (shelf) by default.

Global

- **EQ Type:** Brown (E-series) or Black (G-series). Brown is warmer and slightly softer; Black is brighter with extended low and high response.
- **Bypass:** Reports zero latency to the host while bypassed.
- **Input Gain / Output Gain:** -12 to +12 dB. Use Input Gain to drive the saturation stage harder; Output Gain to compensate.
- **Saturation:** 0 to 100%, default 0. British-style harmonic distortion. Stays clean at 0; adds warmth at 10 to 25%; gets obviously colored above 50%.
- **Oversampling:** 2x or 4x, default 2x. 4x reduces aliasing in the saturation stage at the cost of more CPU.

- **Spectrum Pre/Post:** Default Post. The spectrum analyzer view shows audio after the EQ; flip to Pre to see the input signal.
- **Auto Gain Compensation:** On by default. Keeps output loudness matched to input so EQ comparisons are level-matched.

Tips and Traps

- **Brown vs Black is not a "better" choice.** They are voiced differently. Brown is the smoother, more musical option for vocals and acoustic sources. Black is more aggressive and works well on drums, bass, and full mixes. Try both on the same source and pick what your ears prefer.



- **Bell vs shelf changes the character a lot.** A shelf at 100 Hz at +4 dB lifts everything below 100 Hz; a bell at 100 Hz at +4 dB only lifts a narrow range around 100 Hz. The wrong choice can sound boxy or thin. - **Saturation is gain-driven.** With Input Gain at 0 dB and Saturation at 50%, you may hear very little change. Push Input Gain to +6 to +9 dB and use Output Gain to compensate; the saturation will be much more apparent.



- **Auto Gain Compensation lies a little.** It level-matches RMS, but EQ changes can shift perceived loudness even at matched RMS. Trust your ears, and disable Auto Gain temporarily if you want to confirm an absolute level change. - **The HPF and LPF are sharp.** 18 dB/oct on the HPF will be obvious; do not set the HPF too high on bass instruments.

Presets Explained

4K-EQ ships with 14 factory presets across 7 categories. Each one is a starting point; expect to nudge frequencies and gains based on your source.

Vocals

- **Vocal Presence.** Brown mode, +4 dB HM at 3.5 kHz, +2 dB HF shelf at 8 kHz, gentle 300 Hz cut, HPF at 80 Hz. The classic vocal-forward setting.

Drums

- **Kick Punch.** +4 dB LF shelf at 50 Hz, -2.5 dB at 200 Hz, +3 dB at 2 kHz with tight Q. Body, mud cut, click.
- **Snare Crack.** Bell-shaped boost at 250 Hz for body, 5 kHz for crack, plus 8 kHz shelf for attack.
- **Drum Bus Punch.** Wide LF lift at 70 Hz, 350 Hz cut, 3.5 kHz boost, gentle 10 kHz shelf, 25% saturation, Black mode.

Bass

- **Bass Warmth.** +4 dB shelf at 80 Hz, slight 400 Hz cut, broad 1.5 kHz bell. For round, full bass.
- **Bass Guitar Polish.** Tighter low end (60 Hz shelf), 250 Hz cut, presence boost at 1.2 kHz, slight 4.5 kHz top.

Guitar

- **Acoustic Guitar.** Slight 100 Hz cut for boxiness, 2.5 kHz boost for fingernoise clarity, 12 kHz shelf for sparkle.

Keys

- **Piano Brilliance.** 80 Hz shelf for foundation, 500 Hz cut to clear the mids, 2 kHz and 8 kHz boosts for definition and shine.

Mix Bus

- **Bright Mix.** Wide 60 Hz shelf, gentle 2.5 kHz scoop, 10 kHz shelf with 20% saturation. Adds clarity without changing the mix.
- **Glue Bus.** Similar to Bright Mix with slightly more low end and saturation. The "console color" preset.

Creative

- **Telephone EQ.** Aggressive HPF and LPF (300 Hz and 3 kHz) with a 1 kHz bell boost. Lo-fi telephone effect.
- **Air & Silk.** Top-end-only setting: 7 kHz boost, 15 kHz shelf. Adds air without touching the body.

Mastering

- **Master Sheen.** Subtle 5 kHz and 16 kHz boosts, 10% saturation. Adds polish without changing the mix.
- **Master Bus Sweetening.** Gentle 50 Hz lift, 600 Hz cut, 4 kHz and 15 kHz boosts, 15% saturation, slight input cut. The most subtle preset; use as a starting point for mastering.

Troubleshooting

The EQ has no audible effect. Check that **Bypass** is off and the band you are adjusting has its **Gain** set to something other than 0 dB. The HPF and LPF must be **Enabled** to do anything; they are off by default.

Boosting any band makes the mix louder. That is expected when **Auto Gain Compensation** is off. Turn it on to compare EQ shapes at matched levels. If Auto Gain is on and you still hear level changes, the EQ is shaping the spectrum in a way that affects perceived loudness even at matched RMS.

The saturation knob seems to do nothing. The saturation stage is gain-driven; it only colors when the signal is hot enough to drive it. Turn up **Input Gain** by 6 to 9 dB and compensate with **Output Gain**, or feed the plugin a hotter input signal.

TapeMachine

Overview



Figure 14: TapeMachine main UI

TapeMachine emulates a professional reel-to-reel tape machine. Two machine models (Swiss and American) with four tape formulations (456, GP9, 911, 250), three speeds (7.5, 15, 30 IPS), and two EQ standards (NAB, CCIR) cover most of what you would find in a 1970s to 1990s pro studio. On top of that you get separate wow and flutter controls, switchable noise, oversampled saturation, and a four-position signal path so you can hear just the electronics, just the tape, the full chain, or true bypass.

Use it where you would use a real tape machine: subtle bus glue, vocal warmth, drum bus character, gritty character on guitars, or full lo-fi treatment. The machine model and speed affect the high-frequency response (head bump and roll-off); the tape formulation affects the saturation curve and noise floor; bias and calibration determine how hot you can drive before the sound breaks down.

It is not a chorus, flanger, or pitch-shift plugin. The wow and flutter are subtle modulations of the playback speed and behave like real tape; if you want extreme pitch effects, use a dedicated modulation plugin.

Quick Start

1. Insert TapeMachine on a track or bus. Vocal busses, drum busses, and the master are all common targets.
2. Leave the defaults for your first listen: **Swiss**, **15 IPS**, **Type 456**, **Repro** signal path, **NAB** EQ, **Auto Calibration** on, **Saturation** 4%.
3. Compare bypassed against active. You should hear a subtle high-frequency softening and a touch of harmonic warmth.
4. Push **Input Gain** by 4 to 6 dB to drive the saturation harder. Compensate with **Output Gain** to keep levels matched.
5. Try the other machine: switch **Tape Machine** to **American** for a different head response and more pronounced character.
6. If you want the full vintage effect, raise **Wow** to 10 to 15%, **Flutter** to 5 to 8%, and toggle **Noise Enabled** on with **Noise Amount** around 10%.

You should hear the difference immediately on transients (slightly softer attack), on highs (gentle roll-off above 12 to 15 kHz), and on overall tone (subtle low-mid warmth from the head bump). With Saturation above 10% and Input Gain pushed, the harmonic content becomes audible.

Workflows

Subtle warmth on a vocal

Source: A clean studio vocal that needs analog character without coloration. **Goal:** Slight warmth and softening, no obvious tape sound.

Settings:

- **Tape Machine:** Swiss
- **Tape Speed:** 30 IPS (cleanest, widest frequency response)
- **Tape Type:** GP9 (low noise, high-output)
- **Signal Path:** Repro
- **EQ Standard:** AES (30 IPS forces this)
- **Input Gain:** +2 dB
- **Saturation:** 8%
- **Wow / Flutter:** 0 / 0 (we want clean speed)
- **Noise Enabled:** Off
- **Output Gain:** -1 dB

Why this works. 30 IPS gives the flattest response and the lowest noise; the high-frequency roll-off is barely audible. GP9 tape is voiced for transparency. A small amount of saturation and input gain delivers the subtle nonlinearity that makes audio "sit" better in a mix without obvious distortion. With wow and flutter at zero, only the EQ shape and saturation contribute. This is the "Mastering Touch" preset's approach.

Drum bus character with 70s grit

Source: Stereo drum bus, mixed and balanced. **Goal:** Pre-internet-era warmth and a touch of grit.

Settings:

- **Tape Machine:** American



Figure 15: American machine selected

- **Tape Speed:** 15 IPS
- **Tape Type:** 456
- **Signal Path:** Repro
- **EQ Standard:** NAB
- **Input Gain:** +6 dB
- **Saturation:** 35%
- **Bias:** 42% (slightly under-biased for added harmonic content)
- **Wow:** 12%
- **Flutter:** 6%
- **Noise Enabled:** On, **Noise Amount:** 10%
- **Output Gain:** -3 dB

Why this works. American is the more colored of the two machines. 15 IPS at NAB gives the classic 1970s head response with audible high-frequency roll-off. Type 456 saturates earlier than the modern formulations. Driving the input by 6 dB pushes the saturation into clearly audible territory. Under-biasing (Bias below 50%) thins the saturation slightly and adds harmonic distortion. Audible wow and flutter complete the era. This is the "70s Rock" preset.

For a less aggressive version, drop Saturation to 20% and Input Gain to +3 dB.

Cassette tape lo-fi effect

Source: Anything you want to sound like it is playing back from a worn cassette. **Goal:** Audibly degraded, narrow-bandwidth, modulated.

Settings:

- **Tape Machine:** American
- **Tape Speed:** 7.5 IPS (worst high-frequency response)
- **Tape Type:** Type 250 (highest noise, most coloration)
- **Signal Path:** Repro
- **EQ Standard:** NAB
- **Input Gain:** +4 dB
- **Saturation:** 50%
- **Bias:** 35% (heavily under-biased, distorted)
- **Highpass Frequency:** 80 Hz
- **Lowpass Frequency:** 8000 Hz (thin top end)
- **Wow:** 30%
- **Flutter:** 20%
- **Noise Enabled:** On, **Noise Amount:** 30%
- **Output Gain:** 0 dB

Why this works. 7.5 IPS at the most aggressive tape (Type 250) with heavy under-bias and 50% saturation produces the thick, distorted sound of a cassette running too hot. Audible wow and flutter pitch-shift the playback, completing the cassette illusion. The 8 kHz low-pass narrows the bandwidth dramatically. This roughly matches the "Cassette Deck" preset.

Mastering bus polish

Source: A finished stereo mix. **Goal:** Inaudible analog flavor, no obvious processing.

Settings:

- **Tape Machine:** Swiss
- **Tape Speed:** 30 IPS
- **Tape Type:** GP9
- **Signal Path:** Repro
- **EQ Standard:** AES
- **Input Gain:** +2 dB
- **Saturation:** 6%
- **Bias:** 50% (perfectly biased)
- **Auto Calibration:** On
- **Wow / Flutter:** 0 / 0
- **Noise Enabled:** Off
- **Auto Compensation:** On
- **Oversampling:** 4x
- **Output Gain:** -1 dB

Why this works. Conservative everything: cleanest machine, fastest speed, lowest-noise tape, perfectly biased, no modulation, no noise. Just enough saturation and input gain to add second-harmonic content. 4x oversampling keeps the saturation aliasing-free. A 1 dB cut on the output prevents any chance of clipping into the next plugin. The result should be inaudible on its own; the difference is only obvious on bypass A/B. This matches the "Master Bus Glue" preset.

Parameter Reference

Machine and tape

- **Tape Machine:** Swiss (default) or American. Swiss is the cleaner, more linear machine; American has more pronounced head response and saturation character.
- **Tape Speed:** 7.5, 15 (default), or 30 IPS. Faster speed gives flatter frequency response, less noise, and less saturation per dB of input. 7.5 is the most colored and noisy; 30 is the cleanest.
- **Tape Type:** 456 (default), GP9, 911, or 250. Different tape formulations have different saturation curves, noise floors, and tonal characters. 456 is the classic general-purpose formulation; GP9 is a modern low-noise; 911 is the popular European; 250 has the most character (and the most noise).
- **Signal Path:** Repro (default), Sync, Input, or Thru. Repro is the full signal chain (electronics into tape into reproduce head); Sync uses the record head as the playback head (slightly different EQ); Input is the electronics only with no tape (no saturation or modulation); Thru is true bypass.
- **EQ Standard:** NAB (default), CCIR, or AES. American (NAB) and European (CCIR) tape EQs differ above 1 kHz; AES is mandatory at 30 IPS. The plugin will switch to AES automatically when you pick 30 IPS.

Drive and tone

- **Input Gain:** -12 to +12 dB. Drives the tape stage. Higher values produce more saturation and harmonic distortion.
- **Saturation:** 0 to 100%, default 4%. The depth of the tape saturation curve. Default is barely audible; 20% is "pleasant"; 50% and above is distortion-heavy.
- **Bias:** 0 to 100%, default 50%. Tape bias level. 50% is perfectly biased (cleanest). Below 50% is under-biased and adds distortion. Above 50% is over-biased and dulls the highs.

- **Calibration:** 0 dB (default), +3, +6, or +9 dB. The reference operating level. Higher calibration means the tape is set to handle louder signals; useful when feeding hot mixes.
- **Auto Calibration:** On (default) or Off. When on, bias and calibration auto-adjust based on tape type and speed for optimal recording. Turn off to set bias manually for creative under- or over-biasing.

Filtering and modulation

- **Highpass Frequency:** 20 to 500 Hz, default 20. The corner of the input high-pass filter.
- **Lowpass Frequency:** 3 to 20 kHz, default 20 kHz. The corner of the input low-pass filter.
- **Wow:** 0 to 100%, default 7. Slow pitch drift (around 0.3 to 0.8 Hz). Adds vinyl-like wobble at higher values.
- **Flutter:** 0 to 100%, default 3. Faster pitch modulation (3 to 7 Hz). Adds tape machine character at higher values.
- **Noise Amount:** 0 to 100%, default 0. Tape noise floor level.
- **Noise Enabled:** Off (default) or On. Turn on to introduce noise; Noise Amount is the level.

Output

- **Output Gain:** -12 to +12 dB. Final makeup gain.
- **Auto Compensation:** On (default) or Off. Compensates output level when Input Gain changes, so A/B comparisons are level-matched.
- **Oversampling:** 1x, 2x, or 4x (default). Reduces aliasing from the saturation stage.
- **Bypass:** Reports zero latency to the host while bypassed.

Tips and Traps

- **Auto Calibration is your friend until you want it not to be.** With Auto Calibration on, switching tape type or speed re-biases the tape for you. Turn it off when you want to creatively under-bias (40 to 45% for grit) or over-bias (55 to 60% for a darker, dulled tone).

- **30 IPS forces AES.** This is hardware-accurate behavior; real machines also lock the EQ standard at 30 IPS. Pick 15 or 7.5 IPS if you want NAB or CCIR.

- **Signal Path is not Bypass.** "Thru" is true bypass (signal passes unchanged). "Input" runs the electronics but no tape (no saturation or modulation). "Sync" uses the record head for playback (slightly different EQ). "Repro" is the full chain. Use Sync to compare what the engineer hears during tracking versus the printed tape.

- **The default Saturation of 4% is conservative.** If you cannot hear any change versus bypass, raise it to 15 to 25% and pull Input Gain up by a few dB.

- **Wow and Flutter are subtle by default.** 7% Wow and 3% Flutter approximate a well-maintained machine. Push to 15 to 30% for the obviously-vintage sound; below 5% you may not hear them on most material.

- **Noise is a creative tool, not a problem to fix.** When you want the analog floor sound, enable Noise and set Noise Amount to 5 to 15%. For clean work, leave it disabled (the default).



Figure 16: Bias control with Auto Calibration off



Figure 17: Signal Path dropdown



Figure 18: Preset menu open

Presets Explained

TapeMachine ships with 15 presets across 5 categories. Each is a starting point; expect to nudge Input Gain and Saturation to taste.

Subtle

- **Gentle Warmth.** Swiss, 30 IPS, GP9, low saturation. Almost-inaudible analog flavor for transparent work.
- **Transparent Glue.** Similar to Gentle Warmth but slightly more saturation; for bus duty.
- **Mastering Touch.** The cleanest setting in the plugin: 30 IPS, GP9, AES, perfectly biased, no modulation. For mastering bus or any "I want a hint of analog" job.

Warm

- **Classic Analog.** 15 IPS, Classic, NAB, moderate drive. The textbook warm tape sound.
- **Vintage Warmth.** American, 15 IPS, more pronounced head bump, slight wow. For sources that need to feel older.
- **Tube Console.** Heavier saturation with audible second-harmonic content; pairs well with a bus EQ.

Character

- **70s Rock.** American, 15 IPS, Classic, under-biased (42%), audible wow and flutter, noise on. The drum-bus and rock-mix preset.
- **Tape Saturation.** Heavy drive, full saturation. Think parallel-tape bus sound.
- **Cassette Deck.** 7.5 IPS, narrow bandwidth, audible modulation. Lo-fi territory.

Lo-Fi

- **Lo-Fi Warble.** Heavy wow and flutter, narrow bandwidth, audible noise. Creative-effect preset.
- **Worn Tape.** Even more degraded; for atmospheric or sound-design work.
- **Dusty Reel.** The most extreme; obvious tape grit, modulation, noise. Use sparingly.

Mastering

- **Master Bus Glue.** Cleanest mastering setting; minimal saturation, no modulation, no noise. Subtle bus glue.
- **Analog Sheen.** Adds a touch of harmonic content to brighten a master without obvious processing.
- **Vintage Master.** A more colored mastering option, with a touch of head bump and very subtle modulation.

Troubleshooting

The plugin sounds the same with bypass on and off. With **Saturation** at 4% and **Input Gain** at 0 dB, the difference is genuinely subtle. Push Saturation to 25% and Input Gain to +6 dB to confirm the plugin is doing something, then dial back to taste.

Switching to 30 IPS changed my EQ Standard automatically. That is intentional. Real tape machines lock to AES EQ at 30 IPS; the plugin matches. Pick 15 or 7.5 IPS to use NAB or CCIR.

The Bias knob does nothing. Auto Calibration is on by default and overrides manual bias. Turn off **Auto Calibration** in the calibration section, then Bias is yours to drive.

The CPU usage is high. 4x oversampling is the default and is the heaviest setting. Drop to 2x for tracking; the audible difference is small but the CPU savings are significant. 1x is appropriate only if you cannot afford the 2x cost; aliasing in the saturation stage becomes audible on bright sources.

I hear pitch drift even with Wow and Flutter at zero. Confirm Signal Path is Repro or Sync; if you set everything to zero but still hear drift, restart the plugin (a stuck modulator is rare but possible).

DuskVerb

Pre-release. DuskVerb is currently in pre-release status. Parameter ranges, preset names, and engine behavior may change before the 1.0 release. This manual reflects the current 0.6.x line; check the website for updates if you are reading an older copy.

Overview

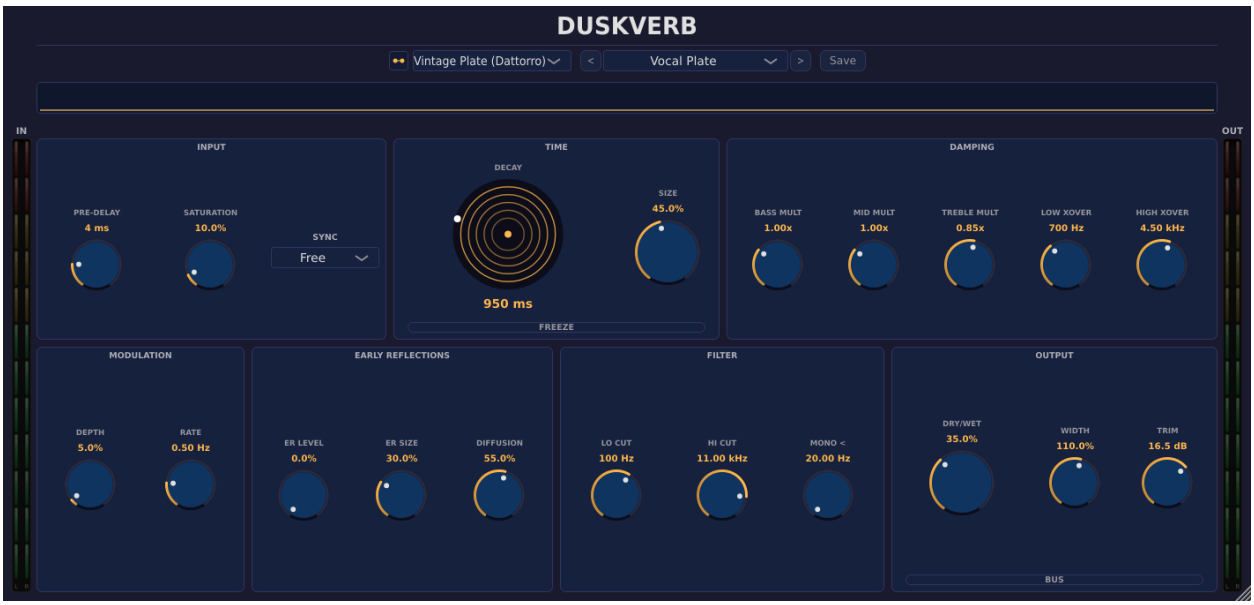


Figure 19: DuskVerb main UI

DuskVerb is an algorithmic reverb with eleven distinct engines under one user interface. Each engine targets a different reverb territory: **Plate** is the classic dense plate that works on almost everything; **Vintage Plate** is the darker, steely vintage-plate character; **Smooth Plate** is a smooth six-allpass plate; **Chamber** is a studio chamber and tight-room voice; **Spring** is the surf-guitar spring sound; **Gated** produces the gated 1980s drum sound and other non-natural curves; **Shimmer** adds pitched sparkle above and warmth below the tail; **Reverse** is the swelling reverse effect; **Hall** is a realistic per-octave-tuned hall; **Tiled Room** is a bright tiled room; and **Dense Hall** is a lush, densely diffused hall.

Use it where you would use any reverb. Vocals, drums, guitars, and full-mix space all have an engine in DuskVerb that is voiced for them. The 20 factory presets are anchored to specific hardware references (classic digital halls, a steel plate, a vintage nonlinear reverb, and others) and serve both as starting points and as a tutorial in what each engine can do.

It is not a convolution reverb (use Convolution Reverb for IR-based work), and it is not a delay/multitap effect. It is an eleven-in-one algorithmic reverb that lets you pick the right engine for the job.

Quick Start

1. Insert DuskVerb on a return bus or directly on a track. For most workflows, an aux/return at 100% wet is the cleanest setup.

2. Pick an **Algorithm** from the top dropdown. If you are not sure, start with **Plate**; it works on almost everything.



3. Open the preset menu. Pick one that matches your source (Vocal Plate for vocals, Small Drum Room for drums, and so on). Presets are laid out in a two-column menu grouped by category. 4. Adjust **Decay Time** to taste. The Decay knob is calibrated so the position reflects the actual tail length; longer decays sustain more, shorter decays sit more discreetly. 5. Use **Pre-Delay** (0 to 250 ms) to push the reverb tail later, which keeps the dry signal forward and the reverb perceived as space rather than smear. 6. The **Dry/Wet** knob controls the mix. On a return bus, leave at 100%. On an insert, dial back to 20-40% wet.

You should hear a tail when you stop the source. If the tail is too dark, raise **Treble Multiply** (or nudge the **Tone** macro brighter); too bright, lower it. If the tail rings or sounds metallic, drop **Diffusion** below the preset value or pick a different engine.

Workflows

Vocal plate on a lead vocal

Source: A lead vocal that needs space without sounding wet. **Goal:** Smooth, classic plate tail that thickens the vocal without obscuring it.

Settings (or load the **Vocal Plate** preset and tweak from there):

- **Algorithm:** Plate
- **Decay Time:** 1.8 s
- **Pre-Delay:** 30 ms
- **Size:** 0.6
- **Mod Depth:** 0.15
- **Mod Rate:** 1.0 Hz
- **Treble Multiply:** 0.8 (slight high-frequency rolloff in the tail)
- **Lo Cut:** 200 Hz (keeps the tail out of the bass range)
- **Hi Cut:** 8000 Hz

- **Dry/Wet:** 100% (use as a send return)

Why this works. Plate reverbs are dense from the first reflection; they do not have audible early reflections. Pre-delay of 30 ms separates the tail from the dry signal so the vocal stays forward. Lo Cut at 200 Hz prevents the tail from muddying the low end. Subtle modulation (depth 0.15, rate 1 Hz) keeps the plate from sounding static. The 1.8-second decay is a typical pop-vocal length.

Tight drum room

Source: Drum bus or a snare track that needs room without long tail. **Goal:** Short, punchy ambience that adds size without smearing transients.

Settings (or load the **Small Drum Room** preset):

- **Algorithm:** Chamber
- **Decay Time:** 0.6 s
- **Pre-Delay:** 0 ms
- **Size:** 0.4
- **Diffusion:** 0.5 (less than fully diffuse keeps transients distinct)
- **Early Ref Level:** 0.7 (early reflections are most of what you hear)
- **Treble Multiply:** 0.95 (keep highs in the room)
- **Lo Cut:** 80 Hz
- **Dry/Wet:** 25% (insert) or 100% on a return

Why this works. Short decay plus low diffusion plus prominent early reflections gives a "room" sound rather than a "reverb" sound. The drums still hit hard; the room just adds a bit of three-dimensional space. The **Small Drum Room** and **Medium Drum Room** presets use these proportions.

Realistic concert hall on piano

Source: Solo piano or piano in a sparse mix. **Goal:** A long, lush hall tail that sounds like a real space.

Settings (or use the **Cathedral Large Hall** preset):

- **Algorithm:** Hall
- **Decay Time:** 4.5 s
- **Pre-Delay:** 60 ms
- **Size:** 0.85
- **Diffusion:** 0.85
- **Early Ref Level:** 0.4
- **Early Ref Size:** 0.7
- **Bass Multiply:** 1.2 (emphasize the low frequencies in the tail, characteristic of large rooms)
- **Mid Multiply:** 1.0
- **Treble Multiply:** 0.7 (concert halls absorb highs)
- **Lo Cut:** 60 Hz
- **Hi Cut:** 12000 Hz
- **Dry/Wet:** 100% on a return at -10 to -6 dB send level

Why this works. The Hall engine tunes decay per octave, so it produces realistic late reverberation with audible per-frequency decay differences. Bass multiply above 1 emphasizes the long bass tail

typical of real halls. Treble multiply below 1 captures the high-frequency absorption you hear in rooms with absorptive surfaces. Long pre-delay (60 ms) maintains piano clarity. For a denser, smoother variant of the same space, try the **Dense Hall** engine.

Gated 80s snare

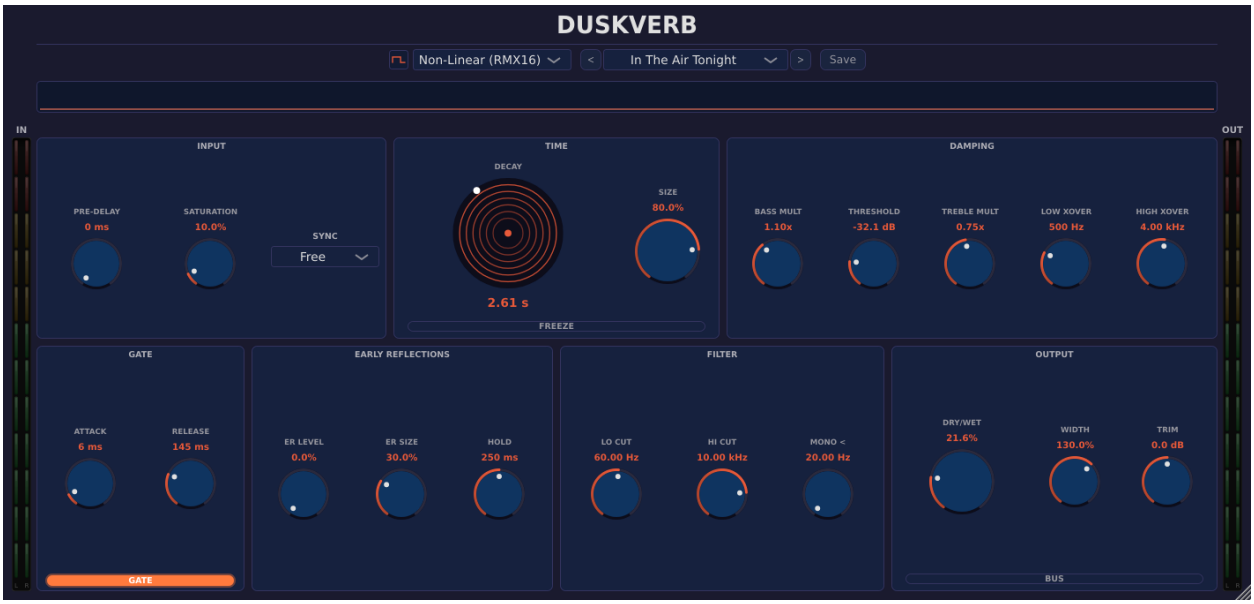


Figure 20: Gated engine with Gate enabled

Source: Snare drum that needs the classic 1980s gated sound. **Goal:** A non-natural reverb shape that cuts off after a fixed time.

Settings (or load the **1981 Gated Snare** preset):

- **Algorithm:** Gated
- **Decay Time:** 1.2 s (the gate length, not natural decay)
- **Pre-Delay:** 0 ms
- **Size:** 0.6
- **Treble Multiply:** 1.0
- **Gate:** Enabled (default; this is the parameter that makes the engine non-linear)
- **Dry/Wet:** 35% (insert)

Why this works. The Gated engine mimics a classic 1980s digital nonlinear reverb algorithm: the reverb has constant level for a fixed time (set by Decay Time) and then cuts to silence rather than fading naturally. This produces the 1980s snare sound that defined a decade of records. For the swelling backward version instead, load **Reverse Taps** on the **Reverse** engine.

Parameter Reference

Algorithm

- **Algorithm:** Selects which DSP engine processes the audio. Eleven choices: Plate, Vintage Plate, Smooth Plate, Chamber, Spring, Gated, Shimmer, Reverse, Hall, Tiled Room, Dense Hall. Switching engines crossfades over a few hundred milliseconds.

Macros

The bottom macro row applies three global shapers on top of whatever engine and preset you have loaded. At their defaults they are neutral, so a preset sounds exactly as designed until you move them.

- **Tone:** -1 to +1. Spectral tilt from dark to bright. Negative values roll the tail down (lower Treble Multiply and Hi Cut); positive values open it up. A fast way to match a reverb to a track without touching the per-band controls.
- **Character:** 0 to 1. Movement and grit. Adds modulation depth and saturation together, taking a static tail toward a livelier, slightly driven one.
- **Duck:** 0 to 1. Wet ducking depth, sidechained off the dry input. Pulls the reverb down while the source plays and lets it bloom in the gaps, keeping dense mixes clear.

Mix and routing

- **Dry/Wet:** 0 to 100%. Wet/dry mix at the plugin output. 100% on returns; lower on inserts.
- **Bus Mode:** On or off. When on, the plugin assumes a 100% wet send/return setup and bypasses the dry path entirely (slight CPU savings).
- **Bypass:** Reports zero latency to the host while bypassed.

Time and size

- **Pre-Delay:** 0 to 250 ms. Delay before the reverb tail begins. Longer pre-delay separates the dry signal from the reverb.
- **Pre-Delay Sync:** Free, 1/32, 1/16, 1/8, 1/4, 1/2, 1/1. When set to a note value, pre-delay locks to the host tempo.
- **Decay Time:** 0.2 to 30 s. Total tail length. The knob is calibrated so its position tracks the actual decay time across engines. The Hall engine reaches the highest decays; the Gated engine treats this as a gate length.
- **Size:** 0 to 1. Perceived room size. Engine-dependent: in Chamber, low values are cabinet-sized and high values are arena-sized.

Modulation

- **Mod Depth:** 0 to 1. Amount of pitch modulation in the tail. Subtle values (0.1 to 0.2) keep the tail moving; high values (0.5 plus) produce chorus-like effects.
- **Mod Rate:** 0.1 to 10 Hz. Speed of the modulation. Slower rates (around 0.5 Hz) sound natural; faster rates produce flanger-like motion.

Frequency shaping

- **Bass Multiply:** 0.3 to 2.5. Decay-time multiplier for low frequencies. Above 1 makes the bass decay longer than the rest (typical of large halls). Below 1 shortens the bass tail (typical of small studios with absorbers).
- **Mid Multiply:** 0.3 to 2.5. Decay multiplier for the mid range.
- **Treble Multiply:** 0.1 to 1.5. Decay multiplier for the high frequencies. Below 1 produces high-frequency damping (the “warm” tail).
- **Low Crossover:** 200 to 4000 Hz. Frequency where Bass Multiply gives way to Mid Multiply.
- **High Crossover:** 1000 to 12000 Hz. Frequency where Mid Multiply gives way to Treble Multiply.

- **Lo Cut / Hi Cut:** Filters the input into the reverb. Lo Cut keeps low frequencies dry; Hi Cut keeps highs dry. Useful for keeping the reverb out of the kick and the air bands respectively.

Color and density

- **Saturation:** 0 to 1. Drives a soft saturation stage in the reverb feedback path. Adds warmth and harmonic content.
- **Diffusion:** 0 to 1. Density of the reverb tail. High diffusion is dense and smooth; low diffusion sounds grainier and more echo-like.
- **Early Ref Level:** 0 to 1. How loud the early reflections are relative to the main tail.
- **Early Ref Size:** 0 to 1. The simulated room size for the early reflections.

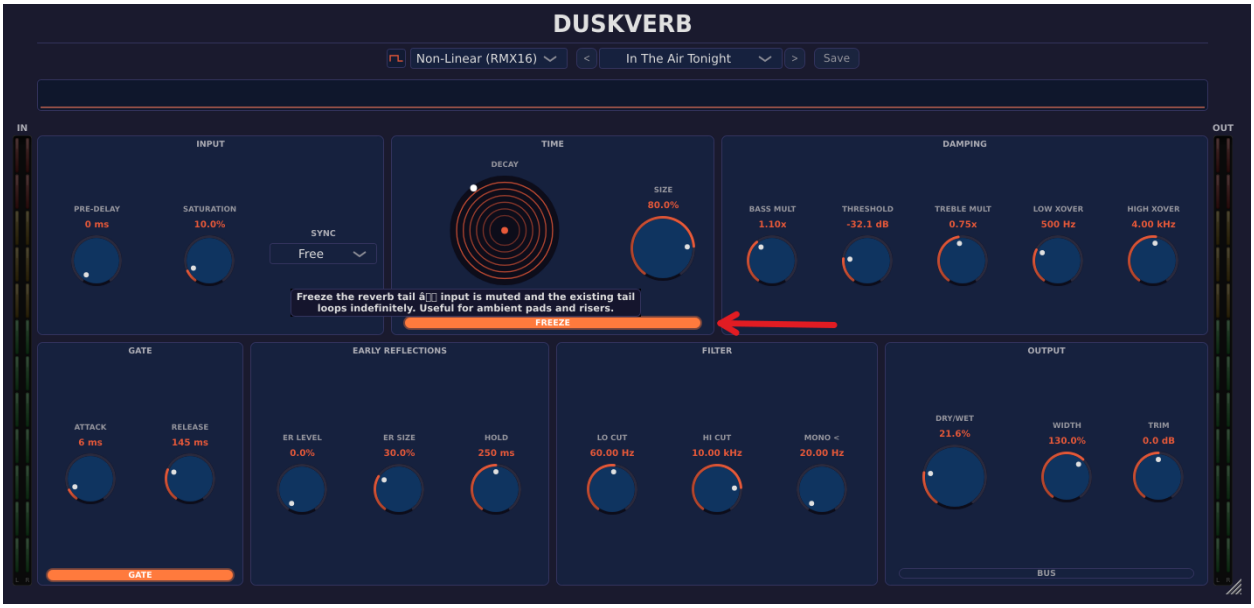
Output and stereo

- **Width:** 0 to 2. Stereo width of the reverb output. 1 is unity; 2 is double-wide (M/S based).
- **Freeze:** Off or On. When on, the reverb tail cycles infinitely.
- **Gate:** Off or On. Enables the Gated engine's gating behavior.
- **Mono Below:** 20 to 300 Hz. Frequencies below this cutoff are summed to mono in the reverb output. Default 20 Hz (effectively bypass); typical settings 80 to 150 Hz to keep low-frequency reverb mono-compatible.
- **Gain Trim:** -48 to +48 dB. Final output level adjustment.

The wet tail preserves the dry source's left/right placement instead of collapsing to the center: a signal panned left decays as a left-leaning tail, a right-panned source keeps its tail on the right, and centered input stays centered. Each preset's stereo behavior is calibrated against its hardware anchor reference, so the amount of preserved lean matches the character of the unit the preset emulates; a few presets keep an intentionally neutral image where their reference does (Black Hole, for example, renders a centered tail like the unit it recreates). Width still applies on top of this, so you can widen or narrow the preserved image to taste.

Tips and Traps

- **Pre-Delay is your most important parameter for clarity.** Without pre-delay, the reverb tail starts on the same sample as the dry signal and the result smears. 20 to 60 ms of pre-delay keeps vocals and drums distinct from their reverb.
- **The macros are the fast path.** Tone, Character, and Duck reshape any preset without diving into the per-band controls. Reach for them first when a preset is close but not quite matched to the track.
- **Hardware anchors are real.** The presets are tuned to specific hardware references. "Vintage Vocal Plate" leans on the darker, steely character of a classic plate; "Blade Runner 224" captures a long-decay vintage digital hall sound; "Cathedral Large Hall" is a long classic digital hall. If you are familiar with the source hardware, the preset name tells you what to expect.
- **Engine switching is not parameter-preserving.** Each engine has its own internal state. Switching engines while a tail is decaying produces a crossfade; do not expect identical-sounding results across engines at the same parameter values.
- **Freeze is loud.** Freeze captures the current reverb tail and loops it indefinitely. Levels can build dramatically; pull Gain Trim back before enabling Freeze on a busy mix.



- **Mono Below preserves bass mono compatibility.** Stereo reverb on bass frequencies often phases on mono fold-down. Set Mono Below to 80 to 120 Hz on mastering or mix-bus reverb.
- **The Gated engine treats Decay Time as a gate length.** It does not behave like other engines for that parameter. Refer to the gated-snare workflow above.

Presets Explained



Figure 21: Preset menu by category

DuskVerb ships with 20 factory presets, laid out in a two-column menu grouped by category. Each is hardware-anchored and serves as a starting point.

Plates

Four plate presets: **Vocal Plate** is the bright, smooth plate for lead vocals; **Vintage Vocal Plate** is the darker, steelier character of a classic plate; **Drum Plate** is the punchy plate for snares and drums; **Vintage Gold Plate** is the long, lush plate that helped define 1980s record production.

Halls

Four halls, from bright studio (**Bright Hall**) through vocal-tuned (**Vocal Hall**) to massive (**Cathedral Large Hall**, a long classic digital concert hall). **Blade Runner 224** captures a long-decay, extended-tail vintage digital hall sound. Pick by length and brightness.

Chambers

79 Vocal Chamber is a studio chamber voiced for vocals (Chamber engine); **Large Chamber** is a bigger, denser chamber space (Dense Hall engine). Both sit between a room and a hall in size.

Rooms

Four rooms covering tight (**Small Drum Room**) through medium (**Medium Drum Room**, **Live Room**) to bright and reflective (**Tiled Room**). Use **Small Drum Room** as your default for drum-bus ambience.

Springs

Surf '63 Spring is the Dick Dale "Misirlou" spring sound; aggressive and bouncy, for surf and reggae guitar work.

Gated and Reverse

1981 Gated Snare is the classic gated-reverb snare on the Gated engine. **Reverse Taps** is the swelling backward-reverb effect on the Reverse engine.

Ambient and Shimmer

Ambience is a soft atmospheric wash for pads and sound design. The shimmer presets use the overhauled Shimmer engine, which adds a post-loop high-frequency air voice for sparkle above the tail and a down-octave voice for warmth below it: **Deep Blue Day** (after the Brian Eno track) is the lush, warm shimmer; **Black Hole** is the huge, sustaining ambient shimmer.

Troubleshooting

The tail sounds metallic or rings. Drop **Diffusion** to around 0.5 to 0.7 if the tail is buzzing or sounds like a ring modulator. Try a different engine; the Hall and Dense Hall engines are the smoothest at long decays, while the Plate engines are naturally denser.

The reverb is too loud at low frequencies. Raise **Lo Cut** to 100 to 200 Hz so the reverb input is high-passed before reaching the engine. Bass frequencies in reverb tails build up quickly and muddy a mix.

Switching engines drops the tail. That is intentional. Each engine has its own internal state; the crossfade between engines is brief (a few hundred milliseconds) and the new engine starts with a clean tail. Plan engine switches between sections, not within a phrase.

The sound is the same regardless of which preset I load. Confirm you are loading from the **Preset** menu and not just changing the **Algorithm**. Loading a preset sets all parameters including the algorithm; changing only the algorithm leaves the other parameters at their previous values.

My host shows extra latency on insert. DuskVerb has a small fixed latency from its early-reflection delay line. The host applies plug-in delay compensation automatically; check that PDC is enabled in your DAW preferences if other tracks sound out of sync.

Spectrum Analyzer

Overview

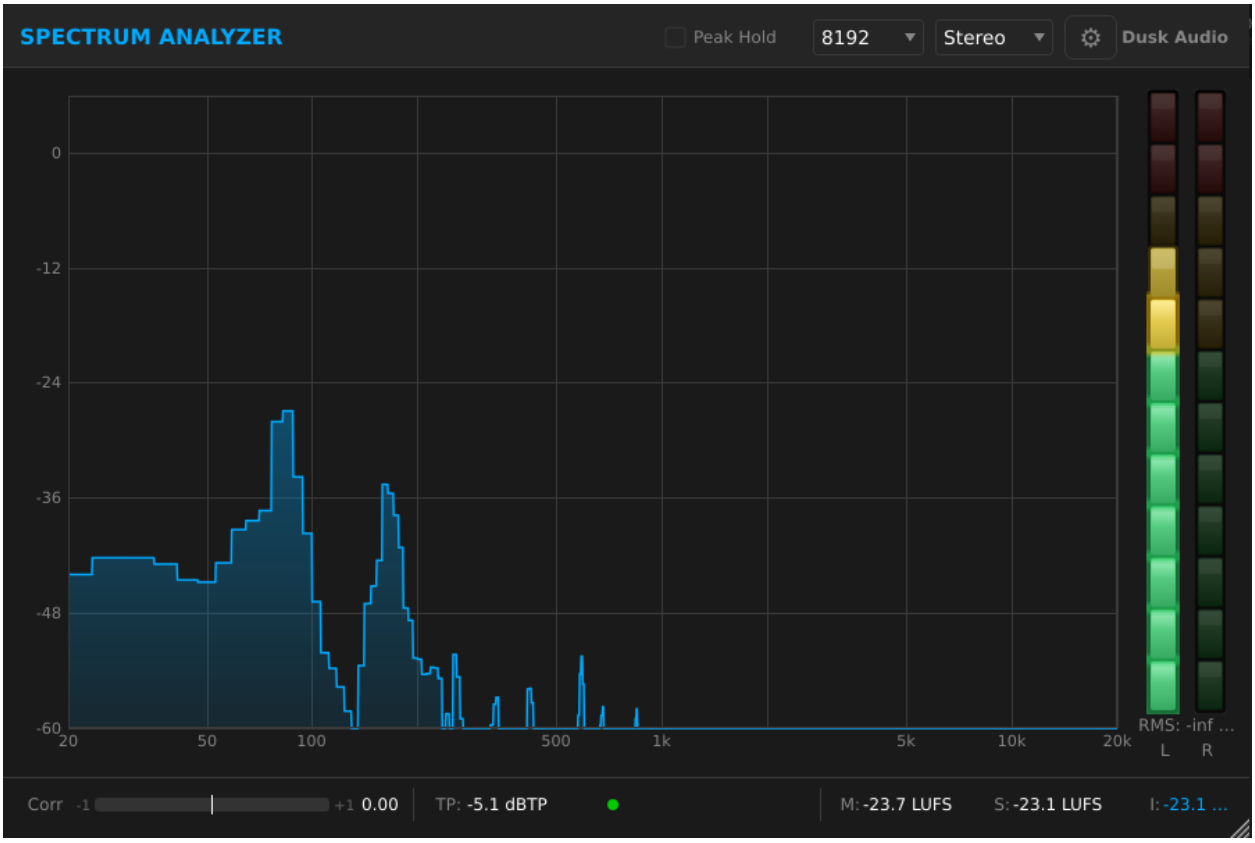


Figure 22: Spectrum Analyzer main UI

Spectrum Analyzer is a measurement plugin, not a sound-shaping plugin. Drop it on any track or bus to see what is actually there: a real-time FFT spectrum, four kinds of loudness reading (Momentary, Short-term, Integrated, LRA), an ITU-R BS.1770-4 compliant true-peak meter, a stereo correlation meter, and K-System headroom display.

Use it to verify what your ears are telling you. The spectrum shows tonal balance and resonance; LUFS shows perceived loudness for streaming-platform targets; true-peak catches inter-sample peaks that a regular sample-peak meter misses; correlation flags phase issues before they cost you a mono fold-down.

It is not a corrective EQ, and it is not a sonic-fingerprint matching tool. It is a precise set of meters that gives you the numbers you need to decide what to do next.

Quick Start

1. Insert Spectrum Analyzer on the track or bus you want to measure. Mastering bus is a common place to put it; it works equally well on a single instrument.

2. Play the project. The spectrum draws across the main display. Loudness readings appear next to it.

3. Click the gear icon (top right) to open the settings overlay. The defaults (4096 FFT, 0.5 smoothing, 0 dB/oct slope, K-14 metering) work for most program material.

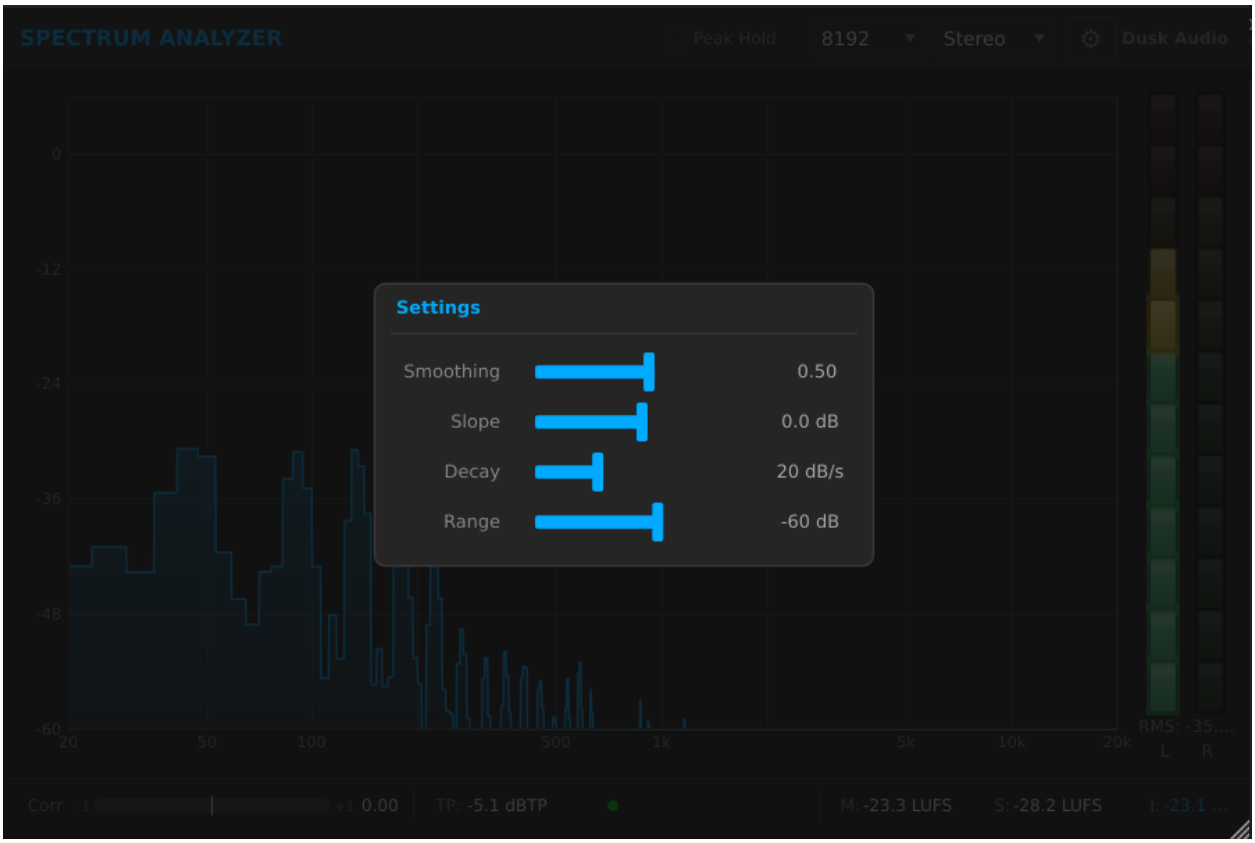


Figure 23: Settings overlay open

4. Look at the Integrated LUFS readout once the section you are checking has played for at least 10 seconds. That is your loudness number.

5. Watch the True Peak indicator. If it lights red, your master is clipping inter-sample; pull the output back at least 1 dB.

6. Use the Channel Mode dropdown in the header to switch between Stereo, Mono, Mid, and Side views.

You should see frequency content where you expect it (kicks under 100 Hz, vocals around 1 to 3 kHz, cymbals above 8 kHz). The correlation bar should sit between 0 and +1 on most material; negative values mean the stereo image is fighting itself.

Workflows

Tonal balance check on a mix

Source: A near-final mix bus, played at performance level. **Goal:** Confirm the mix has the spectral shape you intended.

Settings:

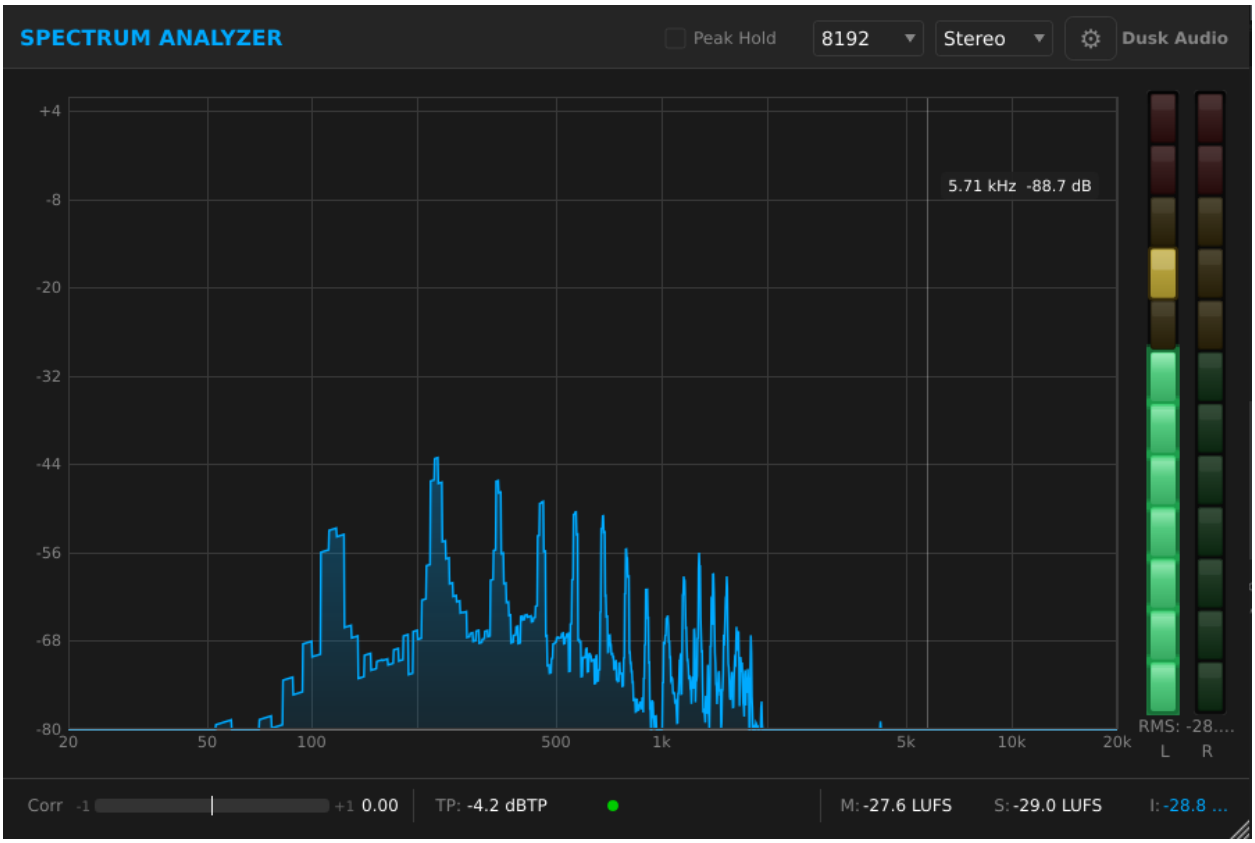


Figure 24: Tilted display with 4.5 dB/oct slope

- **Channel Mode:** Stereo
- **FFT Resolution:** 4096
- **Slope:** 4.5 dB/oct
- **Smoothing:** 0.7
- **Display Min:** -60 dB
- **Display Max:** +6 dB
- **Peak Hold:** On
- **Peak Hold Time:** 5 sec

Why this works. A 4.5 dB/oct slope tilts the display so a balanced "pink-noise" mix appears roughly flat across the spectrum. With Slope at 0, low end always dominates the visualization and you cannot eyeball mid and high content at a glance. Higher smoothing (0.7) damps out per-frame noise so you see steady tonal shape, not transients. Peak Hold with a 5-second decay shows you the loudest frequencies during the chorus.

Look for: a roughly even hill across 60 Hz to 10 kHz, with a small bump in the kick range (50 to 100 Hz) and a gentle roll-off above 12 kHz. Big resonant spikes (a single peak that sticks 6 dB above its neighbors) usually mean a problem frequency you can fix with a narrow EQ cut.

Streaming loudness target



Figure 25: LUFS readouts

Source: A finished master, ready to upload. **Goal:** Confirm Integrated LUFS hits the streaming target (Spotify, YouTube, Amazon Music, and Tidal: -14 LUFS integrated; Apple Music: -16 LUFS integrated; podcasts: -16 LUFS).

Settings:

- **Channel Mode:** Stereo
- **K-System Type:** K-14
- **Display Min:** -60 dB
- **Display Max:** 0 dB
- Other settings: defaults

Why this works. Set Spectrum Analyzer post-fader, post-everything, on the master bus. Play the entire song from start to end without stopping (LUFS Integrated only updates while audio is above the gate threshold; pausing pollutes the reading). The Integrated reading at the end is your loudness number.

If Integrated LUFS reads -10, your master is louder than the streaming target and the platform will turn it down. If it reads -18, you have headroom to push the level up. The True Peak meter must stay at or below -1 dBTP for safe transcoding to lossy formats. Watch the LRA (Loudness Range) reading; for most modern pop and rock, LRA between 5 and 10 LU is typical. Above 15 indicates very dynamic material.

Mid/Side imaging check

Source: A stereo mix that feels "narrow" or "off-center". **Goal:** Identify what is in the middle versus the sides.

Switch the Channel Mode dropdown between Mid and Side while the mix plays. The spectrum updates to show only that component.

- **Mid mode:** kick, snare, bass, lead vocal, anything panned center.
- **Side mode:** room mics, reverb returns, doubled guitars, anything panned wide.

What to look for. If the Side spectrum has heavy low-frequency content (below 150 Hz), the mix has phasey low end; consider high-passing the side channel or summing low end to mono. If the Mid spectrum shows nothing above 5 kHz, the highs are all in the sides; that can sound airy on speakers but vanishes on mono playback.

Phase correlation check

Source: Anything stereo, especially overhead drum mics, room mics, or stereo synths. **Goal:** Catch phase issues before mono fold-down loses the signal.

Watch the correlation bar at the top of the meter strip. Positive (green) values mean the channels mostly agree; this is good. Values near zero (yellow) mean stereo content with low correlation; this is fine for ambient material. Values below zero (red) mean the channels are partially out of phase; if you sum to mono, those frequencies cancel.

If you see persistent negative correlation, swap the polarity of one channel of the offending stereo source and check again. Briefly negative values during transients are normal for very wide stereo material.

Parameter Reference

Spectrum Analyzer has 10 user-facing parameters, all exposed through the settings overlay (gear icon) except Channel Mode which sits in the header. Double-click any knob in the overlay to type an exact value directly instead of dragging. The LUFS, True Peak, and correlation meters are always running and have no parameters of their own.

Display

- **FFT Resolution:** 2048, 4096 (default), 8192. Higher resolution gives finer low-frequency detail at the cost of slower update rate. 2048 is the most responsive; 8192 reveals individual harmonics down to about 30 Hz. 4096 is the right starting point.
- **Smoothing:** 0.0 to 1.0, default 0.5. Damps the per-frame fluctuations of the spectrum line. 0 is maximally responsive (jittery); 1 is heavily smoothed (sluggish). Use 0.7 to 0.9 for tonal-balance work; 0.2 to 0.4 for catching transient resonances.
- **Slope:** -4.5 to +4.5 dB/oct, default 0.0. Tilts the display. +4.5 makes pink-noise read flat; great for mixing reference. 0 shows the raw spectrum.
- **Decay Rate:** 3 to 60 dB/sec, default 20. How fast the spectrum line falls when audio energy drops. Slower decay (lower number) holds peaks visible longer.
- **Display Min:** -100 to -30 dB, default -60. Bottom of the visible vertical range.
- **Display Max:** 0 to +12 dB, default +6. Top of the visible vertical range.

Peak Hold

- **Peak Hold:** On / Off, default Off. Overlays a "peak hold" trace that captures the highest level seen at each frequency.
- **Peak Hold Time:** 0.5 to 10 sec, default 2.0. How long the held peaks stay visible before they decay.

Channel routing

- **Channel Mode:** Stereo, Mono, Mid, or Side. Stereo overlays the L and R spectra; Mono sums both; Mid shows the (L+R)/2 sum; Side shows the (L-R)/2 difference.

Metering

- **K-System Type:** K-12, K-14 (default), K-20. Sets the reference level for the K-System scale. K-14 is the typical mastering reference; K-20 is broadcast/film; K-12 is loudness-targeted music.

Tips and Traps

- **Pre-fader vs post-fader matters.** Spectrum Analyzer measures whatever signal reaches it. Insert it pre-fader to measure the source independent of automation; post-fader to measure what the listener hears.
- **LUFS Integrated needs a continuous play.** The Integrated reading uses gating to ignore silence and very-low-level passages. If you stop and start playback, the gate window resets and the reading drifts. Play through the section in one go.
- **True Peak with 4x oversampling is on by default.** This catches inter-sample peaks that a regular sample-peak meter misses. The cost is small CPU; leave it on for mastering.
- **The correlation meter can mislead on quiet sources.** When the signal is below about -50 dBFS, correlation readings get noisy and can swing wildly. Trust correlation only when there is real audio playing.
- **Slope shifts the curve, it does not change the audio.** This is a display setting only. Do not confuse it with an EQ.
- **Mid/Side mode is for monitoring, not for routing.** Switching to Mid does not change the audio leaving the plugin; it changes what the spectrum draws.
- **The plugin does not write to the audio path.** Spectrum Analyzer is a pure analyzer; output equals input regardless of settings.

Recommended Setups

Spectrum Analyzer does not ship with factory presets, but a few starting-point configurations are worth saving as DAW-level "saved states" or default presets you can recall.

Mixing reference

Slope 4.5, Smoothing 0.8, Decay 20, FFT 4096, Channel Mode Stereo, Peak Hold on with a 3-second decay. Drop this on the mix bus while building a song; the tilted display lets you eyeball whether the mix is generally balanced or skewing one way.

Mastering target

Slope 0, Smoothing 0.5, Display Min -100, Display Max 0, K-System K-14, FFT 8192. The longer FFT gives the low-end resolution you need to spot 50/60 Hz hum and sub-bass issues. K-14 is the standard mastering reference scale.

Imaging analysis

Channel Mode Side, Slope 4.5, Smoothing 0.7. Quickly switches the display to the side component so you can see what is contributing to stereo width. Pair with a second instance set to Mid mode if your DAW supports parallel routing.

Live tracking

Smoothing 0.2, Decay 40, FFT 2048, Peak Hold off. Maximally responsive setup for catching transients and quick spectral changes during recording. The shorter FFT keeps latency-related drawing artifacts to a minimum.

Troubleshooting

The spectrum is completely flat. No audio is reaching the plugin. Check that you have inserted it in the signal path (not on a return) and that the track is playing. If the signal is very quiet (below -60 dB), it can fall below the bottom of the default display range; lower Display Min to -100 dB.

LUFS reads -inf or never settles. Integrated LUFS uses gating; if the audio is mostly below -70 LUFS the gate filters everything out. Make sure the track is at a reasonable level and play through a longer section. The reading needs at least a few seconds of audio above -50 LUFS to give a useful number.

The correlation bar swings between green and red constantly. That is normal on busy stereo material. Watch the average position over a few seconds, not the instantaneous value. Persistent red on quiet material can indicate phase issues with stereo doubling or short-delay effects.

Chord Analyzer

Overview

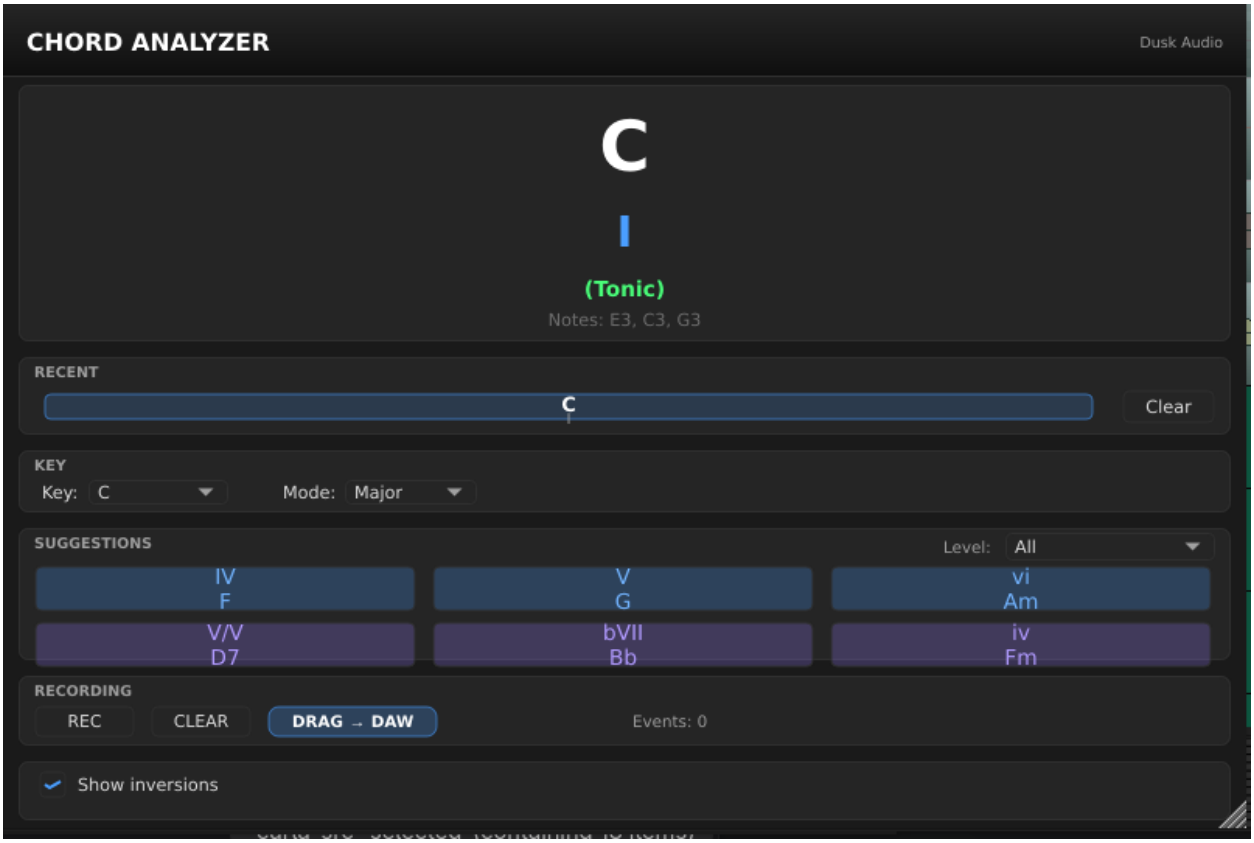


Figure 26: Chord Analyzer main UI showing a detected chord

Chord Analyzer is a MIDI plugin that watches the notes coming into your DAW and tells you what chord you are playing. It detects the chord root, quality (major, minor, 7th, sus, etc.), bass note for slash chords, and inversion. It also shows theory-aware suggestions: which chords are typical in the key you have selected, and which substitutions are available.

Use it for transcription (figure out what a recording is doing), for learning (see chord names while you play), for songwriting (browse suggested next chords), and for live performance (display the chord on a screen so the rest of the band can follow).

It is not a chord-generation tool (it does not play chords for you), and it is not a MIDI processor (it passes notes through unchanged). It analyzes; it does not modify.

The plugin ships in three variants because different DAWs handle MIDI-only plugins differently. Picking the right variant for your DAW is the most important first step.

Quick Start

1. Identify the right variant for your host:
    - **Chord Analyzer MIDI** for desktop DAWs that recognize MIDI-only plugins (Bitwig, Reaper, Logic, Cubase, Ardour, Carla).

- **Chord Analyzer** (the instrument version) for DAWs that need a plugin to claim audio output (Ableton Live, FL Studio).
- **Chord Analyzer Headless** for headless LV2 hosts that have no plugin GUI (Zynthian and similar).

2. Insert the plugin in front of your synth or instrument track. Detailed routing per-DAW is in the next section.
3. Set the **Key Root** and **Key Mode** at the top of the plugin window to match the song you are playing in. C Major is the default.
4. Play notes on your MIDI controller. The detected chord appears in the main display, with inversions and theory suggestions below.

You should see the chord name update as you play (for example, holding C, E, and G shows "C major"; adding B shows "C7"). If the chord display stays empty, check that MIDI is reaching the plugin and that you are holding at least three notes.

Workflows

The right setup depends on your DAW. The plugin always sits in front of your synth or instrument; what differs is how the DAW routes MIDI through it. Pick your DAW from the list below.

Bitwig Studio

1. Add your synth to a MIDI track as normal.
2. In the device chain, click + before your synth.
3. Search for **Chord Analyzer MIDI** in the **Note FX** category and add it.
4. MIDI passes through automatically; your synth still receives all the notes.

Reaper

1. Open the FX chain on your MIDI/instrument track.
2. Click **Add** and search for **Chord Analyzer MIDI** (VST3).
3. Drag it above your synth in the chain.
4. MIDI passes through to your synth automatically.

Logic Pro

1. On your instrument track, click the **MIDI FX** slot (above the Instrument slot).
2. Choose **Chord Analyzer MIDI** under Dusk Audio.
3. The plugin receives MIDI from the track and passes it through to your instrument.

Cubase / Nuendo

1. On your instrument track, add **Chord Analyzer MIDI** as a **MIDI insert**.
2. The plugin receives MIDI from the track automatically.
3. Your instrument remains unchanged.

If Chord Analyzer MIDI does not appear in the MIDI Inserts slot after updating, force a plugin rescan in Cubase's Plug-in Manager so the host picks up the updated plugin category.

Ableton Live

Ableton does not list MIDI-only plugins in MIDI tracks the same way other DAWs do, so use the **instrument version** with an Instrument Rack:

1. Load your synth on a MIDI track as normal.
2. Select your synth and press **Cmd+G** (Mac) or **Ctrl+G** (Windows) to group it into an **Instrument Rack**.
3. Click the **Show Chain List** button (three horizontal lines on the left side of the rack).
4. Right-click in the empty space below your synth's chain and choose **Create Chain**.
5. Drag **Chord Analyzer** (the instrument version, not MIDI) from the browser into the new empty chain.
6. Both chains receive the same MIDI; your synth produces sound, the Chord Analyzer displays chords.

FL Studio

Use the **instrument version** with Patcher:

1. Add a **Patcher** instance to your channel rack.
2. Inside Patcher, add both **Chord Analyzer** and your synth as nodes.
3. Route the MIDI input to both plugins in parallel.
4. Route only your synth's audio output to the Patcher output.
5. Both plugins receive MIDI; your synth produces sound, the Chord Analyzer displays chords.

Desktop LV2 hosts (Ardour, Carla)

Use **Chord Analyzer MIDI**; it declares no audio ports and includes the full custom visualizer in the host plugin window.

1. Add **Chord Analyzer MIDI** to the MIDI chain before your synth.
2. MIDI passes through to the next plugin in the chain.
3. Open the plugin window to see the full chord display.

Headless LV2 hosts (Zynthian)

Use **Chord Analyzer Headless**; it exposes the detected chord through native LV2 output control ports so the host can render the values in its own parameter view.

1. Install the bundle from the `chord-analyzer-headless-*.zip` (separate download).
2. Add **Chord Analyzer Headless** to your MIDI chain before your synth.
3. The detected root, quality, bass, and inversion appear as live values in your host's plugin parameter view.
4. Four suggestion slots appear below them, each a root and a quality, carrying the first four chords the plugin's own suggestion list would show for what you are playing.

The headless bundle publishes nineteen ports:

Index	Symbol	Direction	What it carries
0	`midi_in`	in	MIDI, passed through unchanged
1	`midi_out`	out	The same MIDI, forwarded to the next plugin
2	`key_root`	in	Key Root, C through B
3	`key_mode`	in	Key Mode, Major or Minor
4	`suggestion_level`	in	Suggestion Level, Basic through All

Index	Symbol	Direction	What it carries
5	`show_inversions`	in	Show Inversions
6	`detected_root`	out	Detected Root, - when nothing is playing
7	`detected_quality`	out	Detected Quality
8	`detected_bass`	out	Detected Bass
9	`detected_inversion`	out	Detected Inversion
10	`respect_sustain`	in	Respect Sustain
11, 13, 15, 17	`suggestion_N_root`	out	Root of suggestion N, - for an empty slot
12, 14, 16, 18	`suggestion_N_quality`	out	Quality of suggestion N

Suggestion slots use the same value scale as the detection ports, so a host can render all of them with one widget. They follow **Key Root**, **Key Mode** and **Suggestion Level** as well as the notes you play: at Basic Only a chord with three suggestions fills three slots and leaves the fourth reading –.

Parameter Reference

Chord Analyzer exposes five user-facing parameters plus four read-only detection outputs.

Key context

- **Key Root:** 12 choices, C through B (with enharmonic spellings shown). Default C. Sets the tonal center for theory-aware suggestions. The detector itself is key-agnostic (it identifies chord names regardless of key), but the suggestion display uses this to highlight chords that fit your key.
- **Key Mode:** Major or Minor, default Major. Combined with Key Root, defines the key for suggestions. Set this to match the song you are playing or transcribing.

Detection settings

- **Suggestion Level:** Basic Only, Basic + Intermediate, or All (+ Advanced). Default All. Controls how many tiers of suggested chord substitutions appear below the detected chord. Basic shows only diatonic triads (the seven chords in the key). Intermediate adds 7th chords and common borrowed chords. Advanced adds tritone substitutions, secondary dominants, and modal interchange options.
- **Show Inversions:** On by default. When on, the display tells you which inversion of the detected chord you are playing (root position, first inversion, second inversion, third inversion for 7ths). Turn off if you want a less cluttered display.
- **Respect Sustain:** On by default. When on, MIDI CC 64 (sustain pedal) holds the detected chord on screen until you release the pedal, even if you let go of the keys. Useful for transcription workflows where you want to lift your hands to type or take notes. Turn off if your controller sends sustain CCs you do not want the plugin reacting to.

Detection outputs (read-only)

These parameters expose the detection result for host automation, screen-recording overlays, and the Headless variant's parameter-driven display:

- **Detected Chord:** The whole chord name exactly as the plugin's own display shows it, slash bass and all (`Cadd#11/F#`). Use this when your host can put a parameter's text on a widget, such as Gig Performer, and you want the chord on screen without opening the plugin. It follows **Show Inversions**: turn that off and the slash bass disappears here too. Not available in the Headless variant, whose LV2 control ports carry numbers only.
- **Detected Root:** Current chord root (12 values).
- **Detected Quality:** Current chord quality (major, minor, 7, maj7, m7, sus2, sus4, dim, aug, etc.).
- **Detected Bass:** Bass note for slash chords (like C/E or G/B).
- **Detected Inversion:** Which inversion is currently being played, named after the degree in the bass: `Root`, `1st` (3rd in the bass), `2nd` (5th), `3rd` (7th), `4th` (9th), `5th` (11th), `6th` (13th). A bass note the chord does not contain, such as the F# under `Cadd#11/F#`, reads `Slash` rather than a numbered inversion.

You cannot set these from the host; they update automatically as the plugin detects chords.

The Headless variant adds four suggestion slots to these, as output control ports only. See the port table under **Headless LV2 hosts** above.

Tips and Traps

- **The variant matters more than the version.** Most "I cannot find the plugin" issues are users looking for the MIDI variant in Ableton (which only sees the instrument variant) or the instrument variant in Logic (which puts the MIDI variant in the MIDI FX slot).
- **The plugin needs at least three notes to detect a chord.** Holding a single note shows nothing; holding two shows an interval but not a full chord name. Three or more notes give a full chord identification.
- **Detection runs on whatever notes are currently held.** If you arpeggiate a chord one note at a time without holding the previous notes, the display flickers between intervals. Use the sustain pedal (with **Respect Sustain** on) to hold notes for chord detection while you arpeggiate.
- **The Key setting does not affect detection.** It only affects which suggestions appear. C Major and A minor produce identical chord names; the suggestions shown below differ.
- **The Headless variant has no GUI.** It exposes detection through host parameters only. If you open it in a desktop DAW, you will see only the parameter list, no visualizer. Use the MIDI or instrument variant if you need the visualizer.
- **MIDI passes through unchanged.** Notes, velocities, channel messages, sustain, modulation, all forwarded to the next plugin in the chain. Chord Analyzer does not insert, suppress, or alter any messages.

Working with Suggestion Levels

Chord Analyzer does not ship with factory presets, but the **Suggestion Level** parameter behaves like a preset for the suggestion display.

Basic Only

Shows only the seven diatonic triads of your selected key. Useful if you are learning theory and want to see only the "in-key" chord options. If you play a chord that is not in your key, it is still detected and displayed; only the suggestions below restrict to in-key options.

Basic + Intermediate

Adds 7th chords (Imaj7, ii7, IV7, V7, vi7, etc.) and common borrowed chords (a iv minor in major, a III major in minor). This is the right setting for most pop, rock, and jazz songwriting work.

All (+ Advanced)

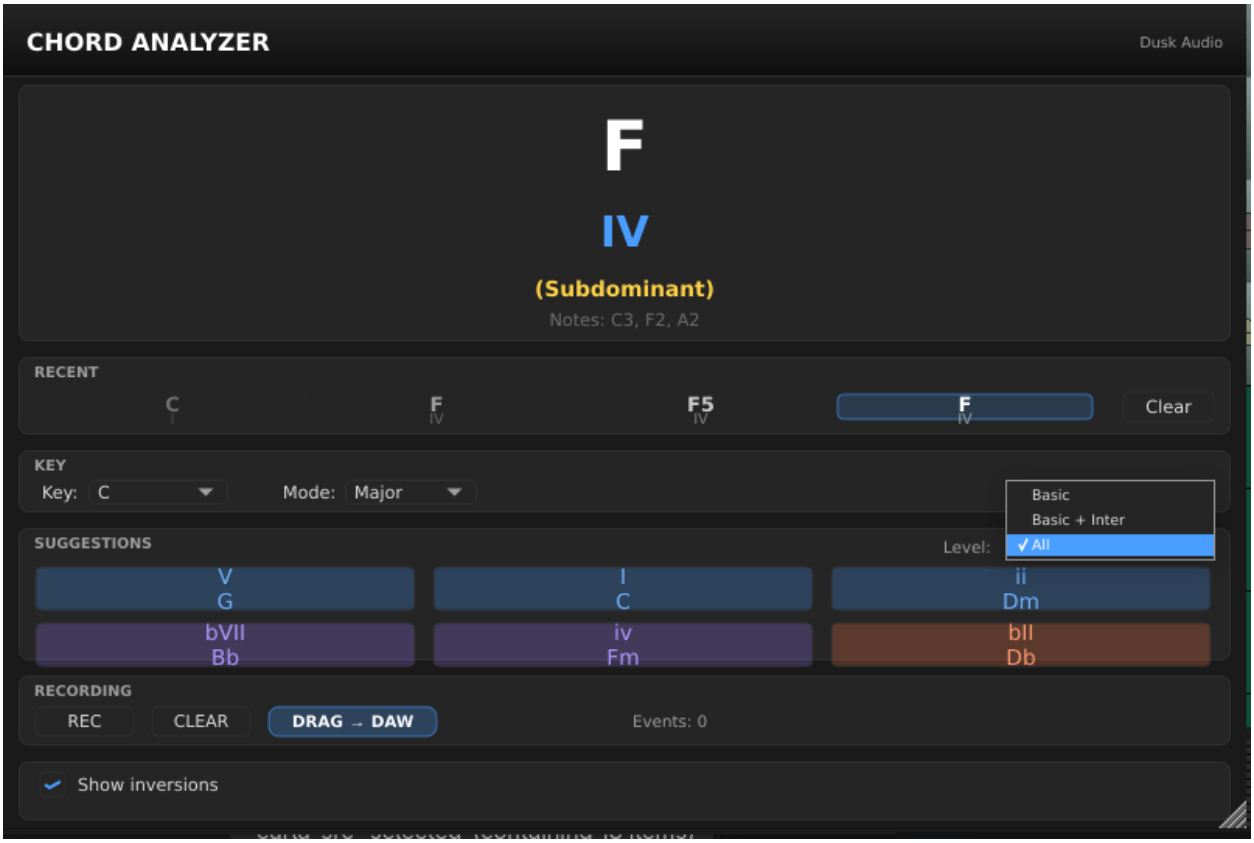


Figure 27: Suggestions panel with All tier visible

Adds tritone substitutions (where bII7 can replace V7), secondary dominants (V7/V, V7/vi, etc.), and modal interchange options drawn from parallel modes. The default; useful for jazz transcription and substitution exploration.

If the suggestion list feels overwhelming, drop to Basic + Intermediate. If you are working through a Real Book chart and the substitutions feel limiting, stay on All.

Troubleshooting

The chord display is empty even though I am playing notes. Confirm the plugin is receiving MIDI. In most DAWs, the MIDI variant must be inserted before the synth on the same track. In Ableton and FL Studio, the instrument variant must be in a parallel chain receiving the same MIDI. If MIDI is reaching it but no chord shows, you are probably holding fewer than three notes; the plugin needs at least a triad to identify a chord.

My DAW does not show the plugin in its menu. Ableton and FL Studio do not list the MIDI variant in their MIDI plugin menus; use the instrument variant with the routing described above. Logic Pro lists the MIDI variant only in the **MIDI FX** slot, not in the regular instrument

menu. If you cannot see either variant anywhere, your DAW has not yet rescanned plugins; force a plugin scan in the host preferences.

MIDI is not reaching my synth after I insert Chord Analyzer. The plugin passes MIDI through unchanged in all desktop variants. If your synth stops receiving notes, the plugin is being inserted as the only destination rather than as a pass-through. Check your DAW's MIDI routing; in Ableton's Instrument Rack approach, both chains receive the same MIDI in parallel, not sequentially.

The detected chord disappears when I let go of the keys. That is expected behavior unless you have **Respect Sustain** on and are pressing the sustain pedal. With the pedal pressed, the detected chord persists until you release the pedal.

DuskVerb 2

Overview

DuskVerb 2 combines the original DuskVerb reverb engines, its own nonlinear engine and updated voicing with a redesigned interface. Its 24 factory presets provide starting points for plates, rooms, halls, springs and nonlinear bursts. Engine selection changes the reverb topology; some controls change labels or availability to match the selected engine.



Figure 28: DuskVerb 2 on the Vocal Plate preset: graphite faceplate, amber engine and decay display, output history, and controls grouped in signal order

The amber display shows the engine, decay setting and output history. The large Decay knob sits beside it; Input, Early Reflections, Modulation, Damping, Macro, Filter and Output controls fill the rows below.

Quick Start

1. Select a factory preset from the top preset menu. The arrows move through presets.
2. Adjust **Decay** and **Size** to fit the source and arrangement.
3. Set **Mix** for an insert, or enable **Bus** for a fully wet return. Turn on **Mix Lock** to keep Mix and Bus where they are while you audition presets.
4. Use **Pre-delay** to separate the source from the reverb. Choose a rhythmic division in **Sync** to follow the host tempo.
5. Shape the tail with the **Damping** multipliers and crossover controls. Use **Tone** for a broader adjustment, then set **Trim**.

6. Click **Save** to store your sound as a user preset.

Controls

Section	Purpose
Input	Pre-delay, tempo sync and saturation.
Filter	Low Cut, High Cut, Mono Below and Mono Depth.
Decay / Size	Tail length and space size.
Output	Dry / Wet, Width, Trim and fully wet Bus mode.
Early Reflections	Level, size and diffusion.
Damping	Three decay multipliers and two crossover frequencies.
Modulation	Depth and rate; labels follow the engine.
Macro	Tone, Character and Duck.

Sync retains the free pre-delay setting, and Bus retains the Dry / Wet setting.

Freeze holds the reverb on supported engines; it is hidden for Reverse and Nonlinear. The gated engine also exposes its **Gate** switch. Controls unused by the selected engine are dimmed. **Tonal Correction** appears only where it changes the sound: the Vocal Plate, Ambience and Tiled Room sounds on the Hall and Tiled Room engines (and on the older hall engines a restored session may still use).

The display at the top left shows the engine (click its name to choose another), the decay time (click it to type a value), the **Freeze**, **Sync** and **Bus** indicators, and **Output History**. Output History shows recent output level; it is not a measured RT60 graph, and the decay readout is the parameter setting, not a guarantee of a measured acoustic decay time. The side meters show input and output levels.

Freeze can sustain a tail indefinitely on supported engines. Choose the desired end of an offline bounce explicitly when using Freeze. CLAP and VST3 report an infinite frozen tail; AU uses the framework's one-hour fallback because its tail property has no infinite sentinel. Unfrozen tail estimates include long decay and damping settings.

Editing Values

- Drag a knob to adjust it; hold Shift for finer movement.
- Use the mouse wheel to step a value.
- Click a numeric readout or double-click a knob to type a value. Enter applies it; Escape cancels. Invalid input stays available for correction.
- Right-click a knob for its reset and value-entry menu.
- Ctrl-click or Alt-click resets a knob on Windows/Linux. On macOS, use Alt-click or Cmd-click.
- Drag the editor corner to resize. The minimum size is 1050 × 700.
- Click **?** for the in-plugin control reference. The amber display also shows this instance's approximate CPU load.

Presets and Comparison

INIT restores defaults. Factory recall applies the factory sound while preserving the settings intentionally excluded by the original preset format. An asterisk beside the preset name indicates

an edited sound. **Mix Lock**, below the Bus switch, keeps Dry / Wet and Bus unchanged on preset recall and INIT; A/B still restores the complete sound.

A and **B** switch between complete comparison snapshots. **COPY** copies the active sound into the other slot without switching. Save sounds you want to keep as user presets; comparison slots are a working audition tool.

The Save dialog requires explicit replacement confirmation for an existing name. Names that map to the same filename, such as `Alpha/Beta` and `Alpha-Beta`, receive distinct filenames. Right-click a user preset to delete it. The preset menu also offers import of original DuskVerb preset files. Malformed imports are rejected without partially applying their values.

User presets are stored in the `DuskAudio/DuskVerb2/presets` subdirectory of the following configuration folder:

- Linux: `$XDG_CONFIG_HOME`, or `~/.config` when XDG_CONFIG_HOME is unset.
- macOS: `~/.config`.
- Windows: `%APPDATA%`, with LOCALAPPDATA as fallback.

Safe publication of new presets requires a filesystem with hard-link support. A save failure is reported rather than replacing an unrelated preset.

Compatibility

DuskVerb 2 has its own plugin identity and can coexist with the original DuskVerb. Importing an original preset does not convert a DAW project or replace an original plugin instance. Keep the original plugin available for projects that use it.

The 1.0 Windows VST3 identity matches Linux and macOS. Windows projects saved with an earlier DuskVerb 2 preview may need the preview instance replaced: export its user preset first, insert 1.0, import the preset, and check automation and routing. Keep the preview available until those projects are migrated. Original DuskVerb has a separate identity and is unaffected.

The plugin supports mono, mono-to-stereo and stereo operation where the host exposes those layouts. Select the appropriate host routing before playback.

Ring Out User Manual

Overview

Ring Out finds the frequencies at which a sound system starts to feed back and notches them out. You insert it on a monitor send or the master bus, arm it, and push the gain up the way you would when ringing out by hand; every time a tone starts to build, Ring Out places a narrow cut at exactly that frequency, deep enough to stop it, and moves on to the next one. After a handful of notches you take the gain back down and have a few dB more headroom before the system rings.

It shines at exactly that job: the pre-show ring-out of wedges, in-ears driven from a floor mic, and a PA in a lively room. Twenty notches, each with its own cut, frequency and Q, can also be placed and adjusted by hand, on the graph or in the notch bank, so it doubles as a surgical notch EQ.

It is not a tool to leave hunting during the show. The detector tells feedback from complex programme well, but it cannot tell a ringing wedge from a guitarist's deliberate sustain, and a notch placed on the wrong thing stays until you remove it.

Quick Start

Install from the Dusk Audio website and insert Ring Out as the last thing on the monitor send (or on the master bus for front of house). Set the channel, aux and amplifier gains the way the show will run them.

Three controls to learn first:

1. **LEARN** arms the detector for one minute. The round button turns amber, shows the time left, and the ring around it drains while it listens.
2. **OUTPUT** is how you provoke the system. With LEARN armed, raise it slowly. When a frequency starts to ring you will see a peak climb on the analyser and, a fraction of a second later, an amber notch open under it with a ripple around its node, and the ringing stop.
3. **CLEAR** removes every notch if you want to start again.

Stop after four to six notches, bring OUTPUT back to 0 dB, and talk into the microphone. The system now sits a few dB further from feedback than before, and the notches are narrow enough that the tone of the mix is barely changed.

Workflows

Ringing out a vocal wedge

Source: a dynamic vocal microphone on a stand, feeding a floor wedge through an aux send, singer not yet on stage.

Target: the wedge loud enough for the singer to hear, with the usual three to five ring frequencies removed.

Settings: SENSITIVITY Low, OUTPUT 0 dB, no notches. Press LEARN. Raise OUTPUT about 1 dB every two seconds. Each time a notch appears, pause a moment and keep going. After five or six notches, or when the detector has been quiet for a while at a level well beyond where you will run the show, press LEARN again to stop, and return OUTPUT to 0 dB.

Why: the slow raise lets each tone build on its own, so each notch lands on one frequency instead of the detector having to pick apart several ringing at once.

Catching one more ring during line check

Source: a system already rung out, but one frequency starts to sing when the singer cups the microphone.

Target: one extra notch, nothing else touched.

Settings: hold **HOLD TO CATCH** while the singer repeats the move. CATCH searches for as long as you hold it and stops the moment it has placed or deepened one notch; the button turns green to tell you. Release it. If the tone persists, hold CATCH again: a second pass on the same frequency deepens the existing notch by 3 dB rather than placing a duplicate.

Why: CATCH never runs away. One press, one notch, and nothing is placed on the band's downbeat.

Pulling a notch by hand

Source: a room resonance you already know, say 125 Hz in a small club.

Target: a 6 dB cut at 125 Hz, Q 4, before anything rings.

Settings: click the + on the first empty strip of the notch bank. The new notch arrives at 24 Hz, 0 dB, Q 2.5 and is selected: its strip is outlined, its node on the graph wears a white ring and a callout with its values. Drag the node to the right until the callout reads about 125 Hz and down to about -6 dB, then roll the mouse wheel over it until Q reads 4.0. For exact values, use the FREQ, CUT and Q read-outs under the bank: the + and - ends step the value, and a double-click lets you type it.

Why: a notch placed by hand behaves exactly like one the detector placed; the engine will deepen it later if that frequency rings anyway.

Trading gain for depth with LINK

Source: a rung-out system, show about to start, the band wants the wedges 3 dB louder.

Target: 3 dB more wedge level with the same margin against feedback.

Settings: click the chain between CUT TRIM and OUTPUT so it joins and turns amber (**LINK**). Turn OUTPUT up to +3.0 dB; CUT TRIM follows to -3.0 dB, so every notch cuts 3 dB deeper at the same time. Both read-outs turn amber while they are away from their defaults.

Why: the feedback margin you built lives in the notches. If the whole system goes up 3 dB, the notches have to go down 3 dB to keep the loop gain where it was.

Checking the damage

Source: a mix that sounds thinner than before the ring-out.

Target: find the notch that hurts and soften it.

Settings: the notch bank shows every notch at once, so scan the faders: a long amber bar is a deep cut. Click a strip, or use the < and > buttons under the bank, to select a notch and see it on the graph. A notch in the 200 Hz to 500 Hz region with a deep cut and a low Q is the usual culprit. Pull its fader up a few dB, or roll the wheel over its node to raise its Q a step or two, and listen. To compare, double-click its number badge (or press ON under the bank) to switch it off and on.

Why: the detector errs on the side of stopping the ringing. Once the pressure is off, a narrower or shallower notch often still holds.

Parameter Reference

Detect

LEARN, press to arm, press again to stop. Arms the detector for 60 seconds, after which it switches itself off; the amber ring around the button drains over the minute and the button shows the time left. Arming from the editor also puts Q SCALE and CUT TRIM back to their defaults, so the notches you are about to place are judged at face value; a host or controller trigger leaves those trims unchanged. The armed state is never saved with a session: a project you reload is always idle. Common mistake: leaving it armed during the show.

CATCH (the HOLD TO CATCH button), momentary. Searches while held, stops after one notch has been placed or deepened, then turns green until you release it. Press again for another. From a mapped controller a tap starts the same search; it ends with its notch or after ten seconds, so a button that sticks cannot keep it hunting.

SENSITIVITY, Low or High, default Low. How readily the detector calls a tone feedback. Low needs the tone to stand clearly above everything else, have no harmonics and persist for about 130 ms; High accepts quieter and shorter tones. Use Low for ringing out before the show; use High on a system that only rings briefly before someone pulls the fader. Common mistake: High on a stage with sustained synth pads, which places notches on the pads.

The **status line** under SENSITIVITY says what the detector is doing: LEARNING, CATCH: LISTENING, CATCH: NOTCH PLACED, LEARN ENDED when the minute runs out, or how many notches are in place. RING NOT COVERED shows in red while it listens if a ring keeps going and the detector has nothing left to do about it (see Tips and Traps).

Notch bank

Twenty strips, one per notch. Each strip shows the notch's number badge (amber when the notch is on, outlined when it is off), its frequency, a **CUT** fader, its cut in dB and its Q. Click a badge or a strip to select the notch; double-click the badge to switch the notch off or on again. Drag a fader to set the cut (hold Shift for fine steps) or roll the mouse wheel over it for 0.5 dB steps. The + on the first empty strip adds a notch. **CLEAR**, above the bank, removes every notch.

Under the bank, the selected notch's exact values:

FREQ, 24 Hz to 20 kHz, default 24 Hz. The + and - ends step by 1 Hz below 500 Hz, 10 Hz to 1 kHz and 100 Hz above. A notch the detector placed shows its exact frequency (66.3 Hz, for example); stepping from there keeps that precision.

CUT, -20 to 0 dB in 0.1 dB steps, default 0 dB. The depth of the notch. Deeper is not better: 6 to 8 dB stops most ringing, and 20 dB with a wide Q removes a noticeable slice of the mix.

Q, 0.5 to 20 in 0.1 steps, default 2.5. Higher is narrower. The detector places notches between Q 3 and Q 8 and only widens them if a tone keeps ringing through a 20 dB cut.

Each read-out steps with its + and - ends, drags up and down, and takes a typed value on a double-click. **ON** switches the selected notch in or out; **DELETE** removes it and the strips to its right move left to close the gap; < > select the previous or next notch.

Graph

The graph shows the input analyser in silver against the right-hand dBFS scale, and the combined response of all notches in amber, hanging from its 0 dB line against the left-hand scale, which runs from 0 to -24 dB and extends in 12 dB steps when the notches go deeper, down to -48 dB; anything

deeper than that is drawn flat on the floor. Each notch is a numbered node at the bottom of its cut, after CUT TRIM; a switched-off notch sits outlined on the 0 dB line.

The graph is also an editor. Drag a node across to change its frequency and up or down to change its cut (hold Shift for fine steps); a switched-off notch drags in frequency only. A drag rounds the frequency to three significant figures; use the FREQ read-out for anything finer. The scale holds still while you drag, and a node pulled below the floor waits there with a small arrow until you let go. Roll the mouse wheel over a node to change its Q (0.5 per step, 0.1 with Shift); double-click a node to switch it off or on. Clicking a node, or anywhere in a notch's frequency column, selects it, and the selected node shows a callout with its number, frequency, cut and Q; with CUT TRIM set, the callout also gives the depth after the trim. A notch the detector has just placed or deepened ripples amber for a moment, as does its badge in the bank.

All notches and output

Q SCALE, 0.2 to 10, default 1.0. Multiplies the Q of every notch; 0.5 makes every notch twice as wide. Reset when LEARN is armed from the editor.

CUT TRIM, -24 to +24 dB, default 0 dB. Added to every notch's cut. Negative deepens all notches, positive shallows them; a notch never turns into a boost, so +24 dB simply flattens everything. Reset when LEARN is armed from the editor.

Both trims show an amber arc from their default to their value and an amber read-out while they are away from the default; untouched, they show no arc.

LINK, on or off: the chain between CUT TRIM and OUTPUT, joined and amber when on. Ties CUT TRIM to OUTPUT in opposite directions when you turn either knob in the editor: raise OUTPUT by 3 dB and CUT TRIM drops by 3 dB. Host automation of one knob moves only that knob.

OUTPUT, -24 to +24 dB, default 0 dB. The output level, and the control you raise to provoke the system during setup.

The **IN** and **OUT** meters read peak level from -80 to 0 dBFS with a peak-hold mark; click a meter to reset its hold.

Tips and Traps

Raise the gain slowly while LEARN listens. Two frequencies ringing at once still get caught, but one at a time gives cleaner notches.

A tone that keeps ringing after a notch has been placed is not ignored: the detector deepens that notch in 3 dB steps to -20 dB and then widens it toward Q 0.7. If you find a notch at -20 dB and a low Q after a session, the system was a long way over the edge at that frequency; look at the gain structure or the microphone placement rather than asking the plugin for more.

More than about twelve notches, or notches at -16 dB and beyond, mean the same thing. With all twenty notches in use the detector will still deepen a notch whose band a new ring falls into, or widen the nearest one toward a ring just outside it, but it will not touch an unrelated one, and the bank has no + left. When a ring keeps going and the detector has nothing left to do about it, the status line shows RING NOT COVERED in red while it is listening: the notches it could use are at their limits, and the fix is in the gain structure.

The detector ignores tones with harmonics, tones that decay, and tones that are themselves a harmonic of something louder. A pure test oscillator held steady will be treated as feedback; so will a flute note held long enough. Stop LEARN before a soundcheck starts in earnest.

Ring Out adds no latency and runs the same notches on both channels of a stereo insert.

The notch table is saved with your session and in user presets, so a ring-out done at soundcheck survives a project reload.

Presets Explained

Default is the only factory preset: every control at its default and no notches. It is also what INIT loads. The header arrows step through the preset list without wrapping and never reload the preset already shown, and they do nothing while an edited, unsaved table is on screen; choosing Default from the list, or INIT, is how you deliberately clear a ring-out.

User presets, saved with SAVE in the header, hold the five settings (SENSITIVITY, Q SCALE, CUT TRIM, LINK, OUTPUT) and the complete notch table. Save one per venue or per wedge mix and recall it at the next show as a starting point; a quick LEARN pass then catches whatever the room does differently that night.

Troubleshooting

LEARN is armed but nothing is being placed. The system is not ringing yet, or it rings so briefly that Low sensitivity will not commit. Raise OUTPUT further, or try SENSITIVITY High. Also check that the plugin is in the path that feeds back: it has to be between the microphone and the loudspeaker.

Notches appear on the music, not on feedback. SENSITIVITY is on High during the performance, or LEARN was left armed. Switch to Low, stop LEARN, remove the stray notches with DELETE, and hold CATCH for a moment when you actually hear a ring.

The ringing came back after a session. Something changed the loop gain: a microphone moved, a monitor was turned up, a fader was pushed. Hold CATCH while it rings to deepen the existing notch, or run LEARN again for a minute. If notches sit at -20 dB and the ringing persists, the problem is the gain structure, not the notches.

Automation lanes have new names. Ring Out 1.1 renamed its parameters to match the editor: Sense is now Sensitivity, Setup is Learn, Add is Catch, Reset is Clear, Global Q is Q Scale, Global Amp is Cut Trim and Gain Out is Output. Existing sessions and automation load unchanged; only the names your host shows are different.

Sunset Circuits

Overview

Sunset Circuits is six vintage synthesizers in one instrument. A row of six backlit mode rockers across the top picks the engine, and each mode is a different classic circuit with its own voice architecture, its own signature controls, and its own panel color. Switching modes is a creative decision, not just a preset change; the whole personality of the instrument changes with it.

The six modes cover the ground that defined subtractive and FM synthesis:

- **Cosmos** is an early-80s Japanese six-voice poly built on digitally controlled oscillators, with a resonant low-pass filter and a built-in bucket-brigade chorus.
- **Oracle** is a late-70s American five-voice poly with two analog oscillators, a self-oscillating filter, and poly-modulation routing.
- **Mono** is an aggressive 70s monophonic voice with two oscillators plus a sub, a fat driven filter, ring modulation, and hard sync.
- **Modular** is a 70s semi-modular patchable voice with three oscillators, a transistor-ladder filter, sample and hold, and a spring reverb.
- **Prism** is a mid-80s four-operator FM digital engine with eight routing algorithms and per-operator envelopes.
- **Acid** is the bass box: one oscillator through a screaming diode-ladder filter, with accent and slide and a sixteen-step pitch pattern sequencer.

The same bone structure is present in every mode. The oscillator and mixer panels sit on the left, the filter and two envelopes fill the center, the two LFOs and the mod matrix are on the right, and the sequencer and effects run along the bottom with a playable keyboard beneath them. Because the layout never moves, you keep your bearings when you switch circuits; only the colors, the mode sub-panel, and a handful of mode-specific controls change.

Sunset Circuits ships with 54 factory presets spread across all six modes, plus a user preset library you build yourself. It is a full instrument, not an effect: insert it on an instrument track, feed it MIDI, and play.

Getting Started

Formats and installation

Sunset Circuits builds as VST3, CLAP, and LV2 on every platform, plus an Audio Unit on macOS. The binaries share the base name `sunset-circuits`.

Linux

```
VST3: ~/.vst3/sunset-circuits.vst3
CLAP: ~/.clap/sunset-circuits.clap
LV2:  ~/.lv2/sunset-circuits.lv2
```

Windows

```
VST3: C:\Program Files\Common Files\VST3\sunset-circuits.vst3
CLAP: C:\Program Files\Common Files\CLAP\sunset-circuits.clap
LV2:  C:\Program Files\Common Files\LV2\sunset-circuits.lv2
```

macOS



Figure 29: Sunset Circuits in Cosmos mode: the six mode rockers across the top, oscillator and mixer panels on the left, filter and the amp and filter envelopes in the center, the two LFOs and the mod matrix on the right, and the sequencer, effects and keyboard along the bottom

```
VST3: ~/Library/Audio/Plug-Ins/VST3/sunset-circuits.vst3
CLAP: ~/Library/Audio/Plug-Ins/CLAP/sunset-circuits.clap
LV2:  ~/Library/Audio/Plug-Ins/LV2/sunset-circuits.lv2
AU:   ~/Library/Audio/Plug-Ins/Components/sunset-circuits.component
```

After installing, rescan plugins in your DAW. If the instrument does not appear, confirm the file landed in the folder above and that your host scans that folder.

The six mode rockers

The six mode rockers at the top center are the most important control in the instrument. Click one to switch circuits. The panel color crossfades over about a quarter second, the mode sub-panel (the small panel on the right, below the LFOs) morphs to show that mode's signature controls, and a few controls in the standard panels appear or disappear depending on which mode is active. The mode is a normal automatable parameter, so you can switch circuits from the host if you want.

Switching modes is click-safe. The old circuit's voices are crossfaded out over 12 ms while the new one comes up, so a mode change under a held chord is a fast dip rather than a click, and the delay and reverb tails ring on through it. The crossfade lands on an exact sample rather than waiting for the next buffer boundary, so a mode switch sounds the same at a 64-sample buffer and at 4096.

Loading a factory preset also sets the mode, because the mode is stored with the preset. If you switch modes by hand without loading a preset, the other parameters keep their current values, so the sound may not match what you expect from that circuit until you load a preset or dial it in.

Playing

The full-width keyboard at the bottom is playable with the mouse and sends MIDI notes to the engine. It spans 21 white keys (three octaves). Notes arriving from your DAW light up on the keyboard as they play. For real performance, drive the instrument from a MIDI controller or a piano roll; the on-screen keyboard is for quick auditioning.

Velocity comes from where you strike the key. The click's vertical position inside the key maps to velocity from 30 at the top edge to 120 at the bottom: strike high on the key for soft, low on the key for hard. Each key measures against its own height, so black keys and white keys feel the same. Dragging along the keyboard glissandos, and the velocity is re-read on each new key, so a drag that wanders down the keys crescendos. The bottom of the range clears the Acid accent threshold (velocity above 100), so in Acid mode a hard strike accents the way it would on the hardware.

The OCT- and OCT+ buttons shift the played octave by twelve semitones per click, from a bottom key of C0 up to a bottom key of C6. The default is C3.

The pitch and mod wheels sit to the left of the keys. The pitch wheel is bipolar and springs back to center when you let go; the mod wheel latches where you leave it and also responds to the scroll wheel. Both grab absolutely, so clicking part-way up a wheel jumps it there. They are the on-screen stand-ins for the two controllers a hardware synth puts left of the keybed, and the only way to exercise the Pitch Bend and Mod Whl mod-matrix sources without external hardware.

Reading the panel

Most knobs carry a persistent value read-out underneath, so you can see what a patch is set to without touching anything. The read-outs need room to be legible: below about 72 percent of the design size (roughly 893 by 562 pixels) they are suppressed and a hover bubble takes over, showing the value of whatever knob the pointer is on. Make the window larger if you want the values on

screen permanently. The Delay and Reverb panels work slightly differently: their knob labels swap to values while the pointer is inside the panel, and the panel latches into value mode while you are dragging a knob.

The window is freely resizable and keeps its aspect ratio; the whole panel scales to fit. The smallest size is half the design size, 620 by 390. The text is drawn from a font atlas baked when the plugin window opens, so if you move the window to a display with a different pixel density the text can look slightly soft until you close and reopen the plugin. Rebuilding the atlas live is planned for a future release.

MIDI implementation

Sunset Circuits listens on **all MIDI channels** (omni), the usual convention for an instrument plugin, since your host already routes MIDI to the instrument per track.

Message	What it does
Note On / Note Off	Plays the instrument. A Note On with velocity 0 counts as a Note Off.
Pitch Bend	Bends pitch over the range set by PB Range, and is available as a mod source.
CC 1 (Mod Wheel)	Feeds the Mod Whl mod source.
CC 64 (Sustain Pedal)	Holds notes after you release the keys, until you lift the pedal.
CC 121 (Reset All Controllers)	Resets the four performance controllers: lifts the sustain pedal, zeroes the mod wheel, zeroes channel aftertouch, and re-centers pitch bend. It does not touch any synth parameter, so your patch is unchanged.
Channel Pressure	Feeds the Aftertouch mod source for the whole instrument.
Poly Key Pressure	Feeds the Aftertouch mod source for that one note's voice only. Ignored while the arpeggiator is running, where the sounding notes are the arpeggiator's own; use channel pressure there.
Program Change	Recalls a factory preset by number (0 to 53); higher numbers are ignored.
CC 120 / CC 123	All Sound Off / All Notes Off. Stops everything, including a latched arpeggiator and a running Acid pattern, and releases the pedal.

The sustain pedal works in every mode. In Mono and Acid it holds the last note you played, and with the arpeggiator or the Acid sequencer running it holds the pattern's notes exactly as keeping the keys down would, so the pattern keeps playing until the pedal comes up. The difference from Arp Latch is that the pedal is momentary: latch holds until you switch it off, the pedal holds only while it is down.

CC 120 and CC 123 are a true panic. They clear the arpeggiator and Acid held-note sets as well as the sounding voices, so a latched pattern stops rather than carrying on with nothing to release

it, and the pedal is lifted so the next note you play cannot be stranded by a pedal the panic left down.

One caveat applies to program changes in the LV2 version only. A preset recalled by a MIDI program change is applied to the sound immediately, but with the plugin window closed the host is not told which knobs moved, so your DAW may save the old values with the session. The VST3 and CLAP builds are unaffected, and so is LV2 with the window open. If you drive presets by MIDI in an LV2 host, open the plugin window once before saving.

The Six Modes

Cosmos: six-voice DCO poly

Cosmos uses a JUNO-inspired charcoal panel, red synth-section bars, blue performance-section bars, yellow/orange chorus switches, and a green display.

Cosmos is modeled on an early-80s Japanese polysynth with digitally controlled oscillators. It is six-voice polyphonic. The DCO combines saw, pulse, and an octave-down sub with fixed mixer levels: adding a waveform adds to the sound without turning the other sources down. The sub follows Oscillator 1 pitch modulation and detune. Cosmos uses white noise. The 24 dB per octave low-pass reaches self-oscillation near the top of the resonance control, paired with a separate high-pass control for thinning the low end.

The signature of Cosmos is the built-in bucket-brigade chorus. The mode sub-panel is titled **BBD CHORUS** and offers four settings, **OFF**, **I**, **II**, and **I+II**, which set the `Chorus Mode`. Chorus I sweeps at about 0.40 Hz; Chorus II is faster and slightly deeper at about 0.83 Hz. Each uses opposing left/right delay sweeps, so the chorus remains audible when summed to mono. I+II adds both for a denser sound. Fixed headroom ahead of the global effects accommodates the additive mixer and chorus. This chorus is separate from the global chorus in the effects section.

Mode-specific controls visible in Cosmos: the **Sub** oscillator (wave and level), the filter **HP** knob, the **Cross Mod** knob (oscillator 2 modulating oscillator 1 at audio rate), and the four chorus buttons.

Oracle: five-voice poly with poly-mod

Oracle is modeled on a late-70s American five-voice polysynth with two true analog oscillators per voice and a self-oscillating low-pass filter. Where Cosmos is clean and stable, Oracle is characterful and alive. The filter reaches self-oscillation at high resonance, so it can sing on its own.

The signature of Oracle is poly-modulation. The mode sub-panel is a six-knob **POLY-MOD / SYNC** section: filter envelope and oscillator 2 can each reach oscillator 1 pitch, oscillator 1 pulse width, and the filter.

- **FEnv to OscA**: the filter envelope modulates oscillator 1 pitch, for pitched attack transients and bell tones.
- **FEnv to PW**: the filter envelope modulates oscillator 1 pulse width, for moving attacks and animated pulse tones.
- **FEnv to Filter**: the filter envelope is added on top of the filter cutoff, for extra bite.
- **OscB to OscA**: oscillator 2 modulates oscillator 1 pitch at audio rate, for clangorous inharmonic timbres.
- **OscB to PW**: oscillator 2 modulates oscillator 1 pulse width, for animated pulse textures.



Figure 30: Oracle mode: two oscillator panels with no sub oscillator, the auxiliary oscillator slot inactive, and the POLY-MOD and SYNC sub-panel with its filter envelope and oscillator 2 routing knobs

- **OscB to Filter**: oscillator 2 modulates filter cutoff at audio rate, for growling and metallic filter FM.
- **Sync**: oscillator 1 hard-syncs oscillator 2, creating swept harmonic edges when their pitches differ.

Mode-specific controls visible in Oracle: the six poly-mod knobs and the **Sync** button. There is no sub oscillator and no filter high-pass in this mode; the third panel on the left reads **AUX OSC** with a note that Oracle uses its two-oscillator poly-mod path instead.

Mono: aggressive monophonic

Mono is modeled on a 70s Japanese monophonic synth voiced for aggression: two oscillators plus a sub, a fat driven filter, ring modulation, and hard sync. It is a single voice (one note at a time), which is exactly right for basses and leads that need weight and glide.



Figure 31: Mono mode: a grey panel with two oscillators plus the sub oscillator, and the RING and SYNC sub-panel carrying the ring modulation knob and the hard sync button

Mode-specific controls visible in Mono: the **Sub** oscillator (wave and level), and the mode sub-panel's **RING / SYNC** section with the **Ring Mod** knob (ring modulation between oscillators 1 and 2) and the **Hard Sync** button (oscillator 2 hard-syncs to oscillator 1 for tearing, vocal-formant timbres).

Modular: semi-modular

Modular is modeled on a 70s semi-modular synthesizer: three oscillators, a transistor-ladder filter, linear FM between oscillators, a sample and hold, and a real dispersive spring reverb. It is two-

voice, so you can play the occasional dyad, but it is at its best on evolving drones, sci-fi effects, and sequences.



Figure 32: Modular mode: a green panel with a third oscillator panel, the audio patch and filter sub-panel carrying sample and hold, oscillator cross routing and ring modulation, and the reverb section showing its fixed spring send

Mode-specific controls visible in Modular: the third oscillator (**Osc 3** wave and level) and, beside it, the **Osc 1 to Osc 2** knob that sets the linear FM amount. The mode sub-panel is titled **AUDIO PATCH / FILTER** and carries the patch points as four knobs:

- **S&H**: the sample and hold clock rate in Hz, which feeds the S&H modulation source.
- **Osc 2 to Osc 1**: cross modulation depth between the first two oscillators.
- **Osc 3 to VCF**: how far oscillator 3 modulates the filter cutoff, for audio-rate filter FM.
- **Ring**: ring modulation depth.

Below them sit a **SYNC** button (oscillator 2 hard-syncs to oscillator 1) and an **EARLY / LATE** selector that picks which revision of the transistor ladder the filter models. The late revision deliberately stops opening at around 12 kHz, the way the later hardware did; the early revision stays open across the full range. When Modular is active the effects reverb panel reads **FIXED SPRING** with its send percentage, to indicate the spring tank is engaged for that mode.

Prism: four-operator FM

Prism is a mid-80s four-operator FM digital engine. It is eight-voice polyphonic. Instead of oscillators and a filter shaping a rich waveform, FM builds timbre by having sine-wave operators

modulate each other's phase. The result is the glassy electric pianos, chiming bells, punchy basses, and metallic brass that defined 80s digital synthesis.

When Prism is active, the left column re-skins into an **operator matrix**: four stacked operator strips replace the standard oscillator panels. Each strip has these controls:

- **Ratio**: the operator's frequency as a multiple of the played note, stepped through a musical snap list.
- **Fine**: fine detune of the operator in cents, for slight beating and inharmonicity.
- **Level**: the operator's output. For a carrier this is volume; for a modulator this is modulation depth (FM brightness).
- **Vel**: how strongly note velocity affects that operator's level, so harder playing gets brighter.
- **Key Scale**: how the operator's level changes across the keyboard, for rolling a bright tine or edge off the top of the range.
- **A / D / S / R**: a full envelope per operator, so each partial can have its own attack and decay.

Operator 4 additionally carries the **Feedback** knob, which feeds the operator back into its own phase for growl and edge. The filter section stays in circuit in Prism, so you can still filter the FM tone if a preset opens the cutoff.

The mode sub-panel is the **ALGORITHM** widget: a numbered list of the eight routings down the left, and a large diagram of the selected one filling the rest of the panel. Click a name to switch. Hovering a name shows its number and title as a tooltip. See the FM guide below for what each algorithm does.

Acid: bass box and pattern sequencer

Acid is the bass box: one oscillator (saw or square) through a three-pole diode-ladder low-pass filter that screams near self-oscillation, with an accent circuit and note-to-note slide, driven by a sixteen-step pattern sequencer. It is monophonic. This is the sound of a thousand acid basslines.

Acid keeps the dark chassis and marks itself with an orange accent, but its left column is rearranged around the dedicated engine: **OSC 2** reads "single-oscillator engine", **AUX OSC** reads "no auxiliary oscillator", and an **ACID CHARACTER** panel takes their place with the Vintage knob, the oversampling selector, and the voice architecture buttons. On the right, **ACID ENGINE** draws the mono signal path of oscillator into an 18 dB filter into the VCA, and **ACID ROUTING** states that the modulation matrix is not in circuit here. The oscillator waveform combo defaults to saw and is intended to be saw or square, though the other waves remain available.

The shared-knob mapping in Acid. Rather than adding a wall of new controls, Acid re-purposes the standard panels you already know:

- **Filter Cutoff** sets the base brightness of the line, the knob you sweep for the classic wah.
- **Filter Resonance** is the squelch. Push it toward the top (around 0.8 and up) for the resonant whistle; near 0.95 the filter is at the edge of self-oscillation, which is the scream.
- **Filter Env Amt** is how far each note's envelope opens the filter, the pluck of the sound.
- **Amp Decay** (with Amp Sustain at zero) sets how quickly each step decays, from short and staccato to long and connected.



Figure 33: Prism mode: the left column replaced by the four-operator matrix, each operator carrying ratio, fine, level, velocity, key scaling and its own envelope, with the algorithm list and the active routing diagram on the right



Figure 34: Acid mode: a single-oscillator engine with oscillator 2 bypassed, the acid engine signal path of oscillator into an 18 dB filter into the VCA, accent and slide controls, and the pattern sequencer with its gate, pitch, accent and slide lanes

- The two Acid globals live in the mode sub-panel: **Acid Accent** sets how much an accented step boosts level, resonance, and envelope depth (the "wow" that makes accents jump out), and **Acid Slide** sets the glide time for slid steps.

The pattern itself lives in the sequencer, which expands to four lanes in Acid mode. See the sequencer chapter below for the gate, pitch, accent, and slide lanes.

Per-mode visibility summary

Control	Cosmos	Oracle	Mono	Modular	Prism	Acid
Standard OSC 1 / OSC 2 panels	yes	yes	yes	yes	hidden (operator matrix)	yes (saw/square)
Osc 3 wave and level	no	no	no	yes	no	no
Sub wave and level	yes	no	yes	no	no	no
Filter HP	yes	no	no	no	no	no
Cross Mod	yes	no	no	no	no	no
Poly-mod (6 knobs)	no	yes	no	no	no	no
Ring Mod	no	no	yes	yes	no	no
Hard Sync	no	yes	yes	yes	no	no
FM Amount	no	no	no	yes	no	no
S&H rate	no	no	no	yes	no	no
Ladder revision (Early / Late)	no	no	no	yes	no	no
Chorus I / II / I+II	yes	no	no	no	no	no
Operator strips and algorithm	no	no	no	no	yes	no
Acid globals and 4 sequencer lanes	no	no	no	no	no	yes

Controls that are irrelevant to a mode are hidden, not greyed out. A hidden widget only removes the on-screen control; the underlying parameter always stays in the host's parameter list, so automation lanes never break.

FM Guide (Prism)

FM synthesis in Prism uses four sine-wave **operators**. An operator that reaches the output is a **carrier** (you hear it). An operator that instead modulates another operator's phase is a **modulator** (you hear its effect on the carrier's timbre, not the operator itself). The **algorithm** is the wiring diagram that decides which operators are carriers and which modulate which.

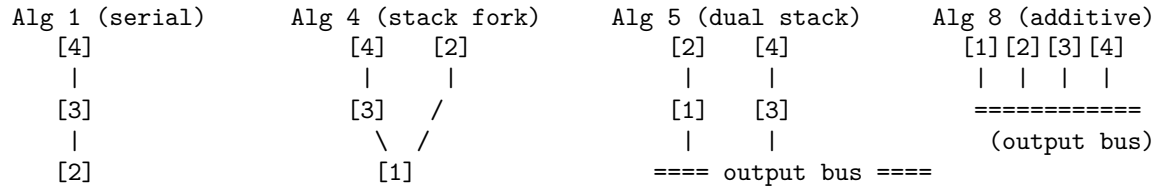
The two levers that shape an FM tone are operator **Level** and operator **Ratio**. Raising a modulator's Level adds harmonics and makes the carrier brighter (this is the FM equivalent of opening a filter). The modulator's Ratio decides where those harmonics land: integer ratios (1, 2, 3) give harmonic, pitched tones, while non-integer ratios (2.76, 5.4) give inharmonic, bell-like and metallic tones. Operator 4's **Feedback** turns operator 4 into a controllable noise and edge source, from a touch of grit to a full growl.

The eight algorithms

Operators are numbered 1 to 4. In the table, `a to b` means operator `a` modulates operator `b`, and the carriers are the operators you hear. Operator 4 is the feedback operator in every algorithm.

#	Name	Routing	Carriers	Character
1	Serial	4 to 3, 3 to 2, 2 to 1	1	One long modulation stack into a single carrier. The brightest and most metallic algorithm; the classic bright bass and lead voice.
2	Stack-2M	4 to 2, 3 to 2, 2 to 1	1	Two modulators feed one, which feeds the carrier. Rich and vocal, good for reeds and complex leads.
3	Split Stack	3 to 2, 2 to 1, 4 to 1	1	A two-operator stack and a second modulator meet at one carrier. Layered and harmonically dense.

#	Name	Routing	Carriers	Character
4	Stack Fork	4 to 3, 3 to 1, 2 to 1	1	A serial branch and a direct modulator meet at one carrier. Bright and responsive without the full serial chain.
5	Dual	2 to 1, 4 to 3	1, 3	Two independent two-operator stacks. Layer a body tone against a tine or edge; the workhorse e-piano algorithm.
6	Three Carrier	4 to 1, 4 to 2, 4 to 3	1, 2, 3	One modulator colors three parallel carriers. Broad, layered tones retain a strong pitched center.
7	Tri+FM	4 to 3	1, 2, 3	One modulated tone plus two clean carriers. Mostly additive with a single FM color; organ-like with an edge.
8	Additive	(none)	1, 2, 3, 4	Four parallel carriers, no modulation. Pure additive synthesis; drawbar-organ and formant tones.



```
   |                  ==== output ====
  [1]
==== output ====
```

The algorithm diagram widget in the Prism sub-panel draws the active routing live, brightens the carriers, and shows the feedback loop on operator 4 with a thickness that tracks the Feedback knob. Click any name in the list beside it to switch algorithms.

Sequencer and Arpeggiator

The sequencer strip along the bottom is an arpeggiator and step sequencer shared by all six modes. In modes Cosmos through Prism it behaves as a classic arpeggiator with sixteen step-mute cells; in Acid it expands into a four-lane pattern sequencer.

Arpeggiator controls

- **Arp On**: enable the arpeggiator or step sequencer.
- **Arp Mode**: the note order the arpeggiator plays: Up, Down, Up/Down, Down/Up, Random, Order (as played), or Chord. Chord is a known limitation in this release: it is selectable and stable, but it currently steps one note at a time exactly as Up does rather than striking every held note on each step. Use a held chord with the arpeggiator off if you want block chords.
- **Arp Octave**: how many octaves the arpeggio spans, 1 to 4.
- **Arp Rate**: the step length as a note division, from 1/1 down to 1/32, including dotted (1/8. and so on) and triplet (1/8T and so on) divisions.
- **Arp Gate**: the length of each step relative to its slot, from staccato to fully connected.
- **Arp Swing**: delays the off-beat steps for a swung feel while keeping the downbeats on the grid.
- **Arp Latch**: holds the pattern after you release the keys, so it keeps playing hands-free.
- **Arp Vel Mode**: where each step's velocity comes from: As Played, Fixed, or Accent.
- **Arp Fixed Vel**: the velocity used when the mode is Fixed.
- **Arp Accent Pattern**: the accent shape, which appears under the velocity selector once the velocity mode is Accent. Downbeat accents every fourth step, Every Other alternates loud and soft, Ramp Up climbs from soft to loud and Ramp Down falls from loud to soft. The shape is laid over the sixteen-cell step grid rather than over the held notes, so it stays put no matter how many keys you hold: Downbeat accents cells 1, 5, 9 and 13, and the two ramps run over eight cells and repeat twice across the row. The control has no effect in the other two velocity modes, which is why it is only shown in Accent. In Acid mode the arpeggiator (and with it this control) is not used at all; the pattern sequencer has its own per-step accent lane, though the parameter itself remains visible to the host.
- **Step cells**: the sixteen cells turn individual steps on or off (step-mutes) in every mode.

Host sync behavior

The arpeggiator always follows the host tempo. The step length is computed from the host BPM and the Arp Rate division, so at 128 BPM a 1/8 step is the correct length automatically, whether or not the transport is running. Auditioning with the transport stopped still steps at the host tempo.

When the DAW transport is playing and reports a valid song position, the arpeggiator phase-locks to the absolute host beat grid. Steps land on the beat rather than on wherever you happened to press the key, and the pattern re-syncs cleanly on a loop wrap so it never drifts over a long section. This is a quantized start: the first step of a held chord snaps to the grid.

When the transport is stopped, the arpeggiator free-runs from the moment you play, still at the host tempo, so you can practice or sound-design without pressing play. Switching between the locked and free states cleanly releases any sounding note so a transport change can never leave a note stuck on.

Acid pattern lanes

In Acid mode the sequencer section retitles itself **PATTERN SEQUENCER** and expands into four labelled lanes that together define the sixteen-step pattern. You hold a single root note and the pattern transposes with it.

- **GATE** (top): the sixteen on/off cells, the same step-mutes used in the other modes. A muted step is a rest.
- **PITCH**: sixteen vertical drag columns, one per step, setting each step's pitch offset from -24 to +24 semitones relative to the held root. Drag a column up or down; the bar fills from the center zero line. This is where the melodic shape of the line lives.
- **ACC**: sixteen cells. An accented step jumps out using the Acid Accent amount (louder, more resonant, more envelope).
- **SLIDE** (bottom): sixteen cells. A slid step glides the pitch into that step over the Acid Slide time, for the legato portamento that defines the acid sound.

The live step position highlights the current column across all four lanes as the pattern plays.

LFOs

There are two LFOs, and every voice owns its own pair, so they are per-note rather than global. Each has a **Shape** (Sine, Triangle, Square, Sample and Hold, or Random Smooth), a **Rate** from 0.01 to 50 Hz, a **Fade In** from 0 to 5 seconds for modulation that arrives after the attack, and a **Sync** switch.

Nothing is hardwired. An LFO does not reach pitch, filter, or amplitude until you route it in the mod matrix, which is also true of the mod wheel. There is no built-in vibrato path to work around.

LFO sync

Sync locks the LFO's phase to the host's song position, not just its speed. With the transport playing and a valid song position, the LFO's phase is computed from the bar position, so it lands the same way on every pass, re-anchors after a seek or a loop wrap, and cannot drift against the bar over a long section. This is the part that a plain rate multiplier never gives you: rate scaling alone gets the average speed right but leaves the phase wherever the last note happened to start it.

Under sync the cycle length comes from the Rate knob rather than from a separate division list. Rate 2.0 is one cycle per quarter note, 1.0 is one cycle per half note, 4.0 is one cycle per eighth note; the relationship is tempo-invariant, so the LFO tracks a tempo change automatically. There is no note-division selector on the LFOs (the arpeggiator and the delay have those).

Two behaviors are worth knowing:

- **A note-on does not retrigger a locked LFO.** That is the point of locking: every voice you play joins the same running phase, so a chord modulates in step instead of each note starting its own sweep. Without sync, a note-on does reset the LFO to the start of its cycle. The fade-in restarts either way.

- **Sync free-runs when the transport is stopped.** You still get a tempo-correct rate for auditioning; the phase lock engages when you press play. The lock is eased in over up to a quarter second rather than snapped, so engaging the transport does not jump a slow modulation.

While phase-locked, **LFO1 Rate / LFO2 Rate modulation routes do not change the cycle length**; the Rate knob and host grid set it. Those routes affect speed again when the transport stops or Sync is off. For Random Smooth, rate modulation still changes the smoothing speed while locked.

Mod Matrix

The mod matrix is an eight-slot modulation router, opened from the **MOD MATRIX** bar on the right (the bar shows how many slots are active). Each slot connects one source to one destination with a bipolar amount, so you can invert a modulation by using a negative amount. All eight slots are always available in every mode.

Sources

Source	What it is
None	Slot inactive.
LFO 1 / LFO 2	The two low-frequency oscillators.
Filt Env	The filter envelope.
Mod Whl	The MIDI modulation wheel.
Aftertouch	Key pressure. Channel pressure moves every voice; polyphonic key pressure moves only the voice playing that note.
Velocity	Note-on velocity.
Key Track	Note pitch across the keyboard.
Random	A new random value per note.
Pitch Bend	The pitch-bend wheel (in addition to its normal pitch effect).
S&H	The sample-and-hold value.

Destinations

Destination	What it modulates
None	Slot inactive.
Osc1 Pitch / Osc2 Pitch	Oscillator 1 or 2 frequency.
Osc1 PW / Osc2 PW	Oscillator 1 or 2 pulse width.
Cutoff	Filter cutoff frequency.
Reso	Filter resonance.
Amp	Voice amplitude.
Pan	Stereo position.
LFO1 Rate / LFO2 Rate	The speed of LFO 1 or 2.
FX Mix	The wet/dry balance of the effects.
Uni Det	The unison detune spread.

A classic starting patch is LFO 1 to Cutoff for a filter wobble, or Key Track to Cutoff so the filter opens as you play higher. Aftertouch to LFO1 Rate gives expressive vibrato that speeds up under pressure.

Presets

The preset menu and the browser

The top bar carries the preset controls: previous and next arrows around the preset name, a **BROWSE** button, and a **SAVE** button. The name itself is a drop-down menu listing the 54 factory presets, then a separator, then your own user presets. The arrows step through factory and user presets as one combined list and wrap around at the ends.

The drop-down is a list you scroll. The browser is the library you read. **BROWSE** opens a full-window preset browser built for finding a patch rather than scrolling past it:

- **Search.** The FIND field is focused the moment the browser opens, so you can just start typing. Matching is case-insensitive on any part of the name.
- **Mode chips.** Seven chips (ALL plus the six mode names) narrow the list to one circuit.
- **Bank chips.** ALL, FACTORY, or USER, so you can look at only your own patches. Each cell is also tagged with its bank and its mode.
- **Keyboard.** Up and Down move a whole row and keep working while you are typing in the search field; Left and Right move one preset when no field has focus. Enter loads the highlighted preset and closes the browser.
- **Esc happens in two stages.** The first press clears a typed search; only once the search is empty does a press close the browser. This means Esc never throws away your place in the list by accident.
- **Mouse.** A single click loads the preset and leaves the browser open, so you can audition down a list while the panel re-skins around you. A double click (or APPLY) loads and closes. The close button in the title bar, the CLOSE button, and a click outside all close the browser without loading.

The footer shows how many presets the current filters match. The browser reads the preset library as it was last scanned; see the note on rescanning below.

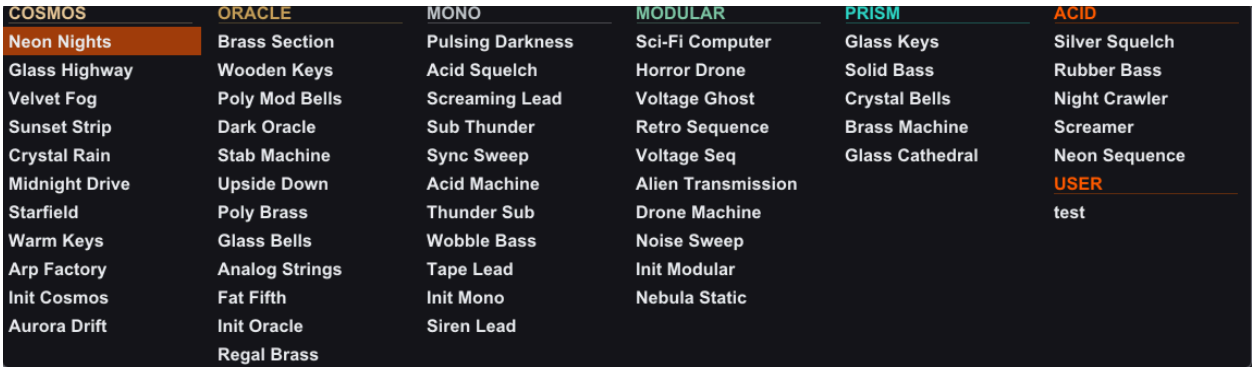


Figure 35: The preset browser: factory presets laid out in a column per mode, Cosmos through Acid, with the user library in its own column

The user preset library

Beyond the 54 factory presets, Sunset Circuits keeps a personal preset library you build yourself, stored as files on disk independent of any DAW session, so your patches follow you between projects and hosts. The library is not saved into the host session; it lives in your own preset folder and is the same library in every project and every DAW.

Saving a preset

Click the **SAVE** button in the top bar, to the right of BROWSE. A name-entry box appears; type a name and confirm (Enter commits, Esc closes). The current state of all parameters is written to a file, and the new preset appears in the preset menu below a separator, in its own user section under the factory presets.

If a preset with that name already exists, the box says so before you commit, and SAVE turns into an OVERWRITE and CANCEL pair so the overwrite is always a second, deliberate click. When the patch you are currently on is a user preset, the same box offers DELETE, which likewise asks for confirmation before it removes the file.

Loading a user preset works exactly like a factory preset: pick it from the menu, find it in the browser, or step to it with the previous and next arrows, and every parameter (including the mode) is restored. The arrows step through the factory and user presets as one combined list and wrap around at the ends.

The preset folder is scanned when the plugin opens and after each save or delete, not on every menu open. If you add or remove `.scpreset` files by hand while the plugin is running, reopen the plugin to see the change.

File location

User presets are plain files in a per-user application-data folder:

```
Linux:   ~/.config/DuskAudio/SunsetCircuits/presets/
macOS:   ~/Library/Application Support/DuskAudio/SunsetCircuits/presets/
Windows: %APPDATA%\DuskAudio\SunsetCircuits\presets\
```

On Linux, if `XDG_CONFIG_HOME` is set it is used in place of `~/.config`. Each preset is one file named after the preset with the extension `.scpreset`.

The .scpreset format

A `.scpreset` file is versioned plain text: a header line, a `name=` line, and one `symbol=value` line per parameter. It is human-readable and safe to back up, copy between machines, or share. Because it is plain text, you can inspect or hand-edit a patch if you want, though the plugin is the intended way to create them.

Loading is deliberately forgiving in one direction and strict in the other. Any parameter the file does not mention keeps its factory default, and any symbol the plugin does not recognize is skipped, so presets stay forward-compatible as the instrument grows. Values are clamped to each parameter's legal range, and a value that is not a finite number is rejected. A file whose header is missing or carries an unknown format version is rejected outright rather than loaded incorrectly. The library loads up to 512 presets in alphabetical order.

Effects

Every mode runs through the same effects chain at the bottom right: Drive, Chorus, Delay, and Reverb, each with its own enable button. The Cosmos mode additionally has its own built-in vintage chorus, separate from the effects chorus.

Drive

A saturation stage with three characters selected by **Drive Type**: **Soft** (gentle tube-like warmth), **Hard** (aggressive clipping), and **Tube** (asymmetric tube drive). **Drive Amount** sets how hard it pushes and **Drive Mix** blends the driven signal against the clean one. Use it for grit on basses and leads, or a light Soft setting to thicken pads.

Chorus

A standard stereo chorus with **Rate**, **Depth**, and **Mix**. This is the global chorus available in every mode. It widens and animates any sound.

Vintage chorus (Cosmos)

Cosmos has its own bucket-brigade chorus built into the circuit, controlled by the three **I / II / I+II** buttons in the mode sub-panel rather than by the effects Chorus. Its stereo delay sweeps and filtered wet paths are calibrated against reference captures. You can run it alongside the effects chorus, but usually one or the other is plenty.

Delay

A stereo delay with **Feedback** and **Mix**, plus **Ping-Pong** and **Tape** options. Ping-Pong bounces the echoes across the stereo field; Tape adds tape-style warmth and saturation to the repeats. The delay can run free or synced:

- **Free**: the **Delay Time** knob sets the delay in milliseconds, up to 2000 ms.
- **Synced** (Delay Sync on): the **Delay Division** knob sets the delay as a note division locked to the host tempo.

The delay buffer holds 2 seconds. In synced mode at a slow tempo, a long division can ask for more than 2 seconds (for example a dotted half note below about 90 BPM); the delay is clamped to the 2-second buffer, so at very slow tempi the longest divisions stop tracking the tempo and hold at 2 seconds. Free mode is already capped at the 2000 ms knob maximum.

Reverb

An algorithmic reverb with **Size** (0 to 1), **Decay** (0.1 to 20 seconds), **Damping**, **Pre-Delay** (up to 200 ms), and **Mix**. It is the same reverb in all six modes.

In **Modular** mode a real dispersive **spring** tank is switched in after it, at a fixed light mix, matching the spring of the semi-modular hardware; the panel shows SPRING to indicate this. Note that the spring is added to the chain rather than replacing the reverb, so in Modular you have both: the main reverb still responds to its own controls, and the spring sits behind it. The spring gives the boingy, dispersive character that a digital reverb cannot, and it is part of what makes Modular sound like a patchable vintage instrument. If you want the spring alone in Modular, turn the main reverb's enable off.

Performance Notes

Oversampling

The **Oversampling** control sets the internal processing rate: **1x**, **2x**, or **4x**. Higher factors reject aliasing in the oscillators, the filter, and the FM operators at the cost of more CPU. The voices render at the oversampled rate; the effects run at the host rate.

- **1x**: lowest CPU, most aliasing on bright and high-pitched sounds. Fine for pads and bass.
- **2x**: the default and the recommended everyday setting. A good balance of clean high end and reasonable CPU.
- **4x**: the cleanest high end, for exposed leads, bright FM tones, and high notes where aliasing would be audible. Use it when you can afford the CPU.

Factory presets ship at 2x.

Reported latency

Oversampling adds a small, fixed processing latency from the decimation filter, which the plugin reports to the host so plugin delay compensation keeps everything aligned:

Oversampling	Reported latency
1x	0 samples
2x	12 samples
4x	14 samples

These are host samples and are independent of the sample rate. If other tracks sound slightly out of time when you change oversampling, confirm plugin delay compensation is enabled in your DAW.

CPU

CPU cost scales with the number of sounding voices, the oversampling factor, and how many effects are engaged. The oscillators, filter, and FM operators run at the oversampled rate, so 4x costs roughly twice what 2x costs for the same patch; the effects run once at the host rate regardless.

The figures below are single-core measurements taken at 48 kHz with a 512-sample buffer, expressed as a percentage of one CPU core. They come from an offline profile of the engine's processing loop on an Intel Core i7-8809G (4 cores, 3.1 GHz base), so treat them as a guide; a faster or slower machine will scale the numbers accordingly, and a smaller buffer raises the per-block overhead slightly.

Patch	Oversampling	CPU (one core)
8-voice Prism FM with drive, chorus, delay, and reverb	4x	about 50%
6-voice Cosmos with 8x unison, dual chorus, and all effects	2x	about 22%
4-voice Oracle pad with reverb	2x	about 15%
Acid sequence with drive, delay, and reverb	2x	about 3%

The first row is the worst case among ordinary patches: a full 8-voice FM patch, every effect on, at the highest oversampling factor. Even there it uses about half of one core, so a typical session runs several instances comfortably. If you need more headroom, drop from 4x to 2x oversampling first, since that is where most of the voice cost lives, then thin out effects you are not using.

One case does run out of headroom: Prism with heavy unison at 4x. Prism's cost is per operator bank, and unison multiplies the bank count rather than sharing one. A patch that keeps sixteen operator banks running at 4x sits at essentially the whole of one core on the reference machine above, and the transient peak while a bulk of voices retires is in the same region. It is a known limit rather than a bug, and it is a wall you have to go looking for; nothing in the 54 factory presets reaches it. If a Prism patch of yours crackles, take it to 2x, or reduce the unison count before you reduce polyphony. The other five circuits do not have a comparable ceiling.

Parameters and automation

Every control is a normal host parameter: 223 automatable parameters, plus two output-only level meters the host can read but not write, for 225 host parameters in total (229 ports in the LV2 build, counting audio, MIDI, and the latency report). Oversampling is one of the 223, so it automates like anything else. Controls that a mode hides are still present in the host's parameter list, so an automation lane never breaks when you switch circuits.

Continuous parameters are smoothed, so automating or dragging them does not produce zipper noise. The smoother settles with an 8 ms time constant, short enough that a knob still feels immediate. Filter cutoffs are smoothed in the log-frequency domain rather than in Hz, so a sweep sounds even across the range instead of racing through the top octaves. The smoothed set covers the level, filter and effect-mix controls; the rest step per block, which for most of them is inaudible. A few are deliberately left unsmoothed because smoothing them would be worse than the step: oscillator and unison detune, pulse width, and the LFO and chorus rates all set a phase increment or an edge position, where a ramp is a pitch glide nobody asked for. Stepped controls (waveforms, modes, sequencer cells, mod-matrix routings) switch outright, as they should.

Loading a preset snaps rather than glides. The plugin tells the engine explicitly that a program change happened, so all 223 values land at once and a preset sounds correct on its first note instead of sliding into place over the first fraction of a second. For hosts that replay a patch as a burst of ordinary parameter writes rather than a program change, the engine detects the burst and snaps anyway.

Known limits

- **Prism with heavy unison at 4x oversampling** can exceed one core. See the CPU section above.
- **Aliasing on the voice path** is measured and bounded, not eliminated. At 2x the worst alias images sit around -47 dBc, which is inaudible on normal material; at 1x they are considerably higher and 1x is best reserved for pads and bass. 4x is there for exposed bright leads and high FM tones.
- **The effects run at the host rate**, downstream of the voice decimation, so they are not covered by the Oversampling control. Drive in particular is a nonlinearity at host rate: pushed hard on an already bright, wide-open source it can fold a little energy back down the spectrum, and raising Oversampling will not help because the voice path was never the problem. If a driven lead sounds gritty in a way you do not want, back off Drive Amount or the Drive Mix, or darken the source with the filter before the drive rather than after. Nothing in the factory bank is pushed into that region.

- **Chord arpeggiator mode** currently plays like Up. See the arpeggiator section.
- **The font atlas is baked when the plugin window opens**, so text can look slightly soft after you move the window to a display with a different pixel density, until you close and reopen the plugin.
- **MIDI program change under LV2 with the window closed** does not tell the host which parameters moved. See the MIDI implementation section.

Factory Presets

Sunset Circuits ships with 54 factory presets. They are grouped by mode below for reading, but they are **not** stored in mode order: the menu, the previous and next arrows, and MIDI program numbers 0 to 53 run through the bank in its own order, which interleaves the modes. Use the browser's mode chips when you want to see one circuit's presets together.

Two names are easy to misread. **Acid Squelch** and **Acid Machine** are Mono-mode presets that happen to go for an acid sound; the five presets that actually use the Acid circuit are listed under Acid at the end.

Cosmos (six-voice DCO poly)

- **Neon Nights**: warm detuned DCO pad with dual chorus and a long release.
- **Glass Highway**: bright chorused arpeggio with a touch of delay.
- **Velvet Fog**: soft, dark, vintage pad for beds and atmospheres.
- **Sunset Strip**: wide, heavily detuned pad with full dual chorus.
- **Crystal Rain**: sparkly fast arpeggio through delay and reverb.
- **Midnight Drive**: driven chorused pad with tape delay, cinematic and wide.
- **Starfield**: octave-spanning sparkle arpeggio with delay and reverb.
- **Warm Keys**: mellow triangle-and-sine keys with single chorus.
- **Arp Factory**: swung up-down arpeggio across three octaves.
- **Aurora Drift**: huge slow-swelling pad, full chorus, very wide.
- **Init Cosmos**: a clean starting point for the Cosmos engine.

Oracle (five-voice poly, poly-mod)

- **Brass Section**: poly-mod brass with filter-envelope bite.
- **Wooden Keys**: mellow plucked keys, short and woody.
- **Poly Mod Bells**: inharmonic bells driven by oscillator-to-oscillator poly-mod.
- **Dark Oracle**: dark sustained pad with vintage character.
- **Stab Machine**: short percussive poly stab.
- **Upside Down**: the flagship Up/Down latched saw arpeggio, the famous 80s sci-fi title-sequence sound out of the box.
- **Poly Brass**: fuller poly-mod brass ensemble.
- **Glass Bells**: bright long-decay bells with poly-mod and reverb.
- **Analog Strings**: four-voice unison string pad with chorus.
- **Fat Fifth**: unison fifths, wide and thick.
- **Regal Brass**: self-oscillating poly-mod brass with two-voice unison.
- **Init Oracle**: a clean starting point for the Oracle engine.

Mono (aggressive monophonic)

- **Pulsing Darkness**: pulsing arpeggiated sub-bass, dark and vintage.
- **Acid Squelch**: resonant squelching bass line with portamento.

- **Screaming Lead**: loud driven lead with delay.
- **Sub Thunder**: deep sine-and-sub sub-bass.
- **Sync Sweep**: ring-modulated hard-sync sweep.
- **Acid Machine**: glide acid arpeggio, high resonance.
- **Thunder Sub**: square-sub sub-bass, tuned low.
- **Wobble Bass**: LFO-to-cutoff wobble bass.
- **Tape Lead**: glide lead through tape delay.
- **Siren Lead**: hard-sync screaming lead, driven, with synced delay and glide.
- **Init Mono**: a clean starting point for the Mono engine.

Modular (semi-modular)

- **Sci-Fi Computer**: random FM arpeggio with hard sync.
- **Horror Drone**: slow evolving ring-modulated drone with a long reverb tail.
- **Voltage Ghost**: FM and sync sweep drone with tape delay.
- **Retro Sequence**: tape-delay sequence with filter-envelope movement.
- **Voltage Seq**: sample-and-hold sequence across two octaves.
- **Alien Transmission**: ring, FM, and sync texture with a long reverb.
- **Drone Machine**: huge slow FM drone with LFO cutoff movement.
- **Noise Sweep**: filtered-noise sweep for risers and effects.
- **Nebula Static**: sample-and-hold texture modulating cutoff and resonance, with spring and hall reverb.
- **Init Modular**: a clean starting point for the Modular engine.

Prism (four-operator FM)

- **Glass Keys**: dual-stack tine electric piano, the tine rolling off up the keyboard.
- **Solid Bass**: serial FM bass with a touch of feedback grit and a filter-envelope pluck.
- **Crystal Bells**: additive inharmonic bells with staggered decays and a reverb tail.
- **Brass Machine**: serial FM brass with strong feedback growl and an attack swell.
- **Glass Cathedral**: dual-stack FM pad with a long swell, chorus, and big reverb.

Acid (bass box and pattern sequencer)

- **Silver Squelch**: classic high-resonance saw line, accents and slides, rolling 1/16 pattern.
- **Rubber Bass**: square wave, low resonance, tight fast decay for a round rubbery bounce.
- **Night Crawler**: slow dark 1/8 pattern drenched in long slides.
- **Screamer**: near-self-oscillating resonance, maxed accent, drive on, the aggressive scream.
- **Neon Sequence**: resonant saw line with a ping-pong tempo-synced delay groove.

Troubleshooting

A note is stuck on. Sunset Circuits guards against stuck notes: when the arpeggiator switches between its host-locked and free-running clocks, or when you start or stop the transport, any sounding note is released cleanly. If you ever hear a hung note, sending an all-notes-off from your DAW (or stopping and starting the transport) clears it, but this should not happen in normal use.

Switching modes did not change the sound the way I expected. Switching the mode by hand keeps all other parameters where they were, so the new circuit plays with your old settings. Load a preset for that mode, or dial in the mode-specific controls, to hear the circuit as intended. Loading a preset always sets the mode along with everything else.

A mode-specific control disappeared. Controls that do not apply to the current mode are hidden, not disabled. For example, the poly-mod knobs only appear in Oracle, and the operator strips only appear in Prism. The parameters still exist for automation; only the on-screen widgets change per mode.

The synced delay is not tracking a very slow tempo. The delay buffer is 2 seconds. At slow tempi the longest note divisions ask for more than 2 seconds and are clamped to the buffer, so they hold at 2 seconds instead of tracking further. Use a shorter division or the free (milliseconds) mode if you need the delay to follow a very slow tempo exactly.

Bright or high notes sound harsh or buzzy. That is aliasing. Raise the Oversampling from 1x to 2x or 4x. FM tones in Prism and high-resonance filter sweeps benefit the most. If raising oversampling changes nothing, the grit is coming from the effects rather than from the voice: Drive runs at the host rate and is not oversampled, so back off Drive Amount or Drive Mix instead.

The interface looks small, or the knob values disappeared. The UI scales with the window: drag the window corner to resize it and the whole panel scales to fit while keeping its aspect ratio. The smallest size is 620 by 390. The persistent value read-outs need legible room and switch off below about 893 by 562, leaving a hover bubble in their place, so if the numbers under the knobs have vanished the window is simply too small.

The text looks slightly soft. The font atlas is baked once when the plugin window opens, at the pixel density of the display it opened on. Moving the window to a display with a different density, or zooming a long way from the size it opened at, softens the text until you close and reopen the plugin. A live atlas rebuild is planned for a future release.

My host shows extra latency. Oversampling at 2x or 4x reports a small fixed latency (12 or 14 samples) so the host can compensate. Confirm plugin delay compensation is enabled in your DAW if other tracks sound out of sync.

Tape Echo 2 User Manual

Version 1.0.0

Tape Echo 2 is a three-head tape-delay and spring-reverb effect. It models the mechanical timing, record and playback coloration, regeneration, tape wear, splice dropout, input/output amplifiers, and independent echo and spring returns of a classic hardware echo workflow.

Signal flow

The stereo input is summed to the modeled record and spring paths. Input drives both paths. Record Input interrupts the tape-record feed while the spring remains live, which makes dub-style echo tails and reverb-only operation possible. The selected playback heads feed both the echo output and the regeneration loop. Bass and Treble affect only the echo return. Echo and spring then receive their own level and pan settings. Mix crossfades the direct path against the combined echo and spring returns before Output applies the final stereo trim.

The VU and peak lamp monitor the record path after Input and include regeneration. They remain active when Record Input is off and go dark when Power is off.

Control names in this manual are the legends printed on the panel. Seven host automation parameters carry a different name from the legend that drives them, because those names were fixed by the 0.1 series and are kept so old projects keep loading; each one is noted in parentheses where the control is described.

Controls

Echo section

- **Head Select** (automation name: Mode) chooses one of 12 head/spring routings:
  1. Head 1
  2. Head 2
  3. Head 3
  4. Heads 2 + 3
  5. Head 1 + Reverb
  6. Head 2 + Reverb
  7. Head 3 + Reverb
  8. Heads 1 + 2 + Reverb
  9. Heads 2 + 3 + Reverb
  10. Heads 1 + 3 + Reverb
  11. Heads 1 + 2 + 3 + Reverb
  12. Reverb Only
- **Repeat Rate** controls tape speed. Head 1 spans approximately 177 ms at the slow end to 69 ms at the fast end. The corresponding Head 2 range is 337 to 131 ms and Head 3 is 489 to 189 ms.
- **Intensity** controls regeneration. High values can self-oscillate and become substantially louder than the input.
- **Echo Level** (automation name: Echo Volume) sets echo-return level.
- **Reverb Level** (automation name: Reverb Volume) sets spring-return level in modes 5 through 12.
- **Bass** and **Treble** shape the echo return without changing the dry or spring paths.

Record and transport section

- **Input** (automation name: Input Volume) sets record/spring send gain and drives the modeled input amplifier and tape nonlinearity.
- **Record Input** (automation name: Input Send) toggles new material into the tape loop. Switching it off lets existing repeats decay while the dry and spring paths continue.
- **Tape Age** selects **New**, **Used**, or **Old** tape. New retains a very quiet tape/electronics bed; older states progressively add bandwidth loss, noise, transport variation, level wear, and a more audible tape splice.
- **Noise** raises the tape and electronics noise (hum, low machine bed, and hiss together) by up to 60 dB. At minimum the noise sits at its very quiet captured level; Tape Age still scales it, so Old is louder than New at any setting. The noise follows the echo path: Bass, Treble, Echo Level, Echo Pan, Mix, and Output all act on it.
- **Hum** sets the mains hum to **60 Hz** or **50 Hz** to match your region. It is not stored in presets, so recalling a factory or user preset leaves it where you set it. Loading a preset through an LV2 host's own preset menu writes every control and returns it to 60 Hz.
- **Wow & Flutter** extends the intrinsic transport movement from subtle mechanical motion to a creative wobble range.
- **Power** is the host-designated bypass and uses a click-free transition.

Tempo and mix section

- **Tempo Sync** derives delay from the host tempo. The rhythmic value belongs to the leading active head in the selected mode. Head 1 spans roughly 70 to 178 ms, and a note longer than the tape can physically carry is held at that maximum rather than transposed, exactly as the hardware behaves. Longer divisions therefore converge on the same endpoint timing at a given tempo, and slow tempos reach the limit sooner. Use a shorter division, or a multi-head mode, when you need the repeat to track the bar at low tempos.
- While Tempo Sync is enabled, the **Repeat Rate** knob becomes the note selector (automation name: Echo Rate Note). It keeps its panel legend and steps through eleven detents instead of sweeping continuously. Each detent selects a rhythmic value for the leading active head, so the available values change with Head Select. The three head displays show the exact `d` (dotted), `t` (triplet), `+`/`-`, and out-of-range blinking indications used by the reference unit. Changing Head Select keeps the knob at the same detent and updates its note assignment. The older Sync Division parameter remains hidden so projects and automation saved by the 0.1 series continue to load correctly.
- **Mix** crossfades the direct signal against the combined echo and spring returns. At 0% the output is dry-only, 50% preserves both paths at unity, and 100% is fully wet for send/return use.
- **Echo Pan** and **Reverb Pan** place the two wet returns independently.
- **Output** (automation name: Output Volume) provides −20 dB to +20 dB after the complete mix; its midpoint is unity gain.

Factory presets

Preset	Purpose
Default	Neutral Head 1 starting point
Slapback Vocal	Short, full-level vocal slap
Rockabilly Guitar	Bright single-head rockabilly echo

Preset	Purpose
Classic Tape Echo	Medium Head 2 echo
Dub Throw	Long feedback throw with a dark return
Synced 1/8 Dub	Host-synced Head 2 dub echo
Multi-Head Bounce	Heads 2 + 3 with spring and right-panned echo
Orbital Echo	Host-synced Heads 1 + 2 with split wet panning
Full Wash	Old-tape three-head and spring wash
Ambient Trails	Old-tape Head 3 and spring ambience
Worn Tape	Old-tape pitch and tone character
Runaway Drone	High-feedback Heads 1 + 3 texture
Spring Only	Head Select 12 (Reverb Only) spring-reverb return

Factory programs never change host bypass or restore meter outputs. The two synced programs use host tempo; all other programs use the stored free-time rate.

Host and format notes

Tape Echo 2 is available as VST3, CLAP, LV2, and Audio Unit on supported platforms. It accepts mono or stereo program material and produces stereo output. Tempo Sync requires a host that supplies a valid tempo; 120 BPM is used as a safe fallback when tempo information is unavailable.

By default, macOS builds are ad-hoc signed and are not notarized. The first time a downloaded build is loaded, Gatekeeper may report that it cannot verify the developer. Approve the plugin once under System Settings > Privacy & Security ("Open Anyway"), or remove the quarantine attribute from the installed bundle in Terminal, for example: `xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/Components/tape-echo-2.component`. Rescan or restart the host afterward; Logic Pro users should re-run the Audio Unit scan.

When `MACOS_SIGNING_ENABLED=true` and the required Apple signing and notarization secrets are configured, the release workflow instead produces Developer ID signed and notarized AU, VST3, and CLAP bundles. The LV2 bundle is Developer ID signed but is not submitted for notarization.

Parameter IDs from the 0.1 series are preserved. Mix was appended after all previously shipped IDs. The hidden Dry Level parameter retains its original meaning for old project state and automation. Tape Age snaps stored continuous values to the nearest New/Used/Old state. Noise and Hum Frequency were appended in the same way; projects saved before they existed load with the captured noise level and 60 Hz hum, so they sound unchanged.

Gain and safety

Self-oscillation is an intentional part of the model and can exceed 0 dBFS. Lower Intensity or Echo Level before changing modes on loud material, and use Output for final gain staging. Power clears the modeled loop so stale regeneration does not return when the effect is enabled again.

License and support

Tape Echo 2 is released under GPL-3.0-or-later. Third-party component notices are included with release archives. Support and source information are available at https://dusk-audio.github.io/.

TapeMachine 2

Overview

TapeMachine 2 combines two tape-deck models in one stereo effect. **Swiss** provides tracking and mix coloration, while **American** provides a mastering deck topology with additional transport and electronics options. The machine, speed, tape formulation, equalization, bias, and signal path all change the modeled recording and playback chain.



Figure 36: TapeMachine 2 main interface: dual VU meters, the machine and tape selector row, and the gain staging, tape character, transport, and filter controls

The factory bank contains 20 calibrated presets. They are useful starting points, but Input, Bias, transport modulation, filters, and the Advanced reproduce EQ remain fully editable.

TapeMachine 2 has a distinct plugin ID from TapeMachine 1.x, so both can be installed for old-session compatibility.

Quick start

1. Insert TapeMachine 2 on a stereo track or bus.
2. Choose **Classic Rock Crisp** for restrained Swiss color or **Big 456 Master** for American mastering color.
3. Leave **Gain Link** on and raise **Input**. The linked Output control moves in the opposite direction so louder is not mistaken for better.
4. Compare with the top-right **Bypass** button.
5. For stronger color, try **Thick Saturation**, **Drum Bus**, or **Fat 456 Master**. For deliberate degradation, try **Sunbaked Cassette** or **Old Tape**.

Machine and tape controls

Machine

- **Swiss** is the tracking and mix model. It supports 7.5, 15, and 30 IPS.
- **American** is the mastering model. It adds 3.75 IPS, head width, crosstalk, a master wow/flutter switch, and transformer switching.

Tape speed

- **3.75 IPS** is American only; it has narrow bandwidth and the strongest slow-speed character.
- **7.5 IPS** gives pronounced head and bandwidth character.
- **15 IPS** is the general-purpose reference speed.
- **30 IPS** gives tighter low-frequency behavior and extended bandwidth.

Tape type

- **456**: classic general-purpose formulation.
- **GP9**: high-output formulation.
- **900**: clean modern formulation.
- **250**: older, more colored formulation.

Signal path

- **Repro**: complete record/tape/reproduce path.
- **Sync**: playback through the sync-head geometry.
- **Input**: electronics and EQ without tape saturation, modulation, or noise.
- **Thru**: direct signal path for comparison.

EQ and bias

- **EQ Standard** selects NAB or CCIR.
- **Auto Bias** derives bias from the current tape and speed. Turn it off to use the Bias knob manually.

Main controls

Gain staging

- **Input** ranges from -12 to +12 dB and is the primary drive control.
- **Output** ranges from -12 to +12 dB.
- **Gain Link** applies inverse output compensation. With it on, Output acts as an additional makeup trim on top of the inverse Input gain.

Tape character

- **Bias** ranges from 0 to 100%. It is inactive while Auto Bias is on.
- **Noise** controls the modeled deck-specific hiss and hum. Zero disables it.

Transport

- **Wow** controls slow pitch drift.
- **Flutter** controls faster pitch variation.
- American also has a **W&F** master switch that gates both controls.

Filters

- **Highpass** ranges from 20 to 500 Hz; 20 Hz displays as Off.

- **Lowpass** ranges from 3 to 20 kHz; 20 kHz displays as Off.

The nonlinear core is fixed at its fitted 2× oversampling mode. There is no user-facing oversampling selector in TapeMachine 2.

Advanced panel

Click **Advanced** to open the reproduce-head controls:

- **Low**: 80 Hz shelf, ±12 dB
- **Lo-Mid**: 160 Hz peak, ±12 dB
- **Hi-Mid**: 5 kHz peak, ±12 dB
- **High**: 9 kHz shelf, ±12 dB

On American, the panel also exposes:

- **Crosstalk**: adjacent-track bleed modeling
- **XFMR**: output-transformer low-frequency and even-harmonic character

These controls are part of the factory-preset fits. Resetting them to zero can substantially change a preset's match and saturation reading.

Factory presets

American mastering

- Big 456 Master
- Nice 456 Master
- Jazz Vision Master
- Clean 900 Master

American color

- Fat 456 Master
- GP9 Drum Bus
- Bass Bump
- Bright & Sizzly

Swiss mix

- Classic Rock Crisp
- Modern Rock
- Drum Bus
- Hi-Fi Shine
- Lush Film
- Jazz Warmth

Swiss color

- Thick Saturation
- Hip-Hop Punch
- Vocal Presence

Lo-fi

- Sunbaked Cassette: American at 3.75 IPS
- Analog Warmth: American at 3.75 IPS
- Old Tape: Swiss at 7.5 IPS

The preset menu is split into American and Swiss columns. **Init** restores defaults. **Save** stores user presets in `~/.config/DuskAudio/TapeMachine2/presets`.

Practical workflows

Transparent mix color

Start with **Classic Rock Crisp** or **Modern Rock**. Leave wow, flutter, and noise at zero. Keep Gain Link on, then adjust Input while level-matched. Switch to 30 IPS if you want less low-speed character.

Drum-bus saturation

Start with **Drum Bus**, **GP9 Drum Bus**, or **Fat 456 Master**. Preserve the factory Advanced EQ at first. Adjust Input for density, then use Output only as the final makeup trim.

Strong tape color

Start with **Thick Saturation**. Its Input and Hi-Mid reproduce settings were jointly fitted: changing one affects both frequency response and measured THD. Use small moves and level-match before deciding.

Lo-fi treatment

Start with **Sunbaked Cassette** or **Old Tape**. Raise Wow, Flutter, or Noise for more instability; lower the Lowpass for a narrower result. Sunbaked uses the American-only 3.75 IPS mode.

Troubleshooting

Input gets louder but Output turns down. Gain Link is on. This is intentional level compensation. Turn it off for independent input and output gains.

Bias does not respond. Auto Bias is on. Disable it before adjusting Bias.

Wow and Flutter do nothing on American. Enable the W&F master switch.

I cannot select 3.75 IPS on Swiss. That speed is American-only.

I cannot find oversampling. TapeMachine 2 is permanently tuned at 2×. The legacy parameter is hidden and retained only for session-state compatibility.

The factory preset changed after zeroing Advanced EQ. The four reproduce EQ values are intentional parts of each fitted preset.

Formats and requirements

- macOS 10.15 or newer: AU, VST3, CLAP, LV2; universal arm64/x86_64
- Linux (x86_64 and arm64): VST3, CLAP, LV2
- Windows (x86_64): VST3, CLAP
- A Standalone build is not currently shipped

macOS builds are not code-signed or notarized. On first launch Gatekeeper will warn; either right-click the plugin and choose Open, or clear the quarantine flag with `xattr -dr com.apple.quarantine <bundle>`.

Each release archive contains the plugin bundles, the GPL LICENSE, third-party license notices, and the User Manual PDF. Every release also publishes a SHA256SUMS file for integrity checking.